



THE REPUBLIC OF KENYA

COMPETENCY-BASED MODULAR CURRICULUM

FOR

ELECTRICAL ENGINEERING

KNQF LEVEL 6

CYCLE 3

PROGRAMME ISCED CODE: 0713 554A



**TVET CDACC
P.O. BOX 15745-00100
NAIROBI**

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FOREWORD

Provision of quality education and training is fundamental to the Government's overall strategy for socio-economic development. Quality education and training contribute to achievement focused on Kenya's development blueprint and sustainable development goals.

Reforms in the education and training sector are necessary for achievement of Kenya Vision 2030 and meeting the provisions the Constitution of Kenya. The education sector had to be aligned to the Constitution and this resulted in formulation of the Policy Framework for Reforming Education and Training (Sessional Paper No. 1 of 2019). A key feature of this policy is the change in the design and delivery of TVET training. This policy document requires that training in TVET be competency based, curriculum development be industry led, certification be based on demonstration of competence and mode of delivery that allows for multiple entry and exit in TVET programs.

These reforms demand that Industry takes a leading role in curriculum development to ensure the curriculum addresses its competence needs. It is against this background that this Curriculum has been developed.

It is my conviction that this curriculum will play a great role towards development of competent human resource for the Electricity and Energy sector's growth and sustainable development.

**PRINCIPAL SECRETARY
STATE DEPARTMENT FOR TVET
MINISTRY OF EDUCATION**

PREFACE

Kenya Vision 2030 aims to transform Kenya into a newly industrializing middle-income country, providing high-quality life to all its citizens by the year 2030. Kenya intends to create globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through lifelong education and training. TVET has a responsibility to facilitate the process of inculcating knowledge, skills, and worker behaviour necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency-Based Education and Training (CBET).

TVET Act CAP 210A and Sessional Paper No. 1 of 2019 on Reforming Education and Training in Kenya for Sustainable Development emphasized the need to reform curriculum development, assessment, and certification. This called for a shift to CBET to address the mismatch between skills acquired through training and skills needed by industry, as well as increase the global competitiveness of the Kenyan labour force.

This curriculum has been developed in adherence to the Kenya National Qualifications Framework and CBETA standards and guidelines. The curriculum is designed and organized into Units of Learning with Learning Outcomes, suggested delivery methods, learning resources, and methods of assessing the trainee's achievement. In addition, the units of learning have been grouped in modules to concretize the skills acquisition process and streamline upskilling.

I am grateful to all expert trainers and everyone who played a role in translating the Occupational Standards into this competency-based modular curriculum

CHAIRPERSON, TVET CDACC

ACKNOWLEDGEMENT

This curriculum has been designed for competency-based training and has independent units of learning that allow the trainee flexibility in entry and exit. In developing the curriculum, significant involvement and support was received from industry and various organizations.

I appreciate Electrical Engineering National Sector Skills Committee who enabled the development of this curriculum. I recognize with appreciation the role of the SSC in ensuring that competencies required by the industry are addressed in this curriculum.

I also thank all stakeholders in the Electricity and Energy sector for their valuable input and all those who participated in the process of developing this curriculum.

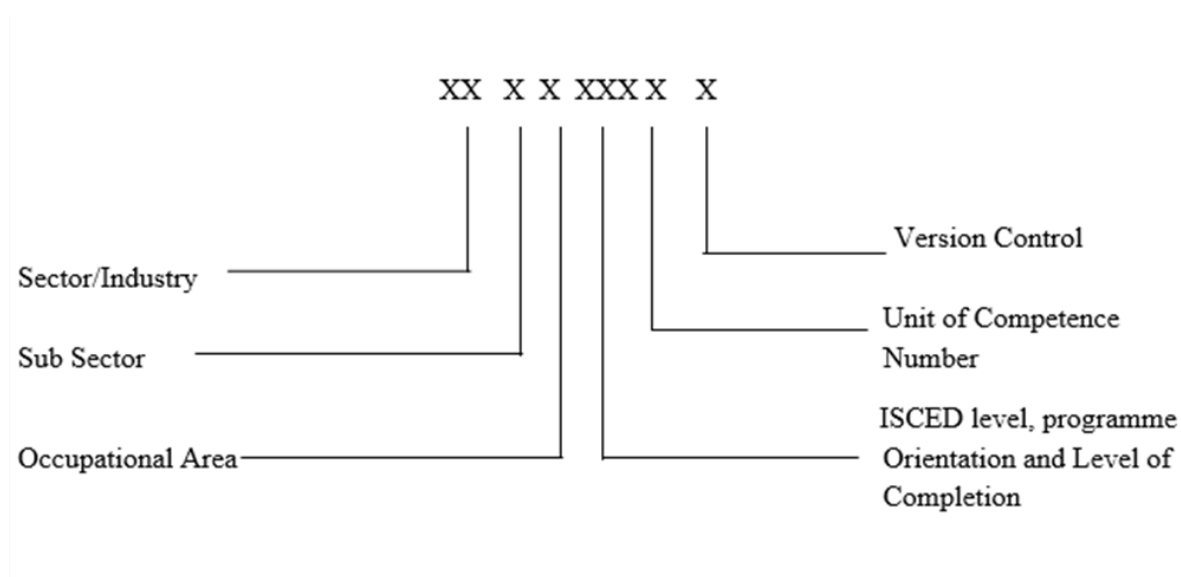
I am convinced that this curriculum will go a long way in ensuring that workers in Electricity and Energy sector will acquire competencies that will enable them perform their work more efficiently.

**SECRETARY/CEO
TVET CDACC**

ABBREVIATIONS AND ACRONYMS

BOQ	Bill of Quantities
EHS	Environment, Health and Safety
IET	Institute of Electrical and electronics Engineers
KP	Kenya Power
SOP	Standard operating procedure
NCA	National Construction Authority
OSHA	Occupational Safety and Health Act
PPE	Personal Protective Equipment
PV	Photo Voltaic
TVET	Technical and Vocational Education and Training

KEY TO UNIT CODE



KEY TO TVET CDACC UNIT CODE

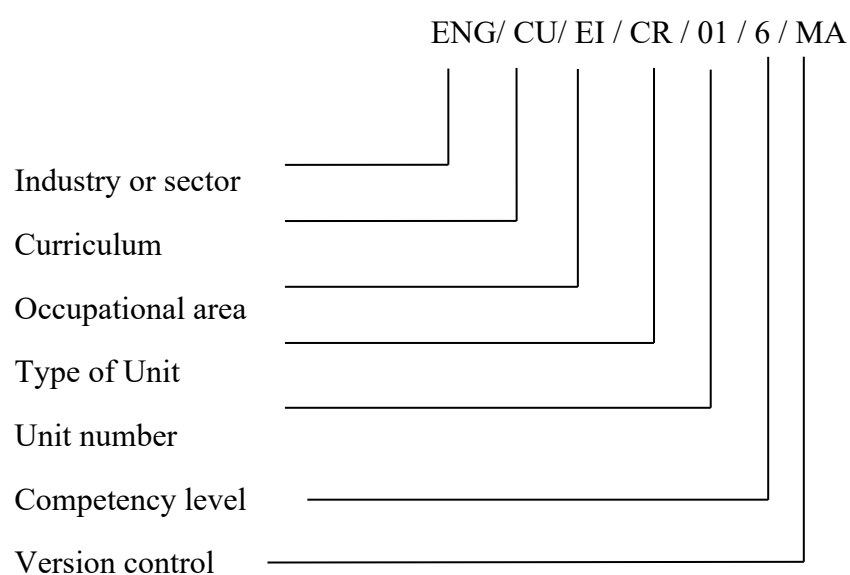


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COURSE OVERVIEW

Electrical Engineering Level 6 curriculum consists of competencies that an individual must achieve to perform electrical engineering activities. It involves PVC sheathed cable system installation, trunking system installation, conduit system installation, single-phase electrical installation, stand-alone solar PV systems, bell and alarm installation, electrical installation, solar PV systems, electrical machines installation, security systems installation, electrical powerlines, power systems, automating electrical systems, and analyzing power electronic circuits.

Table 1: Summary of Units of Learning

MODULE ONE

Unit of Code	Unit Category	Unit	HRS	Credit Factor
0713 251 23A	ENG/CU/EI/CR/01/3/MA	PVC Sheathed Cable System Installation	100	10.0
0713 251 24A	ENG/CU/EI/CR/02/3/MA	Trunking System Installation	100	10.0
0713 251 25A	ENG/CU/EI/CR/03/3/MA	Conduit System Installation	100	10.0
MODULE ONE TOTAL			300	30.0

MODULE TWO

Unit of Code		Unit	HRS	Credit Factor
0713 351 26A	ENG/CU/EI/CR/01/4/MA	Stand-Alone Solar PV Systems	120	12
0714 351 27A	ENG/CU/EI/CR/02/4/MA	Bell and Alarm Installation	120	12
0713 351 28A	ENG/CU/EI/CR/03/4/MA	Electrical Machine Winding	120	12
MODULE TWO TOTAL			360	36

MODULE THREE

Unit Code	Unit Category	Unit Title	Duration in Hours	Credit Factor
0611 451	ENG/CU/EI/BC/01/5/MA	Digital literacy	40	4

02A				
0031 441 01A	ENG/CU/EI/BC/02/5/MA	Communication skills	40	4
0713 441 09A	ENG/CU/EI/CC/01/5/MA	Basic Electrical Principles	100	10
0732 441 13A	ENG/CU/EI/CC/02/5/MA	Technical Drawing	80	8
0713 451 29A	ENG/CU/EI/CR/01/5/MA	Electrical Installation	110	11
MODULE THREE GRAND TOTAL			370	37

MODULE FOUR

Unit Code	Unit Category	Unit Title	Duration in Hours	Credit Factor
0417 441 04A	ENG/CU/EI/BC/03/5/MA	Work Ethics and Practices	40	4
0541 441 05A	ENG/CU/EI/CC/03/5/MA	Engineering Mathematics I	100	10
0714 441 15A	ENG/CU/EI/CC/04/5/MA	Analog Electronics I	70	7
0714 441 17A	ENG/CU/EI/CC/05/5/MA	Digital Electronics I	70	7
0713 451 33A	ENG/CU/EI/CR/02/5/MA	Security Systems Installation	120	12
MODULE FOUR GRAND TOTAL			400	40

MODULE FIVE

Unit Code	Unit Category	Unit Title	Duration in Hours	Credit Factor
0413 451 03A	ENG/CU/EI/BC/04/5/MA	Entrepreneurial Skills	40	4
0714 441 16A	ENG/CU/EI/CC/06/5/MA	Analogue Electronics II	80	8
0541 441 05A	ENG/CU/EI/CC/07/5/MA	Engineering Mathematics II	100	10
0714 441 18A	ENG/CU/EI/CC/08/5/MA	Digital Electronics II	80	8
0713 451 31A	ENG/CU/EI/CR/03/5/MA	Electrical Machines Installation	120	12
0713 451 32A	ENG/CU/EI/CR/04/5/MA	Solar PV Systems Installation	60	6

MODULE FIVE GRAND TOTAL	480	48
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MODULE SIX

Unit Code	Unit Category	Unit Title	Duration in Hours	Credit Factor
0541 541 07A	ENG/CU/EI/CC/01/6/MA	Engineering Mathematics III	100	10
0713 541 10A	ENG/CU/EI/CC/02/6/MA	Electrical Principles I	80	8
0714 541 19A	ENG/CU/EI/CC/03/6/MA	Micro-Control Systems	80	8
0713 551 30A	ENG/CU/EI/CR/01/6/MA	Electrical Powerlines Installation	180	18
MODULE SIX GRAND TOTAL			440	44

MODULE SEVEN

Unit Code	Unit Category	Unit Title	Duration in Hours	Credit Factor
0541 541 08A	ENG/CU/EI/CC/04/6/MA	Engineering Mathematics IV	100	10
0713 541 11A	ENG/CU/EI/CC/05/6/MA	Electrical Principles II	100	10
0714 541 12A	ENG/CU/EI/CC/06/6/MA	Control Systems	100	10
0714 551 36A	ENG/CU/EI/CR/02/6/MA	Electrical Systems Automation	160	16
MODULE SEVEN GRAND TOTAL			460	46

MODULE EIGHT

Unit Code	Unit Category	Unit Title	Duration in Hours	Credit Factor
0111 541 14C	ENG/CU/EI/CC/07/6/MA	Research Project	80	6
0713 551 22C	ENG/CU/EI/CC/08/6/MA	Electrical Project Supervision	60	6
0713 541 20A	ENG/CU/EI/CC/09/6/MA	Electrical Measurement and Fault Diagnosis	80	8
0714 551	ENG/CU/EI/CR/03/6/MA	Power Electronic	150	15

35A		Circuits Fabrication		
MODULE SEVEN GRAND TOTAL			350	
		Industry Training	480	48
LEVEL 6 COURSE TOTAL			3640	364

Entry Requirements

An individual entering this course should have any of the following minimum requirements:

- a) Kenya Certificate of Secondary Education (K.C.S.E.) with Grade C- (minus);
- Or**
- b) Certificate in Electrical Engineering or related level 5.
- Or**
- c) Equivalent qualifications as determined by TVETA

Trainer Qualification

Qualifications of a trainer for this course include:

- a) Possession of at least Electrical Engineering Level 7 or in related trade area;
- b) License by TVETA; and
- c) License by EBK/KETRB

Industry Training

An individual enrolled in this course will be required to undergo Industry training for a minimum period of 480 hours in Electrical Engineering sector. The industrial training may be taken after completion of all units for those pursuing the full qualification or be distributed equally in each unit for those pursuing part qualification. In the case of dual training model, industrial training shall be as guided by the dual training policy.

Assessment

The course shall be assessed formatively and summatively:

- a) During formative assessment all performance criteria shall be assessed based on performance criteria weighting.

- b) Number of formative assessments shall minimally be equal to the number of elements in a unit of competency.
- c) During summative assessment basic and common units may be integrated in the core units or assessed as discrete units.
- d) Theoretical and practical weighting for each unit of learning shall be as follows:
 - I. 10 :90 for units in module I and module II
 - II. 30:70 for units in module III to module V
 - III. 40:60 for units in module VI to module VIII
- e) Formative and summative assessments shall be weighted at 60% and 40% respectively in the overall unit of learning score
- f) For a candidate to be declared competent in a unit of competency, the candidate must meet the following conditions:
 - i) Obtained at least 40% in theory assessment in formative and summative assessments.
 - ii) Obtained at least 60% in practical assessment in formative and summative assessment where applicable.
 - iii) Obtained at least 50% in the weighted results between formative assessment and summative assessment where the former constitutes 60% and the latter 40% of the overall score.
- g) Assessment performance rating for each unit of competency shall be as follows:

MARKS	COMPETENCE RATING
80 -100	Attained Mastery
65 - 79	Proficient
50 - 64	Competent
49 and below	Not Yet Competent

Y	Assessment Malpractice/irregularities
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- h) Assessment for Recognition of Prior Learning (RPL) may lead to award of part and/or full qualification.

Certification

A candidate will be issued with a Certificate of Competency upon demonstration of competence in a core Unit of Competency. To attain the Kenya National TVET Certificate in Electrical Engineering Level 6, the candidate must demonstrate competence in all the Units of Competency as given in the qualification pack. Statement of Attainment certificate may be awarded upon demonstration of competence in certifiable element within a unit.

These certificates will be issued by TVETCDACC

MODULE I

PVC SHEATHED CABLE SYSTEM INSTALLATION

ISCED UNIT CODE: 0713 251 23A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/01/3/MA

UNIT DURATION: 100 hrs

Relationship to Occupational Standards

This unit addresses the unit of competency: install PVC sheathed cable system

UNIT DESCRIPTION

This unit specifies the competences required to install PVC sheathed cable system for a single-phase electrical installation. It involves identifying electrical cables and accessories, making cable joints, interpreting electrical symbols, fixing electrical accessories and performing test and inspection.

Summary of Learning Outcomes

By the end of this unit, the learner should be able to:

S/NO	Learning Outcomes	Duration (Hour)
1.	Identify cables and accessories	10
2.	Make cable joints	5
3.	Interpret electrical symbols	10
4.	Fix electrical accessories	60
5.	Perform Test and Inspection	15
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods:

Learning Outcome	Content	Suggested Assessment Methods
1. Identify cables and accessories	1.1 Workplace safety 1.1.1 Importance of workplace safety 1.1.2 Types of PPEs 1.2 Electrical cable colour coding	• Written tests • Practical assessment • Oral questioning • Third party report • Portfolio of

Learning Outcome	Content	Suggested Assessment Methods
	1.3 Electrical cable types and sizes 1.3.1 Types of cables 1.3.1.1 Single core 1.3.1.2 Twisted 1.3.1.3 Flex 1.3.1.4 Twine with Earth (TWE) 1.3.2 Cables Size 1.3.2.1 1.0 mm² 1.3.2.2 1.5 mm² 1.3.2.3 2.5 mm² 1.3.2.4 4.0 mm² 1.3.2.5 6.0 mm² 1.4 Electrical accessories 1.4.1 Lighting accessories 1.4.1.1 Switches 1.4.1.2 Lamp holders 1.4.1.3 Electrical lamps 1.4.2 Power accessories Sockets 1.4.3 General accessories 1.4.3.1 Joint boxes/junction boxes/inspection boxes 1.4.3.2 Pattress 1.4.3.3 Circuit breakers/MCB 1.4.3.4 C.C.U 1.4.3.5 Cut- outs (1ϕ) 1.4.3.6 Energy meters(1ϕ) 1.4.3.7 Earth rods 1.4.3.8 Metre box 1.4.3.9 Switch fuse (1ϕ)	evidence
2. Make cable joints	2.1 Electrical tools and equipment: e.g. 2.2 Measuring tools e.g. 2.2.1 Tape measure 2.2.2 Tri-square 2.2.3 Steel rule 2.2.4 Spirit level	• Written tests • Practical assessment • Oral questioning • Interviews • Project

Learning Outcome	Content	Suggested Assessment Methods
	2.3 Cutting tools e.g. 2.3.1 Hacksaw 2.3.2 Stripping knife 2.3.3 Side cutter 2.3.4 Pliers 2.4 Fixing tools e.g. 2.4.1 Hammers 2.4.2 Screw drivers 2.4.3 Crimping tool 2.5 Equipment 2.5.1 Multimeter 2.5.2 Insulation resistance meter 2.5.3 Earth resistance meter 2.5.4 Clamp meter 2.5.5 Drill etc. 2.6 Storage of electrical tools 2.7 Electrical cable joints 2.7.1 Britannia 2.7.2 Married 2.7.3 Tee 2.7.4 Straight twist 2.7.5 Scarf 2.8 Housekeeping activities	• Third party report • Portfolio of evidence
3. Interpretation of electrical symbols	3.1 interpret electrical symbols 3.2 interpret wiring diagrams 3.3 Electrical components	• Written tests • Practical assessment • Oral questioning • Third party report • Portfolio of evidence
4. Fix electrical accessories	4.1 Electrical single-phase intake point installation 4.1.1 Cut- outs (1Φ) 4.1.2 Energy meters(1Φ) 4.1.3 C.C.U 4.1.4 Earth rods 4.1.5 Switch fuse (1Φ) 4.2 Electrical final circuits installation 4.2.1 Lighting 4.2.1.1 One way 4.2.1.2 Two way	• Written tests • Practical assessment • Oral questioning • Interviews • Project • Third party report • Portfolio of evidence

Learning Outcome	Content	Suggested Assessment Methods
	4.2.1.3 Intermediate 4.2.1.4 Looping in methods at ceiling rose, joint boxes, switches 4.3 Power 4.3.1 Socket outlets (ring and radial). 4.3.2 Cooker units. 4.3.3 Water heaters. 4.4 Housekeeping activities 4.4.1 General cleanliness	
5. Perform Test and Inspection	1.5 Visual inspection 1.6 Test 1.6.1 Continuity 1.6.2 Polarity 1.6.3 Ring test	• Project • practical • Third party report. • Written assessment. • Oral assessment

Suggested Methods of Instruction

- Discussions
- Project
- Direct instructions
- Demonstration by trainer
- Practice by the trainee
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	B. Scaddan Electrical installation work J. Hyde Electrical installation	5 pcs	1:5

		Principles and Practices		
2.	Installation manuals	IEEE regulation BS3939 NEMA regulations OSHA	5 pcs	1:5
3.	Charts	Single line diagram Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
5.	Lecture/theory room	50m ²	1	1:25
6.	Workshop	150m ²	1	1:25
7.	Site			
C	Consumable materials			
8.	Electrical cables	1.5mm ² (TWE)	5 rolls	1:5
		2.5mm ² (TWE)	5 rolls	1:5
		4.0 mm ² (TWE)	3 rolls	1:10
		6.0 mm ² (TWE)	2 rolls	1:12
		10 mm ² (TWE)	2 rolls	1:12
9.	Insulation tapes		25 pcs	1:1
10.	Wood screws		50 pcs	1:1
D	Accessories			
11.	Switches, sockets, Junction boxes, Consumer units, Lamp holders, Patress boxes, Circuit		25 pcs	1:1

	breakers, Energy meters(1☐), Earth rods, Switch fuse (1☐), wood screws, assorted cable clips			
D	Tools and Equipment			
12.	Striping knives		25 pcs	1:1
13.	Side cutters		25 pcs	1:1
14.	Pliers		25 pcs	1:1
15.	Tape measure		25 pcs	1:1
16.	Try Square		25 pcs	1:1
17.	Spirit level		25 pcs	1:1
18.	Assorted Screw driver		25 pcs	1:1
19.	Assorted hammers		25 pcs	1:1
20.	Crimping tools		5 pcs	1:5
21.	PPEs		25 pcs	1:1
22.	Multimeters		5 pcs	1:5
23.	Clamp meters		5 pcs	1:5
24.	Earth resistance meter		5 pcs	1:5
25.	Drilling machines		5 pcs	1:5
26.	Work stations		25	1:1
27.	Installation boards	1.2 by 1m	13 pcs	1:2

TRUNKING SYSTEM INSTALLATION

ISCED UNIT CODE: 0713 251 24A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/02/3/MA

UNIT DURATION: 100 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Install Trunking system.

Unit Description

This unit covers competences required in installing trunking system. It involves identifying trunking accessories, preparing trunking work pieces, mounting trunking work pieces, installing electrical cables and accessories and perform test and inspection.

Summary of Learning Outcomes

By the end of this unit, the learner should be able to:

S/ NO	Learning Outcome	Duration (Hours)
1.	Identify trunking accessories	10
2.	Prepare trunking work pieces	10
3.	Mount trunking work pieces	30
4.	Install electrical cables and accessories	40
5.	Perform Test and Inspection	10
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Identify trunking accessories	1.1 Workplace safety 1.1.1 Importance of workplace safety 1.1.2 Basic workshop rules 1.1.3 Types of PPEs 1.2 Trunking types 1.2.1 PVC 1.2.2 Metallic 1.2.3 Galvanized iron 1.3 Trunking sizes	• Practical demonstration • Projects • Written tests • Oral test

	1.3.1 16mm x 16mm 1.3.2 25mm x 16mm 1.3.3 25mm x 25mm 1.3.4 40mm x 25mm 1.4 Trunking accessories 1.4.1 Sockets 1.4.2 Switches 1.4.3 Lamp holders 1.4.4 Joint boxes 1.4.5 Pattress 1.4.6 Circuit breakers 1.4.7 C.C.U 1.4.8 Cut- outs (1ϕ) 1.4.9 Energy meters(1ϕ) 1.4.10 Earth rods 1.4.11 Switch fuse (1ϕ) 1.8 Housekeeping (cleaning of the workplace after work)	
2. Prepare trunking work pieces	2.1 Electrical tools and equipment 2.1.1 Cutting tools 2.1.2 Fastening tools 2.1.3 Measuring tools 2.1.4 Marking tools 2.1.5 Measuring instruments 2.2 Measurements 2.3 Housekeeping activities 2.3.1 General cleanliness 2.3.2 Tools and equipment storage	• Practical demonstration • Projects • Written tests • Oral test
3. Mount trunking work pieces	3.1 Marking mounting points 3.2 Trunking angles e.g. 45°,90° 3.3 Mounting Trunking work pieces 3.4 Housekeeping activities 3.4.1 General cleanliness 3.4.2 Tools and equipment storage	• Practical demonstration • Projects • Written tests • Oral test

4. Install electrical cables and accessories	4.1 Electrical cable colour code identification 4.2 Electrical <i>cable sizes</i> identification 4.2.1 1.0mm² 4.2.2 1.5 mm² 4.2.3 2.5 mm² 4.2.4 4.0 mm² 4.2.5 6.0 mm² 4.3 Cable laying 4.3.1 Single core cables 4.4 Electrical single-phase intake point installation 4.4.1 Cut- outs (1Φ) 4.4.2 Energy meters(1Φ) 4.4.3 C.C.U 4.4.4 Earth rods 4.4.5 Switch fuse (1Φ) 4.5 Electrical final circuits installation 4.5.1 Lighting 4.5.2 One way 4.5.3 Two way 4.5.4 Intermediate 4.5.5 Looping in methods at ceiling rose, joint boxes, switches 4.6 Power 4.6.1 Socket outlets (ring and radial). 4.6.2 Cooker units. 4.6.3 Water heaters. 4.7 Housekeeping activities 4.7.1 General cleanliness 4.7.2 Tools and equipment storage	• Practical demonstration • Projects • Written tests • Oral test
5. Test and Inspection	5.1 Visual inspection 5.2 Test 5.2.1 Continuity 5.2.2 Polarity 5.2.3 Ring test	• Project • practical • Third party report. • Written assessment. • Oral assessment

Suggested Methods of Instruction

- Projects
- Written tests
- Oral questions
- Portfolio of evidence
- Third party evidence
- Discussions
- Demonstration by trainer
- Practice by the trainee
- Direct instructions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	B. Scaddan Electrical installation work J. Hyde Electrical installation Principles and Practices	5 pcs	1:5
2.	Installation manuals		5 pcs	1:5
3.	Charts	Single line diagram Circuit diagrams Colour codes	1 pc for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
5.	Lecture/theory room	50m ²	1	1:25
6.	Workshop	150m ²	1	1:25
7.	Laboratory	100m ²	1	1:25
8.	Site			
C	Consumable materials			
9.	Electrical cables	1.5mm ² (red, black green)	5 rolls	1:5

		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
10.	Insulation tapes		25 pcs	1:1
11.	Wood screws		50pcs	1:1
D	Accessories	Switches, sockets, Junction boxes, Consumer units, Lamp holders, Patresses boxes, Circuit breakers	25 pcs	1:1
12.	Trunkings	Mini trunking	25 pcs	1:1
E	Tools and Equipment			
13.	Hacksaws		25 pcs	1:1
14.	Striping knives		25 pcs	1:1
15.	Side cutters		25 pcs	1:1
16.	Pliers		25 pcs	1:1
17.	Tape measure		25 pcs	1:1
18.	Try Square		25 pcs	1:1
19.	Spirit level		25 pcs	1:1
20.	Assorted Screw driver		25 pcs	1:1
21.	Assorted hammers		25 pcs	1:1
22.	PPEs		25 pcs	1:1
23.	Multimeters		5 pcs	1:5

CONDUIT SYSTEM INSTALLATION

ISCED UNIT CODE: 0713 251 25A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/03/3/MA

UNIT DURATION: 100 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: install conduit system

Unit Description

This unit covers competences required in installing conduit system. It involves identifying conduit accessories, preparing conduit work pieces, mounting conduit work pieces, installing electrical cables and accessories and performing test and inspection.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/ NO	Learning Outcome	Duration (Hours)
1.	Identify conduit accessories	10
2.	Prepare conduit work pieces	10
3.	Mount conduit work pieces	30
4.	Install electrical cables and accessories	40
5.	Perform Test and Inspection	10
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Identify conduit accessories	1.1. Workplace safety 1.1.1 Importance of workplace safety 1.1.2 Types of PPEs 1.1.3 Housekeeping (cleaning of the workplace after work)	• Practical demonstration • Projects • Written tests • Oral test

	1.2. Conduit types identification 1.2.1 PVC 1.2.1 Metallic 1.3 Conduit sizes identification 1.3.1 20mm² 1.3.2 25mm² 1.3.3 32mm² 1.4 Conduit accessories identification 1.4.1 Sockets 1.4.2 Switches 1.4.3 Lamp holders 1.4.4 Strip connectors 1.4.5 Ceiling roses 1.4.6 Circular connectors 1.4.7 Junction boxes 1.4.8 Saddles and saddle clips 1.4.9 Bushes 1.4.10 Couplers 1.4.11 Switch boxes 1.4.12 Circuit breakers 1.4.13 C.C.U 1.4.14 Cut- outs (1ϕ) 1.4.15 Energy meters(1ϕ) 1.4.16 Earth rods 1.4.17 Switch fuse (1ϕ) 1.5 Electrical signs and symbols	
2. prepare conduit work pieces	2.1 Electrical tools and equipment 2.1.1 Cutting tools 2.1.2 Fastening tools 2.1.3 Measuring tools 2.1.4 Marking tools 2.1.5 Measuring instruments 2.2 Measurements 2.3 Conduit work pieces cutting 2.4 Housekeeping activities	• Practical demonstration • Projects • Written tests • Oral test

	2.4.1 General cleanliness 2.4.2 Tools and equipment storage	
3. Mount conduit work pieces	3.1. Marking mounting points 3.2. Conduit bends e.g. 45°,90°, offsets 3.3. Mounting conduit work pieces 3.4. Housekeeping activities 3.4.1 General cleanliness 3.4.2 Tools and equipment storage	• Practical demonstration • Projects • Written tests • Oral test
4. Install electrical cables and accessories	4.1 Electrical cable colour code identification 4.2 Electrical <i>cable sizes</i> identification 4.2.1 1.0 mm ² 4.2.2 1.5 mm ² 4.2.3 2.5 mm ² 4.2.4 4.0 mm ² 4.2.5 6.0 mm ² 4.3 Cable laying 4.3.1 Single core cables 4.4 Electrical single-phase intake point installation 4.4.1 Cut- outs (1ϕ) 4.4.2 Energy meters(1ϕ) 4.4.3 C.C.U 4.4.4 Earth rods 4.4.5 Switch fuse (1ϕ) 4.5 Electrical final circuits installation 4.5.1 Lighting 4.5.2 One way 4.5.3 Two way 4.5.4 Intermediate 4.5.5 Looping in methods at ceiling rose, joint boxes, switches 4.6 Power 4.6.1 Socket outlets (ring and radial). 4.6.2 Cooker units.	• Practical demonstration • Projects • Written tests • Oral test

	4.6.3 Water heaters. 4.7 Housekeeping activities 4.7.1 General cleanliness 4.7.2 Tools and equipment storage	
5 Test and Inspection	5.1 Visual inspection 5.2 Test 5.2.1 Continuity 5.2.2 Polarity 5.2.3 Ring test	• Project • practical • Third party report. • Written assessment. • Oral assessment

Suggested Methods of Instruction

- Discussions
- Projects
- Demonstration by trainer
- Practice by the trainee
- Direct instructions
- Field trips
- On-job-training

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	B. Scaddan Electrical installation work J. Hyde Electrical installation Principles and Practices	5 pcs	1:5
2.	Installation manuals	IEEE regulation BS3939 NEMA regulations OSHA	5 pcs	1:5

3.	Charts	Single line diagram Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
	Lecture/theory room	50m ²	1	1:25
	Workshop	150m ²	1	1:25
	Laboratory	100m ²	1	1:25
	Site			
C	Consumable materials			
1.	Electrical wires	1.0mm ² (red, black green)	5 rolls	1:5
		1.5mm ² (red, black green)	5 rolls	1:5
2.	Insulation tapes		25 pcs	1:1
3.	Accessories	Switches, sockets, Junction boxes, Consumer units, Lamp holders, Patrice boxes, Circuit breakers, relays, transformers	25 pcs	1:1
4.	Pipes	PVC conduits, Metallic conduits	25 pcs	1:1
5.	Wood screws		50 pcs	1:1
D	Tools and Equipment			
	Hacksaws		25 pcs	1:1
	Striping knives		25 pcs	1:1
	Side cutters		25 pcs	1:1
	Pliers		25 pcs	1:1

	Tape measure		25 pcs	1:1
	Draw wire		25 pcs	1:1
	Try Square		25 pcs	1:1
	File		5 pcs	1:5
	Spirit level		25 pcs	1:1
	Assorted Screw driver		25 pcs	1:1
	Assorted hammers		25 pcs	1:1
	Crimping tools		5 pcs	1:5
	PPEs		25 pcs	1:1
	Multimeters		5 pcs	1:5
	Clamp meters		5 pcs	1:5
	Earth resistance meter		5 pcs	1:5
	Bending spring		5 pcs	1:5
	Drilling machines		5 pcs	1:5
	Work stations		25	1:1
	Installation boards		13 pcs	1:2
	Bending machine		5 pcs	1:5
	Bench vice		5 pcs	1:5
	Stock and die		5 pcs	1:5
	Oil cans		5 pcs	1:5

MODULE II

STAND-ALONE SOLAR PV SYSTEMS

ISCED UNIT CODE: 0713 351 26A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/01/4/MA

UNIT DURATION: 120 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Install stand-alone solar PV systems

Unit Description

This unit covers competences required in installing stand-alone solar PV systems. Competences include: applying basic electrical concepts, interpreting stand-alone solar PV Installation drawings, installing stand-alone Solar PV components, installing electrical wiring system, testing stand-alone solar PV system installation and maintaining stand-alone Solar PV system

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/ NO	Learning Outcome	Duration (Hours)
1.	Apply basic electrical concepts	5
2.	Interpret stand-alone solar PV Installation drawings	5
3.	Install stand-alone Solar PV Components	30
4.	Install electrical wiring system	40
5.	Test stand-alone solar PV system installation	20
6.	Maintain stand-alone Solar PV system	20
TOTAL		120

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply basic electrical concepts	1.1 The meaning of SI unit 1.2 SI unit of various types of Electrical parameters 1.2.1 Power – Watts (W) 1.2.2 Current – Amperes	• Practical demonstration • Projects • Written tests

	(A) 1.2.3 Resistance – Ohms(Ω) 1.2.4 Voltage – Volts (V) 1.3 Identification of Quantities of Charge, force, work and power 1.4 Ohm’s law 1.5 Calculations involving various Electrical parameters e.g. Power, Current, Voltage, Resistance	• Oral test
2. Interpret stand-alone solar PV Installation drawings	1.1 Drawing standard electrical symbols 1.2 Identification of scope of installation work 1.3 Interpretation of various electrical working drawings 1.4 Drawing of electrical diagrams 2.4.1 Block 2.4.2 Layout/Schematic 2.4.3 Wiring	• Practical demonstration • Projects • Written tests • Oral test
3. Install stand-alone solar PV system	3.1 Solar PV system components termination 3.1.1 DC connections; 3.1.2 PV module 3.1.3 Charge controller 3.1.4 Batteries 3.1.5 Load 3.1.6 AC connection 3.1.7 PV module 3.1.8 Charge controller 3.1.9 Batteries 3.1.10 Inverter 3.1.11 Load	• Practical demonstration • Projects • Written tests • Oral test
4 Install electrical wiring system	4.1 Electrical cable, joints and termination 4.1.1 Cables: 4.1.2 Types; e.g PVC sheathed cables, armoured cables 4.1.3 Cable sizes e.g.	•

	10.0mm², 6.0mm², 4.0mm², 2.5mm², 1.5mm², 1.0mm² 4.1.4 Cable joints: e.g. Britannia, Married, Tee , Straight twist, scarf 4.1.5 Cable terminations: e.g. crimped, cables lugs. 4.2 Cable management system 4.2.1 Cable trays 4.2.2 Cable duct 4.2.3 Bus-bars 4.2.4 Sheath/surface 4.2.5 Conduits 4.2.6 Trunking 4.3 Electrical accessories and fittings e.g. 4.3.1 Control consumer unit 4.3.2 Patrice 4.3.3 Lamp holders 4.3.4 Sockets outlets 4.3.5 Conduits 4.3.6 Mini-trunking 4.3.7 Saddles and space bars 4.3.8 Bushes 4.3.9 Ceiling roses 4.3.10 Couplers 4.3.11 Switches etc. 4.4 Electrical final circuits 4.4.1 Lighting circuits 4.4.2 One way, two way, intermediate 4.4.3 Looping in methods at ceiling rose, joint boxes, switches 4.4.4 Power circuits 4.4.5 Radial circuits, ring circuits 4.4.6 Water heating circuits 4.4.7 Cooker unit 4.5 Wiring systems	
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	4.5.1 Surface 4.5.2 Conduits 4.5.3 Trunks 4.6 Earthing system types 4.6.1 Direct bonding 4.6.2 Components of earthing system e.g. Earth Rod, Earth lead, etc. Earth 4.6.3 continuity conductor	
5 Test stand-alone solar PV system installation	5.1 visual inspections 5.2 Continuity test 5.2.1 continuity test tools and equipment: multimeter, test lamp 5.2.2 Results are documented and interpreted 5.3 Insulation resistance test 5.3.1 insulation resistance test tools and equipment: megger tester 5.3.2 Results are documented and interpreted 5.4 Polarity test 5.4.1 Polarity test tools and equipment: multimeter, test lamp 5.4.2 Results are documented and interpreted 5.5 Earth resistance tests 5.5.1 Earth resistance test tools and equipment: Earth resistance tester 5.5.2 Results are documented and interpreted	• Practical demonstration • Projects • Written tests • Oral test
6 Maintain stand-alone Solar PV system	6.1 Electrical system inspection 6.2 Identification of maintenance tools and equipment. 6.3 Maintenance activities e.g. 6.3.1 Disassembling/Assembling 6.3.2 Cleaning 6.3.3 Tightening	• Written tests • Observation • Oral questioning • Portfolio of evidence • Interviews • Project • Third party report

	6.3.4 Replacement of faulty component etc 6.4 Electrical system tests e.g polarity tests, continuity test, insulation test, earth resistance test, open circuit test, short circuit test e.t.c 6.5 Maintenance activities 6.5.1 Cleaning 6.5.2 Topping up batteries 6.5.3 Loose connections 6.6 Identification of faulty components 6.7 Recording of inspection findings 6.8 Repair/Replacement of faulty components 6.9 Fill in maintenance checklist 6.10 Disposal of waste materials	
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Suggested Methods of Instruction

- Discussions
- Projects
- Demonstration by trainer
- Practice by the trainee
- Direct instructions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
5	Textbooks	Solar Electric Handbook by Solar energy International B. Scaddan Electrical	5 pcs	1:5

		installation work J. Hyde Electrical installation Principles and Practices		
6	Installation manuals	Solar PV manuals	5 pcs	1:5
7	Charts	Single line diagram Solar PV layout Circuit diagrams Colour codes	1 pc for each	1:25
8	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
24.	Lecture/theory room	50m ²	1	1:25
25.	Workshop	150m ²	1	1:25
26.	Laboratory	100m ²	1	1:25
27.	Site			
C	Consumable materials			
28.	Electrical wires	1.5mm ² (red, black green)	5 rolls	1:5
		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
29.	Insulation tapes		25 pcs	1:1
30.	Accessories	Switches, sockets, Junction boxes, Consumer units,	25 pcs	1:1

		Lamp holders, Patrice boxes, Circuit breakers		
31.	Solar Panels		10	1:3
32.	Charge controller		10	1:3
33.	Batteries		10	1:3
34.	Inverter		10	1:3
35.	Mounting racks		10	1:3
36.	Pipes and trunkings	PVC conduits, Steel conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			
37.	Hacksaws		25 pcs	1:1
38.	Striping knives		25 pcs	1:1
39.	Side cutters		25 pcs	1:1
40.	Pliers		25 pcs	1:1
41.	Tape measure		25 pcs	1:1
42.	Try Square		25 pcs	1:1
43.	Spirit level		25 pcs	1:1
44.	Assorted Screw driver		25 pcs	1:1
45.	Assorted hammers		25 pcs	1:1
46.	Crimping tools		5 pcs	1:5
47.	PPEs		25 pcs	1:1
48.	Multimeters		5 pcs	1:5

BELL AND ALARM INSTALLATION

ISCED UNIT CODE: 0714 351 27A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/02/4/MA

UNIT DURATION: 120 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform bell and alarm installation

Unit Description

This unit covers the competencies required in perform bell and alarm installation. Competencies include interpreting bell and alarm installation drawings, installing bell and alarm wiring system, installing bell and alarm components, testing bell and alarm installation and maintaining bell and alarm installation.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/ NO	Learning Outcome	Duration (Hours)
1.	Interpret bell and alarm installation drawings	20
2.	Install bell and alarm wiring system	40
3.	Bell and alarm components	20
4.	Test bell and alarm installation	20
5.	Maintain bell and alarm installation.	20
TOTAL		120

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Interpret bell and alarm installation drawings	1.1 Bell and alarm circuits 1.1.1 Bells circuit diagrams 1.1.2 Definition purpose and applications of bell and alarm circuits 1.2 Definition, purpose and applications of bell and alarm circuits 1.3 Bell and alarm symbols e.g	• Practical demonstration • Projects • Written tests • Oral test

	Bell transformers, Indicator board, Bell bushes, Bell relays, Control panel, Sounders, Buzzers Bell, Hooters, Sirens etc 1.4 Mapping Wiring system 1.5 Types of bell Circuits 1.6 Interpreting bell circuit diagrams	
2 Install bell and alarm wiring system	2.1 Safety measures 2.1.1 Relevant clauses in appropriate Acts e.g. 2.1.2 Occupational safety and health act (OSHA) 2.1.3 Work injury benefits act (WIBA) 2.1.4 Safety Regulations and procedures 2.1.5 PPEs 2.1.6 First Aid 2.1.7 IEE regulations 2.1.8 Common hazards and sources of danger e.g. burns, cuts, electric shock, falling from heights, falling objects, noise, dust, chemicals 2.1.9 Safe and correct handling, use, maintenance and storage of different types of tools. 2.2 Electrical tools, equipment, and instruments 2.2.1 Classification of tools e.g. 2.2.2 Cutting tools e.g. 2.2.3 Hacksaw, Stripping knife, Side cutter etc. 2.2.4 Pliers 2.2.5 Measuring tools e.g.	• Practical demonstration • Projects • Written tests • Oral test

	2.2.6 Tape measure, Tri-square, Steel rule, Spirit level etc. 2.2.7 Fixing tools e.g. 2.2.8 Hammers, Screw drivers 2.2.9 Crimping tool 2.2.10 Equipment e.g stock and die, vice, pipe bending machine 2.2.11 Instruments such as Multimeter, Insulation resistance meter, Earth resistance meter, Clamp meter, Solar Irradiance meter, Hydrometer 2.2.12 Draw wire, Bending spring, Drill etc. 2.3 Use, care and maintenance of tools, equipment, and instruments. 2.4 Electrical cable, joints and termination. 2.4.1 Cables: 2.4.2 Types; e.g. PVC sheathed cables 2.4.3 Cable sizes 2.4.4 Cable joints: Britannia, Married, Tee , Straight twist, scarf 2.4.5 Cable terminations 2.4.6 Bell and alarm accessories and fittings. 2.4.7 Pattresses 2.4.8 Junction boxes 2.4.9 Switches/Pushbuttons 2.4.10 Couplers 2.4.11 Circuit breakers 2.5 Cable management system 2.5.1 Cable trays	
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	2.5.2 Cable duct 2.5.3 Bus-bars 2.5.4 Sheath/surface 2.5.5 Conduits 2.5.6 Trunking 2.6 Installation of bell and alarm systems. 2.6.1 Installation Requirements 2.6.2 Wiring Diagrams 2.6.3 Connection Techniques	
3 Install bell and alarm components	2.1 Bell and alarm components 3.1.1 Bell transformers 3.1.2 Indicator board 3.1.3 Bell bushes 3.1.4 Bell relays 3.1.5 Control panel 3.1.6 Sounders 3.1.7 Buzzers 3.1.8 Bell 3.1.9 Hooters 3.1.10 Sirens 3.2 Electrical power supply installation 3.2.1 240VAC supply 3.2.2 12VAC supply 3.3 Bell and alarm components installation	• Practical demonstration • Projects • Written tests • Oral test
4 Test bell and alarm installation	4.1 Visual inspections of Bell and alarm systems. 4.2 Continuity test 4.2.1 continuity test tools and equipment: multimeter, test lamp 4.2.2 Results are documented and interpreted 4.3 Insulation resistance test 4.3.1 insulation resistance test tools and equipment: megger	• Practical demonstration • Projects • Written tests • Oral test

	tester 4.3.2 Results are documented and interpreted 4.4 Polarity test 4.4.1 Polarity test tools and equipment: multimeter, test lamp 4.4.2 Results are documented and interpreted 4.5 Earth resistance tests 4.5.1 Earth resistance test tools and equipment: Earth resistance tester 4.5.2 Results are documented and interpreted	
5 Maintain bell and alarm installation	5.1 Bell and alarm inspection 5.2 Identification and preparation of maintenance materials, tools and equipment. 5.3 Bell and alarm system Maintenance activities e.g. 5.3.1 Disassembling/Assembling 5.3.2 Cleaning 5.3.3 Tightening Loose connections 5.3.4 Replacement of faulty component etc 5.4 System system tests e.g polarity tests, continuity test, insulation test, earth resistance test, open circuit test, short circuit test e.t.c 5.5 Maintenance records documentation	• Practical demonstration • Projects • Written tests • Oral test

Suggested Methods of Instruction

- Discussions
- Projects
- Demonstration by trainer
- Practice by the trainee
- Direct instructions
- Field trips
- On-job-training

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
5.	Textbooks	B. Scaddan Electrical installation work J. Hyde Electrical installation Principles and Practices	5 pcs	1:5
6.	Installation manuals	IEEE regulation BS3939 NEMA regulations OSHA	5 pcs	1:5
7.	Charts	Single line diagram Circuit diagrams Colour codes	1 pcs for each	1:25
8.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	50m ²	1	1:25

2.	Workshop	150m ²	1	1:25
3.	Laboratory	100m ²	1	1:25
4.	Site			
C	Consumable materials			
6.	Electrical wires	1.0mm ² (red, black green)	5 rolls	1:5
		1.5mm ² (red, black green)	5 rolls	1:5
7.	Insulation tapes		25 pcs	1:1
8.	Accessories	Switches, sockets, Junction boxes, Consumer units, Lamp holders, Patrice boxes, Circuit breakers, relays, transformers	25 pcs	1:1
9.	Pipes and trunkings	PVC conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			
1.	Hacksaws		25 pcs	1:1
2.	Striping knives		25 pcs	1:1
3.	Side cutters		25 pcs	1:1
4.	Pliers		25 pcs	1:1
5.	Tape measure		25 pcs	1:1
6.	Try Square		25 pcs	1:1
7.	Spirit level		25 pcs	1:1
8.	Assorted Screw driver		25 pcs	1:1
9.	Assorted hammers		25 pcs	1:1

10.	Crimping tools		5 pcs	1:5
11.	PPEs		25 pcs	1:1
12.	Multimeters		5 pcs	1:5
13.	Clamp meters		5 pcs	1:5
14.	Earth resistance meter		5 pcs	1:5
15.	Bending spring		5 pcs	1:5
16.	Drilling machines		5 pcs	1:5
17.	Work stations		25	1:1
18.	Installation boards		13 pcs	1:2

ELECTRICAL MACHINE WINDING

ISCED UNIT CODE: 0713 351 28A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/03/4/MA

UNIT DURATION: 120 hours

UNIT DESCRIPTION

This unit specifies the competencies required for winding electrical machines. The competencies include; disassembling rotating electrical machines, installing Machine winding, assembling rotating electrical machines and performing electrical machine testing.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/ NO	Learning Outcome	Duration (Hours)
1.	Disassemble rotating electrical machines	20
2.	Install Machine winding	40
3.	Assemble rotating electrical machines	40
4.	Perform electrical machine testing	20
TOTAL		120

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Disassemble rotating electrical machine	4.1 Safety Measures 1.1.1 Donning of PPEs 1.1.2 Handling of Tools 4.2 Disassembling Tools and Equipment 1.2.1 Fastening Tools e.g. Allen keys, bearing puller, etc. 1.2.2 Testing equipment 4.3 Record Machine data 4.4 Disassembling Rotating Machine parts 1.4.1 End caps 1.4.2 Fan 1.4.3 Rotor	• Practical • project • Portfolio of evidence • Third party report • Written assessment • Oral assessment

	1.4.4 Armature 1.4.5 Stator 1.4.6 Conductors 1.4.7 Bearings 1.4.8 Procedure of disassembling 4.5 Winding data e.g. 1.5.1 Number of slots 1.5.2 Conductor size 1.5.3 Number of coils 1.5.4 Number of turns per coil 1.5.5 Critical dimensions 1.5.6 Bearing data 4.6 Housekeeping 1.6.1 Definition 1.6.2 Activities e.g. General cleanliness, tools, and equipment storage, etc.	
2. Install machine winding	2.1 Safety Measures 2.1.1 PPEs 2.1.2 Handling of Tools 2.1.3 Housekeeping Practices 2.2 Machine winding tools and equipment e.g. 2.2.1 Lapse counter 2.2.2 Wire tensioner 2.2.3 Slot liner 2.2.4 Shrink tubing, etc. 2.3 Installation procedures 2.3.1 Cleaning machine parts 2.3.2 Preparation of windings 2.3.3 laying of windings 2.3.4 Curing of windings e.g. dry, varnish, bake, etc	• Practical • project • Portfolio of evidence • Third party report • Written assessment • Oral assessment

3. Assemble rotating electrical machines	3.1 Electrical Tools and Equipment 3.1.1 Fastening Tools e.g. Allen keys, bearing puller, etc. 3.1.2 Testing equipment 3.2 Assembling Rotating Machine 3.2.1 Machine's Parts 3.2.2 End caps 3.2.3 Fan 3.2.4 Rotor 3.2.5 Armature 3.2.6 Stator 3.2.7 Conductors 3.2.8 Bearings 3.2.9 Procedure of assembling 3.3 Winding data e.g. 3.3.1 Number of slots 3.3.2 Conductor size 3.3.3 Number of coils 3.3.4 Number of turns per coil 3.3.5 Critical dimensions 3.3.6 Bearing data 3.4 Housekeeping 3.4.1 Definition 3.4.2 Activities e.g. General cleanliness, tools, and equipment storage, etc.	• Practical • project • Portfolio of evidence • Third party report • Written assessment • Oral assessment •
4 Perform electrical machine testing	4.1 Test equipment set up 4.2 Test parameters e.g. 4.2.1 Turns ratio 4.2.2 Winding resistance 4.2.3 Voltage 4.2.4 Current, etc. 4.3 Types of tests 4.3.1 Continuity 4.3.2 Insulation resistance 4.3.3 Voltage test 4.3.4 Current balance 4.3 Documentation e.g. compiling test reports, etc. 4.4 Housekeeping activities 4.4.1 General cleanliness	

	4.4.2 Tools and equipment storage	
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Suggested Methods of Instruction

- Practical
- Demonstration
- Direct instruction
- Group Discussion

Recommended Resources for 25 trainees

S/No	Item	Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	Textbook electrical technology by B.L Theraja Handbook of Electric Motors by Hamid A. Toliyat and Gerald B. Kilman Practical Electric motor by Irving Gottlieb AC motor rewinding and Electrical machine repair by Ali H.Ahmad	5 pcs	1:5
2.	Installation manuals	Manufacturer installation manuals	5 pcs	1:5

B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
2.	Workshop	150m ²	1	1:25
C	Consumable materials			
1.	Copper/Aluminum wires	Size: 0.07mm to 5.00mm	5 rolls each	1:5
2.	Insulation materials	Insulation tape, insulation paper, slot liners, inter-turn insulation	25 pcs each	1:1
3.	Accessories	Winding mandrels and forms, wire tensioner, wire guides, insulation separator, binding tape, coil supports and spacers, turns counter	25 pcs each	1:1
D	Tools and Equipment			
1.	Hacksaws		25 pcs	1:1
2.	Striping knives		25 pcs	1:1
3.	Side cutters		25 pcs	1:1
4.	Pliers		25 pcs	1:1
5.	Tape measure		25 pcs	1:1
6.	Try Square		25 pcs	1:1
7.	Assorted Screw driver		25 pcs	1:1
8.	Assorted hammers		25 pcs	1:1
9.	Crimping tools		5 pcs	1:5
10.	PPEs		25 pcs	1:1
11.	Multimeter		5 pcs	1:5

12.	Earth resistance meter		5 pcs	1:5
13.	Work stations		25	1:1
14.	Baking oven		5	1:5
15.	Hot air gun		5	1:5
16.	Bearing puller		5	1:5

MODULE III

DIGITAL LITERACY

ISCED UNIT CODE: 0611 441 02B

TVET CDACC UNIT CODE: ENG/CU/EI/BC/01/5/MA

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Digital Literacy

Duration of Unit: 40 Hours

Unit Description

This unit covers the competencies required to demonstrate digital literacy. It involves operating computer devices, solving tasks using the office suite, managing data and information, performing online communication and collaboration, applying cybersecurity skills, and performing jobs online.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Operate Computer Devices	10
2.	Solve Tasks Using Office Suite	5
3.	Manage Data and Information	10
4.	Perform Online Communication and Collaboration	5
5.	Apply Cyber security Skills	5
6.	Perform Online Jobs	5
TOTAL		40

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Operate computer devices	1.1 Meaning and importance of digital literacy 1.2 Functions and Uses of Computers 1.3 Classification of computers 1.4 Components of a computer system 1.5 Computer Hardware 1.6 The System Unit E.g. Motherboard, CPU, casing	1 Observation 2 Portfolio of Evidence 3 Project 4 Written assessment 5 Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	1.7 Input Devices e.g. Pointing, keying, scanning, voice/speech recognition, direct data capture devices. 1.8 Output Devices e.g. hardcopy output and softcopy output 1.9 Storage Devices e.g. main memory e.g. RAM, secondary storage (Solid state devices, Hard Drives, CDs & DVDs, Memory cards, Flash drives 1.10 Computer Ports e.g. HDMI, DVI, VGA, USB type C etc. 1.11 Classification of computer software 1.12 Operating system functions 1.13 Procedure for turning/off a computer 1.14 Mouse use techniques 1.15 Keyboard Parts and Use Techniques 1.16 Desktop Customization 1.17 File and Files Management using an operating system 1.18 Computer Internet Connection Options 1.18.1 Mobile Networks/Data Plans 1.18.2 Wireless Hotspots 1.18.3 Cabled (Ethernet/Fiber) 1.18.4 Dial-Up 1.18.5 Satellite 1.18.6 Computer external devices management 1.19 Device connections 1.20 Device controls (volume controls and display properties)	6 Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
2. Solve tasks using Office suite	2.1 Meaning and Importance of Word Processing 2.2 Examples of Word Processors 2.3 Working with word documents 2.3.1 Open and close word processor 2.3.2 Create a new document 2.3.3 Save a document 2.3.4 Switch between open documents 2.4 Enhancing productivity 2.4.1 Set basic options/preferences 2.4.2 Help resources 2.4.3 Use magnification/zoom tools 2.4.4 Display, hide built-in tool bar 2.4.5 Using navigation tools 2.5 Typing Text 2.6 Document editing (copy, cut, paste commands, spelling and Grammar check) 2.7 Document formatting 2.7.1 Formatting text 2.7.2 Formatting paragraph 2.7.3 Formatting styles 2.7.4 Alignment 2.7.5 Creating tables 2.7.6 Formatting tables 2.8 Graphical objects 2.8.1 Insert object (picture, drawn object) 2.8.2 Select an object 2.8.3 Edit an object 2.8.4 Format an object 2.9 Document Print setup 2.9.1 Page layout, 2.9.2 Margins set up 2.9.3 Orientation.	1 Observation 2 Portfolio of Evidence 3 Project 4 Written assessment 5 Practical assessment 6 Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	2.10 Word Document Printing 2.11 Meaning & Importance of electronic spreadsheets 2.12 Components of Spreadsheets 2.13 Application areas of spreadsheets 2.14 Using spreadsheet application 2.14.1 Parts of Excel screen: ribbon, formula bar, active cell, name box, column letter,row number, Quick Access Toolbar. 2.14.2 Cell Data Types 2.14.3 Block operations 2.14.4 Arithmetic operators (formula bar (-, +, *, /). 2.14.5 Cell Referencing 2.15 Data Manipulation 2.15.1 Using Functions (Sum, Average, SumIF, Count, Max, Max, IF, Rank, Product, mode etc) 2.15.2 Using Formulae 2.15.3 Sorting data 2.15.4 Filtering data 2.15.5 Visual representation using charts 2.16 Worksheet printing 2.17 Electronic Presentations 2.18 Meaning and Importance of electronic presentations 2.19 Examples of Presentation Software 2.20 Using the electronic presentation application 2.20.1 Parts of the PowerPoint screen (slide navigation pane, slide pane, notes, the ribbon, quick access toolbar, and scroll bars). 2.20.2 Open and close presentations	

Learning Outcome	Content	Suggested Assessment Methods
	2.20.3 Creating Slides (Insert new slides, duplicate, or reuse slides.) 2.20.4 Text Management (insert, delete, copy, cut and paste, drag and drop, format, and use spell check). 2.20.5 Use magnification/zoom tools 2.20.6 Apply or change a theme. 2.20.7 Save a presentation 2.20.8 Switch between open presentations 2.21 Developing a presentation 2.21.1 Presentation views 2.21.2 Slides 2.21.3 Master slide 2.22 Text 2.22.1 Editing text 2.22.2 Formatting 2.22.3 Tables 2.23 Charts 2.23.1 Using charts 2.23.2 Organization charts 2.24 Graphical objects 2.24.1 Insert, manipulate 2.24.2 Drawings 2.25 Prepare outputs 2.25.1 Applying slide effects and transitions 2.25.2 Check and deliver • Spell check a presentation • Slide orientation • Slide shows, navigation 2.26 Print presentations (slides and handouts)	

Learning Outcome	Content	Suggested Assessment Methods
3. Manage Data and Information	3.1 Meaning of Data and information 3.2 Importance and Uses of data and information 3.3 Types of internet services 3.3.1 Communication Services 3.3.2 Information Retrieval Services 3.3.3 File Transfer 3.3.4 World Wide Web Services 3.3.5 Web Services 3.3.6 Automatic Network Address Configuration 3.3.7 News Group 3.3.8 Ecommerce 3.4 Types of Internet Access Applications 3.5 Web browsing concepts 3.5.1 Key concepts 3.5.2 Security and safety 3.6 Web browsing 3.6.1 Using the web browser 3.6.2 Tools and settings 3.6.3 Clearing Cache and cookies 3.6.4 URIs 3.6.5 Bookmarks 3.6.6 Web outputs 3.7 Web based information 3.7.1 Search 3.7.2 Critical evaluation of information 3.7.3 Copyright, data protection 3.8 Downloads Management 3.9 Performing Digital Data Backup (Online and Offline) 3.10 Emerging issues in internet	1 Observation 2 Portfolio of Evidence 3 Project 4 Written assessment 5 Practical assessment 6 Oral assessment
4. Perform online communication and collaboration	4.1 Netiquette principles 4.2 Communication concepts 4.2.1 Online communities	1 Observation 2 Portfolio of

Learning Outcome	Content	Suggested Assessment Methods
	4.2.2 Communication tools 4.2.3 Email concepts 4.3 Using email 4.3.1 Sending email 4.3.2 Receiving email 4.3.3 Tools and settings 4.3.4 Organizing email 4.4 Digital content copyright and licenses 4.5 Online collaboration tools 4.5.1 Online Storage (Google Drive) 4.5.2 Online productivity applications (Google Docs & Forms) 4.5.3 Online meetings (Google Meet/Zoom) 4.5.4 Online learning environments 4.5.5 Online calendars (Google Calendars) 4.5.6 Social networks (Facebook/Twitter - Settings & Privacy) 4.6 Preparation for online collaboration 4.6.1 Common setup features 4.6.2 Setup 4.7 Mobile collaboration 4.7.1 Key concepts 4.7.2 Using mobile devices 4.7.3 Applications 4.7.4 Synchronization	Evidence 3 Project 4 Written assessment 5 Practical assessment 6 Oral assessment
5. Apply cyber security skills.	5.1 Data protection and privacy 5.1.1 Confidentiality of data/information 5.1.2 Integrity of data/information 5.1.3 Availability of data/information 5.2 Internet security threats 5.2.1 Malware attacks 5.2.2 Social engineering attacks 5.2.3 Distributed denial of service (DDoS) 5.2.4 Man-in-the-middle attack (MitM)	1 Observation 2 Portfolio of Evidence 3 Project 4 Written assessment 5 Practical assessment 6 Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	5.2.5 Password attacks 5.2.6 IoT Attacks 5.2.7 Phishing Attacks 5.2.8 Ransomware 5.3 Computer threats and crimes 5.4 Cybersecurity control measures 5.4.1 Physical Controls 5.4.2 Technical/Logical Controls (Passwords, Pins, Biometrics) 5.4.3 Operational Controls 5.5 Laws governing protection of ICT in Kenya 5.5.1 The Computer Misuse and Cybercrimes Act No. 5 of 2018 5.5.2 The Data Protection Act No. 24 Of 2019	
6. Perform Online Jobs	6.1 Introduction to online working 6.2 Types of online Jobs 6.3 Online job platforms 6.3.1 Remotask 6.3.2 Data annotation tech 6.3.3 Cloud worker 6.3.4 Upwork 6.3.5 Oneforma 6.3.6 Appen 6.4 Online account and profile management 6.5 Identifying online jobs/job bidding 6.6 Online digital identity 6.7 Executing online tasks 6.8 Management of online payment accounts.	1 Observation 2 Portfolio of Evidence 3 Project 4 Written assessment 5 Practical assessment 6 Oral assessment
7. Apply job entry techniques	7.1 Types of job opportunities 7.1.1 Self-employment 7.1.2 Service provision 7.1.3 product development 7.1.4 salaried employment 7.2 Sources of job opportunities	1 Observation 2 Portfolio of Evidence 3 Project 4 Written

Learning Outcome	Content	Suggested Assessment Methods
	7.3 Resume/ curriculum vitae 7.3.1 What is a CV 7.3.2 How long should a CV be 7.3.3 What to include in a CV 7.3.4 Format of CV 7.3.5 How to write a good CV 7.3.6 Don'ts of writing a CV 7.4 Job application letter 7.4.1 What to include 7.4.2 Addressing a cover letter 7.4.3 Signing off a cover letter 7.5 Portfolio of Evidence 7.5.1 Academic credentials 7.5.2 Letters of commendations 7.5.3 Certification of participations 7.5.4 Awards and decorations 7.6 Interview skills 7.6.1 Listening skills 7.6.2 Grooming 7.6.3 Language command 7.6.4 Articulation of issues 7.6.5 Body language 7.6.6 Time management 7.6.7 Honesty 7.7 Generally knowledgeable in current affairs and technical area	assessment 5 Practical assessment 6 Oral assessment

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project
- Role play
- Case study

Recommended Resources for 25 Trainees

S/N0	RESOURCES	QUANTITY
1.	computers with the following software: • Windows/Linux/Macintosh Operating System • Microsoft Office Software • Google Workspace Account • Antivirus Software	25
2.	Printers	2
3.	Printing Papers	enough
4.	External storage media	25
5.	Projector	1
6.	Whiteboard	1
7.	Smartboard/Smart TV (Where applicable)	1
8.	Assorted whiteboard markers	enough
9.	Internet connection	enough
10.	5 samples of CVs	5
11.	Assorted whiteboard markers	enough
12.	5 samples of job applications	5

COMMUNICATION SKILLS

ISCED UNIT CODE:0031 441 01B

TVET CDACC UNIT CODE: ENG/CU/EI/BC/02/5/MA

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Communication Skills

Duration of Unit: 40 hours

Unit Description

This unit covers the competencies required to apply communication skills. It involves applying communication channels, written communication skills, non-verbal skills, oral communication skills, and group communication skills.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply communication channels.	10
2.	Apply written communication skills.	10
3.	Apply non-verbal skills.	10
4.	Apply oral communication skills.	5
5.	Apply group communication skills.	5
TOTAL		40

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply communication channels	1.1 Communication process 1.2 Principles of effective communication 1.3 Channels/medium/modes of communication 1.4 Factors to consider when selecting a channel of communication 1.5 Barriers to effective communication	• Oral questions • Written assessment • Observation • Portfolio of Evidence • Practical assessment • Third party report

Learning Outcome	Content	Suggested Assessment Methods
	1.6 Flow/patterns of communication 1.7 Sources of information 1.8 Organizational policies	
2. Apply written communication skills	2.1 Types of written communication 2.2 Elements of communication 2.3 Organization requirements for written communication	• Oral assessment • Written assessment • Observation • Portfolio of Evidence • Practical assessment • Third party report
3. Apply non-verbal communication skills	3.1 Utilize body language and gesture 3.2 Apply body posture 3.3 Apply workplace dressing code	• Oral assessment • Written assessment • Observation • Portfolio of Evidence • Practical assessment • Third party report
4. Apply oral communication skills	4.1 Types of oral communication pathways 4.2 Effective questioning techniques 4.3 Workplace etiquette 4.4 Active listening	• Oral assessment • Written assessment • Observation • Portfolio of Evidence • Practical assessment • Third party report
5. Apply group communication skills	5.1 Establishing rapport 5.2 Facilitating resolution of issues 5.3 Developing action plans 5.4 Group organization techniques 5.5 Turn-taking techniques 5.6 Conflict resolution techniques 5.7 Team-work	• Oral assessment • Written assessment • Observation • Portfolio of Evidence • Practical assessment

Suggested Methods of Instruction

- Discussion
- Roleplaying
- Simulation
- Direct instruction
- Demonstration
- Field trips

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 Desktop computers/laptops	Mobile phones	Flashcards
• Internet connection		Flip charts
• 1 Projector • 1 Printer		2 packets of assorted colors of whiteboard marker pens
• 1 Whiteboard		Printing papers
• Report writing templates		

BASIC ELECTRICAL PRINCIPLES

ISCED UNIT CODE:0713 441 09A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/01/5/MA

UNIT DURATION: 100 HOURS

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply electrical principles

UNIT DESCRIPTION

This unit describes competences required to apply electrical principles in their work. It involves applying Electrical quantities, using cells and batteries, applying magnetism and electromagnetism, applying Electrostatics principles, apply concepts of D.C circuit theory and performing electrical measurements.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Applying Electrical quantities	10
2.	Using cells and batteries	10
3.	Apply concepts of D.C circuit theory	20
4.	Applying magnetism and electromagnetism	20
5.	Applying Electrostatics principles	20
6.	Performing electrical measurements	20
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Applying Electrical quantities	1.1 The meaning of SI unit 1.2 SI unit of various types of Electrical parameters 1.3 Ohm's law 1.4 Calculations involving various Electrical parameters e.g. Power, Current, Voltage, Resistance 1.5 Instruments used in measuring various types of Electrical parameters	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning

2. Using cells and batteries	2.1 Sources of electricity 2.2 electrolysis and its applications 2.3 Simple cells 2.4 Primary and secondary cells 2.5 Types of cells and batteries 2.5.1 Dry cells 2.5.2 Leclanché 2.5.3 Mercury 2.5.4 Lead-acid 2.5.5 Alkaline 2.5.6 Lithium 2.6 E.m.f and internal resistance of cells 2.7 Maintenance of batteries 2.8 Applications of batteries	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
3. Apply magnetism and electromagnetism concepts	3.1 Magnetic and non-magnetic materials 3.2 Concepts of magnetic fields and field distribution 3.3 Concepts of electromagnetism 3.4 Laws of electromagnetic induction 3.5 Concepts of self and mutual induction	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
4. Apply Electrostatics principles	4.1 Electrostatics quantities 4.2 Types of capacitors 4.3 Concept of charge and electrostatic field 4.4 Capacitors in series and parallel 4.5 Measurement of capacitance 4.6 Application of Capacitors 4.7 Testing of capacitor	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
5. Apply concepts of D.C circuit theory	1.1 Resistance and resistivity 1.2 Parallel and series circuits 1.3 Basic electrical laws 1.3.1 Ohms law 1.3.2 Kirchhoff's theorem	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
6. Perform electrical measurements	6.1 Types of transducers 6.2 Types of electrical instruments 6.3 Measurements of electrical quantities using Instruments 6.4 Calculations involving electrical	• Practical • Project • Third party report • Portfolio of evidence

	instruments 6.5 Instrumental/systematic errors 6.6 Calculations involving systematic errors	• Written tests • Oral questioning
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Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	J. Bird Electrical and Electronic Principles V.K. Mehta & R. Mehta Basic Electrical Engineering	5 pcs	1:5
2.	Installation manuals	Electronic components datasheets	5 pcs	1:5
3.	Charts	Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Scientific Calculators		25	1:1
5.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
6.	Lecture/theory room	60m ²	1	1:25
7.	Workshop	150m ²	1	1:25
8.	Laboratory	100m ²	1	1:25
9.	Computer laboratory	100m ²	1	1:25
C	Consumable materials			
10.	Connector wires	Jumper wires,	5 pkts	1:5

11.	Insulation tapes		25 pcs	1:1
12.	Circuit boards	Bread board, copper strip boards	25 pcs	1:1
13.	Assorted electronic components	Resistors, diodes, capacitors, transistors, ICs, Transformers, Inductors, Batteries	25 pcs	1:1
14.	Soldering wires		5 rolls	1:5
D	Tools and Equipment			
15.	Striping knives		25 pcs	1:1
16.	Side cutters		25 pcs	1:1
17.	Pliers		25 pcs	1:1
18.	Assorted Screw driver		25 pcs	1:1
19.	Crimping tools		5 pcs	1:5
20.	PPEs		25 pcs	1:1
21.	Multimeters		5 pcs	1:5
22.	Oscilloscope		5 pcs	1:5
23.	Function generator		5 pcs	1:5
24.	Spectrum analyser		5 pcs	1:5
25.	Variable power supply		5 pcs	1:5
26.	Solder guns		25 pcs	1:1
27.	Hot air gun		5 pcs	1:5
28.	Work stations		25	1:1

TECHNICAL DRAWING

ISCED UNIT CODE: 0732 441 13A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/02/5/MA

UNIT DURATION: 100 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Prepare technical drawings

UNIT DESCRIPTION

This unit covers competences required to prepare technical drawings. Competences include preparing drawing equipment and materials, producing plane geometry drawings, producing pictorial and orthographic drawings of components, managing basic operations in AutoCAD, develop drawings in AutoCAD and producing electrical drawings.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Prepare drawing equipment and materials	10
2.	Produce plane geometry drawings	10
3.	Produce pictorial and orthographic drawings of components	20
4.	Manage basic operations in AutoCAD	20
5.	Develop Drawings in AutoCAD	20
6.	Produce electrical drawings	20
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods:

Learning Outcome	Content	Suggested Assessment Methods
1. Prepare drawing equipment and materials	1.1 Identification and care of drawing equipment 1.1.1 Drawing boards 1.1.2 T and set squares 1.1.3 Drawing set	• Practical • Project • Third party report • Portfolio of evidence

Learning Outcome	Content	Suggested Assessment Methods
	1.2 Identification and care of drawing materials 1.2.1 Drawing papers 1.2.2 Pencils 1.2.3 Erasers 1.2.4 Masking tapes 1.2.5 Paper clips 1.3 Reference to manufacturer's instructions and work place procedures on use and maintenance of drawing equipment and materials	• Written tests • Oral questioning
2. Produce plane geometry drawings	2.1 Types of lines in drawings 2.2 Freehand printing of letters 2.3 Borderlines and title blocks 2.4 Construction of different angles 2.5 Measurement of different angles 2.6 Bisection of different angles and lines 2.7 Construction of geometric figures 2.7.1 Circles 2.7.2 Triangles 2.7.3 Rectangles 2.7.4 Parallelogram 2.7.5 Polygons 2.7.6 Pyramids 2.7.7 Conic sections 2.7.8 Prisms 2.8 Patterns development e.g cones, pyramids, prisms, cylinders	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	2.9 Different types of Tangents 2.9.1 Exterior tangents to a circle 2.9.2 Interior tangents to a circle 2.10 Standard drawing conventions	
3. Produce pictorial and orthographic drawings of components	3.1 Meaning of pictorial and orthographic drawings 3.2 Meaning of symbols and abbreviations 3.3 Free hand sketching of different types of geometric forms, tools and equipment 3.4 Drawing and interpretation of orthographic elevations 3.5 Drawing objects in isometric view 3.6 Drawing objects in oblique view	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
4. Manage basic operations in AutoCAD	4.1 Identification of Key features of CAD software 4.2 AutoCAD visual reference commands 4.3 AutoCAD ribbon tools 4.4 AutoCAD status bar tools 4.5 AutoCAD navigation commands 4.6 AutoCAD drawing work 4.7 Saving of AutoCAD drawing files in proper format	• Practical • Demonstration • Projects • Written tests • Oral test
5. Develop Drawings in AutoCAD	5.1 Setting up of Drawing interface. 5.2 Creating drawing Layouts 5.3 Creation of 2D drawings 5.4 Editing/modification of 2D drawings	• Practical • Demonstration • Projects • Written tests

Learning Outcome	Content	Suggested Assessment Methods
	5.5 Saving AutoCAD drawings 5.6 AutoCAD 2D drawing work	• Oral test
6. Produce electrical drawings	6.1 Electrical symbols and abbreviations 6.2 Meaning of electrical drawings 6.3 Drawing of electrical diagrams e.g. block, schematic, circuit, line and wiring 6.4 Drawing of <i>Electrical and Electronic drawings</i> using an appropriate CAD software 6.5 Simulation of Electrical and electronic drawings 6.6 Installation of Electrical components in a building plan using AutoCAD	• Observation • Oral questioning • Practical tests

Suggested Methods of Delivery

- Projects
- Demonstration by trainer
- Practice by the trainee
- Discussions

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	K.Morling Geometric and Engineering drawing	5 pcs	1:5
2.	Drawing instruments	T-squares, set squares, drawing	25	1:1

		sets, Masking tapes		
3.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
4.	Lecture/theory room	50m ²	1	1:25
5.	Drawing tables		25	1:1

ELECTRICAL INSTALLATION

ISCED UNIT CODE: 0713 451 29A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/01/5/MA

UNIT DURATION: 110 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Perform Electrical Installation

Unit Description

This unit describes competences required for performing electrical installation. Competences include conducting site survey, preparing electrical drawings, performing installation system sizing, installing electrical system, perform electrical installation testing, performing electrical installation maintenance and commissioning electrical installation.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Conduct site survey	10
2.	Prepare electrical drawings,	10
3.	Perform installation system sizing	30
4.	Install electrical system	40
5.	Perform electrical installation testing	5
6.	Perform electrical installation maintenance	10
7.	Commission electrical installation.	5
TOTAL		110

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Conduct site survey	1.1 Relevant clauses in appropriate Acts e.g. 1.1.1 Occupational safety and health act (OSHA) 1.1.2 Work injury benefits act (WIBA) 1.1.3 Environment management and coordination Act (EMCA) 1.2 Relevant regulations and standards: 1.2.1 IEEE regulations 1.2.2 BS 7671 regulations 1.2.3 KEBS standards 1.2.4 KPLC/EPRA by-laws 1.2.5 County by-laws 1.3 Common hazards, their sources of danger and preventive measures e.g. burns, cuts, electric shock, electric fire, falling from heights, falling objects, noise, dust, chemicals etc 1.4 Safe and correct handling, use, maintenance and storage of different types of PPE 1.5 Utility services 1.5.1 Water pipes 1.5.2 Gas pipes 1.5.3 Sewer lines 1.6 Electrical parameters e.g. Voltage level, Earth resistance etc. 1.7 Installation route 1.7.1 Surface 1.7.2 Underground 1.7.3 Overhead 1.7.4 Concealed 1.8 Type of installations 1.8.1 Domestic installations 1.8.2 Industrial installations 1.8.3 Commercial installations 1.9 Type of building e.g. 1.9.1 Permanent building 1.9.2 Semi-permanent buildings	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	1.9.3 Special buildings e.g gas filling stations 1.10 Utilities available 1.10.1 Water 1.10.2 Electricity 1.10.3 Communication e.g. Phones 1.11 Installation conditions e.g. temperature, humidity, moisture 1.12 Taking measurements on site 1.12.1 Length e.g. conduits size 1.12.2 Total area 1.13 Survey report 1.13.1 May include; client/ details, technical lead details, sketches, collected data, available utilities, site photos,	
2. Prepare electrical drawings.	2.1 Meaning of working drawings 2.2 Drawing standard electrical symbols 2.3 Drawing of electrical diagrams 2.3.1 Block 2.3.2 Layout/Schematic 2.3.3 Wiring 2.4 Reading and Interpretation of architectural drawings 2.5 Reading and Interpretation of electrical drawings 2.6 Use of Computer Aided Design (CAD) applications e.g. AutoCAD 2.7 Relevant approval authorities 2.8 Types of wiring system e.g. Surface, Conduits, Trunking, etc.	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning
3. Perform installation system sizing	3.1 Electrical parameters 3.1.1 Voltage level 3.1.2 Earth resistance 3.2 Installation route 3.2.1 Surface	• Practical • Projects • Third party report • Portfolio of evidence

Learning Outcome	Content	Suggested Assessment Methods
	3.2.2 Underground 3.2.3 Overhead 3.2.4 Concealed 3.3 Type of installations 3.3.1 Domestic installations 3.3.2 Industrial installations 3.3.3 Commercial installations 3.4 Type of building e.g. 3.4.1 Permanent building 3.4.2 Semi-permanent buildings 3.4.3 Utilities available 3.4.4 Water 3.4.5 Electricity 3.4.6 Communication e.g. Phones 3.5 Installation conditions e.g. temperature, humidity, moisture 3.6 Taking measurements on site 3.6.1 Length e.g. conduits size 3.6.2 Total area 3.6.3 Temperature 3.6.4 Humidity 3.7 Determination of number of luminaries 3.7.1 Determination of power points 3.7.2 Rating of lighting and power circuits 3.7.3 Diversity factor 3.8 Types of loads e.g. single phase load, three-phase load, etc. 3.9 Determining cable 3.9.1 Types 3.9.2 Ratings 3.9.3 Sizes	• Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	3.9.4 Insulation types 3.10 Protective devices 3.10.1 Types 3.10.2 Ratings 3.11 Accessories 3.11.1 Types e.g. switches, sockets, plugs, CBs, Fuses, distribution board, conduits etc. 3.11.2 Ratings. 3.12 Phase balancing 3.12.1 Meaning 3.12.2 Reasons 3.12.3 Causes of imbalance 3.12.4 Effects of imbalance 3.12.5 Procedure 3.12.6 Techniques 3.13 Preparation of working drawings 3.14 Preparation of BOQ 3.15 Contract documents e.g. contract letter, BOQ, working drawings, etc	
4. Install Electrical System	4.1 Safety procedures e.g. 4.1.1 Use PPE 4.1.2 Common hazards and sources of danger during installation e.g. burns, cuts, electric shock, falling from heights, falling objects, noise, dust, chemical etc. 4.2 Handling tools, equipment and materials 4.3 Cables management systems; 4.3.1 Types e.g. cable ducts, cable trays etc. 4.3.2 Benefits of cable management systems. 4.4 Accessories used in Electrical installation e.g. Lamp holders, Conduits, Ceiling roses, Patress etc. 4.5 Installation procedures e.g Marking out, cutting, bending, threading,	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	chiselling, trenching 4.6 Joints and terminations 4.7 Preparation of cable ways and wiring systems; 4.7.1 Surface 4.7.2 PVC conduit 4.7.3 Metal conduit 4.7.4 Trunking 4.7.5 Ducts 4.7.6 Cable trays 4.7.7 Fixing 4.7.8 Draw Cables and conductors 4.8 Cable lugging, glanding, labeling and dressing 4.9 Earthing and protection systems 4.10 Installation of final circuits 4.10.1 Lighting circuits e.g one-way, two-way, intermediate, dimmer switches, looping-in methods at ceiling rose, etc 4.10.2 Power circuits eg. radial circuits, ring circuits, water heating circuits, cooker circuits etc. 4.11 Operation of electrical system 4.11.1 Thermostat 4.11.2 Three heat switches 4.11.3 Cooker socket 4.11.4 Earthing and protection devices 4.11.5 Relevant IEEE regulations 4.12 Housekeeping 4.12.1 Waste disposal 4.12.2 Cleaning 4.12.3 Tool storage 4.12.4 Reusing and recycling	

Learning Outcome	Content	Suggested Assessment Methods
5. Perform electrical installation testing	5.1 Reading and interpretation of appropriate manuals. 5.2 Identification of test equipment; 5.2.1 Test lamp 5.2.2 Insulation resistance tester 5.2.3 Continuity tester 5.2.4 Earth loop impedance tester. 5.3 Meaning of terms 5.4 Types of tests e.g. 5.4.1 Earth continuity tests 5.4.2 Ring continuity circuit test 5.4.3 Insulation resistance tests 5.4.4 Earth electrode resistance test 5.4.5 Polarity test 5.4.6 Earth loop impedance test 5.5 Inspection 5.6 Testing tools e.g. 5.6.1 Multimeter 5.6.2 Insulation resistance tester 5.6.3 Earth electrode resistance tester 5.6.4 Ohmmeter 5.7 Procedure for conducting identified test; 5.7.1 Polarity 5.7.2 Effectiveness of earthing 5.7.3 Insulation resistance 5.7.4 Ring circuit continuity 5.8 Recording and verification of results against appropriate standards. 5.9 Safety precautions to be observed	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
6. Perform electrical installation maintenance	6.1 Safety procedures 6.2 Meaning of terms. 6.3 Maintenance 6.3.1 Types and procedures 6.3.2 Periodic service 6.3.3 Preventive 6.3.4 Breakdown 6.3.5 Corrective 6.3.6 Scheduling maintenance based on service manuals 6.3.7 Identification and documentation of maintenance tools, materials and equipment 6.3.8 Specifications of identified tools, materials and equipment against safety standards 6.4 Fault diagnosis/troubleshooting 6.5 System isolation points 6.6 System tests e.g. continuity test, functionality test, polarity test etc. 6.7 Maintenance report and records	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning
7. Commission electrical installation	7.1 Commissioning 7.1.1 Meaning 7.1.2 Importance 7.1.3 Formulation of commissioning procedure with representatives of organization, consultant and client 7.1.4 Identification and briefing of the commissioning	• Practical • Projects • Third party report • Portfolio of evidence • Written tests

Learning Outcome	Content	Suggested Assessment Methods
	team 7.2 End-user training 7.3 Operation manuals and brochures 7.4 Preparation of completion documents 7.4.1 Completion certificate 7.4.2 Importance 7.4.3 Components 7.4.4 User manuals 7.5 Functionality test 7.6 Project handover 7.6.1 Meaning of Handover 7.6.2 Handover documents	Oral questioning

Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/N o.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
28	Textbooks	B. Scaddan Electrical installation work J. Hyde Electrical installation Principles and Practices	5 pcs	1:5
29	Installation manuals	IEEE regulation BS3939 NEMA regulations OSHA	5 pcs	1:5
30	Charts	Single line diagram	1 pcs for each	1:25

		Circuit diagrams Colour codes		
31	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
1	Lecture/theory room	60m ²	1	1:25
1.	Workshop	150m ²	1	1:25
2.	Laboratory	100m ²	1	1:25
3.	Site			
C	Consumable materials			
1.	Electrical wires	1.5mm ² (red, black green)	5 rolls	1:5
		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
2.	Insulation tapes		25 pcs	1:1
2	Accessories	Switches, sockets, Junction boxes, Consumer units, Lamp holders, Patrice boxes, Circuit breakers	25 pcs	1:1
3	Pipes and trunkings	PVC conduits, Steel conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			
4.	Hacksaws		25 pcs	1:1
5.	Striping knives		25 pcs	1:1
6.	Side cutters		25 pcs	1:1
7.	Pliers		25 pcs	1:1

8.	Tape measure		25 pcs	1:1
9.	Try Square		25 pcs	1:1
10	Spirit level		25 pcs	1:1
11	Assorted Screw driver		25 pcs	1:1
12	Assorted hammers		25 pcs	1:1
13	Crimping tools		5 pcs	1:5
14	PPEs		25 pcs	1:1
15	Multimeters		5 pcs	1:5
16	Clamp meters		5 pcs	1:5
17	Earth resistance meter		5 pcs	1:5
18	Stocks & Dies		5 pcs	1:5
19	Vices		5 pcs	1:5
20	Oscilloscope		5 pcs	1:5
21	Pipe bending Machine		5 pcs	1:5
22	Bending spring		5 pcs	1:5
23	Drilling machines		5 pcs	1:5
24	Work stations		25	1:1
25	Installation boards		13 pcs	1:2

MODULE IV

WORK ETHICS AND PRACTICES

ISCED UNIT CODE: 0417 441 04B

TVET CDACC UNIT CODE: ENG/CU/EI/BC/03/5/MA

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply work ethics and practices.

Duration of Unit: 40 hours

Unit Description

This unit covers competencies required to apply work ethics and practices. It involves applying self-management skills, promoting ethical work practices and values, promoting teamwork, maintaining professional and personal development, applying problem-solving and promoting customer care.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply Self-Management Skills	10
2.	Promote Ethical Practices and Values	10
3.	Promote Teamwork	5
4.	Maintain Professional and Personal Development	5
5.	Apply Problem-Solving Skills	5
6.	Promote Customer Care.	5
TOTAL		40

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply Self-Management Skills	1.1 Self-awareness 1.2 Formulating personal vision, mission, and goals 1.3 Healthy lifestyle practices 1.4 Strategies for overcoming work challenges 1.5 Emotional intelligence 1.6 Coping with Work Stress. 1.7 Assertiveness versus aggressiveness and passiveness 1.8 Developing and maintaining	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	high self-esteem 1.9 Developing and maintaining positive self-image 1.10 Time management 1.11 Setting performance targets 1.12 Monitoring and evaluating performance targets	
2. Promote Ethical Work Practices and Values	2.1 Integrity 2.2 Core Values, ethics and beliefs 2.3 Patriotism 2.4 Professionalism 2.5 Organizational codes of conduct 2.6 Industry policies and procedures	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Promote Teamwork	3.1 Types of teams 3.2 Team building 3.3 Individual responsibilities in a team 3.3.1 Determination of team roles and objectives 3.3.2 Team parameters and relationships 3.4 Benefits of teamwork 3.5 Qualities of a team player Leading a team 3.6 Team performance and evaluation 3.7 Conflicts and conflict resolution 3.8 Gender and diversity mainstreaming 3.9 Developing Healthy workplace relationships 3.10 Adaptability and flexibility 3.11 Coaching and mentoring skills	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
4. Maintain professional and personal development	4.1 Personal vs professional development and growth 4.2 Avenues for professional	• Observation • Written assessment

Learning Outcome	Content	Suggested Assessment Methods
	growth 4.3 Recognizing career advancement 4.4 Training and career opportunities 4.5 Assessing training needs 4.6 Mobilizing training resources 4.7 Licenses and certifications for professional growth and development 4.8 Pursuing personal and organizational goals 4.9 Managing work priorities and commitments 4.10 Dynamism and on-the-job learning	• Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Apply problem-solving skills	5.1 Causes of problems 5.2 Methods of solving problems 5.3 Problem-solving process 5.4 Decision making 5.5 Creative thinking and critical thinking process in development of innovative and practical solutions	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Promote customer care	6.1 Identifying customer needs 6.2 Qualities of good customer service 6.3 Customer feedback methods 6.4 Resolving customer concerns 6.5 Customer outreach programs 6.6 Customer retention	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion

- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 Trainees

S/NO	RESOURCES	QUANTITY
1.	Computers	25
2.	Stationery	ENOUGH
3.	Charts	25
4.	Video clips	ENOUGH
5.	Audio tapes	ENOUGH
6.	Radio sets	2
7.	TV sets	1
8.	LCD projectors	1

ENGINEERING MATHEMATICS I

ISCED UNIT CODE: 0541 441 05A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/03/5/MA

UNIT DURATION: 80 HOURS

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply engineering mathematics

Unit Description

This unit describes the competencies required to apply a wide range of engineering technician mathematics. Competencies include: applying number systems, applying algebra, applying trigonometry and hyperbolic functions, performing coordinates geometry and carrying out binomial expansions.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply number systems	5
2.	Apply algebra	20
3.	Apply Trigonometry and Hyperbolic functions	20
4.	Perform coordinates geometry	15
5.	Carry out binomial expansions	20
TOTAL		80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply number systems	1.1 Types of numbers 1.1.1 Rounding off numbers to the nearest tens,	• Written assessment • Practical

	1.1.2hundreds, thousands, millions and billions 1.1.3 Whole numbers 1.1.4Odd numbers 1.1.5Even numbers 1.1.6Prime number 1.1.7Ratio and proportions 1.2 Percentages 1.2.1 Word problems involving natural 1.2.2Numbers 1.3 Factors 1.3.1Factors of composite numbers 1.3.2Prime factors 1.3.3Factors in power form 1.4 Divisibility Test 1.4.1GCD 1.4.2Application of GCD/HCF to real life situations 1.5 LCM 1.5.1Multiples of a number 1.5.2 LCM of a set of numbers 1.5.3Application of LCM in real life situations 1.6 Integers 1.6.1 The number line 1.6.2Operation on integers	assessment • Oral Questioning
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	1.6.3 Order of operations 1.6.4 Application to real life situation 1.7 Fractions 1.7.1 Proper, improper fractions and mixed numbers 1.7.2 Conversion of improper fractions to mixed numbers and vice versa 1.7.3 Comparing fractions 1.7.4 Operations on fractions 1.7.5 Order of operations on fractions 1.7.6 Word problems involving fractions in real life situations 1.8 Decimals 1.8.1 Fractions and decimals 1.8.2 Recurring decimals 1.8.3 Recurring decimals and fractions 1.8.4 Decimal places 1.8.5 Standard form 1.8.6 Operations on decimals 1.8.7 Order of operations 1.8.8 Real life problems involving decimals 1.9 Arithmetic operation 1.9.1 Addition 1.9.2 Subtraction	
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	1.9.3 Multiplication 1.9.4 Division 1.10 Squares and square roots 1.10.1 Squares by multiplication 1.10.2 Square roots by factorization 1.10.3 Squares and Square roots using Calculators	
2. Apply algebra	2.1 Indices and logarithms 2.1.1 Indices (powers) and base 2.1.2 Laws of indices (including positive integers, negative integers and fractional indices) 2.1.4 indices) 2.1.5 Powers of 10 and common logarithms 2.1.6 Common logarithms; 2.1.7 characteristics 2.1.8 mantissa 2.1.9 Logarithm tables 2.1.10 Application of common logarithms in 2.1.11 multiplication, division and finding 2.1.12 roots 2.2 Algebra 2.2.1 Algebraic expressions including	• Written assessment • Practical assessment • Oral Questioning

	algebraic fractions 2.2.2 Simplification of algebraic expressions 2.2.3 Factorization by grouping 2.2.4 Removal of brackets 2.2.5 Substitution and evaluation 2.2.6 Problem solving in real situation 2.3 Simultaneous Equation 2.4 Solve simultaneous equation by; 2.4.1 Elimination method 2.4.2 Substitution method 2.4.3 Graphical method 2.4.4 Solve real life problems 2.5 Quadratic expressions and equations 2.5.1 Expansion of algebraic expressions 2.5.2 The three quadratic identities 2.5.3 Using the three quadratic identities 2.5.4 Factorization of quadratic expressions 2.5.5 Solutions of quadratic equations by factor method 2.5.6 Formation and solution of quadratic equations	
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3. Apply trigonometry and hyperbolic functions	3.1 Importance of trigonometry in engineering. 3.2 Trigonometric Ratios and Functions 3.2.1 Definitions of sine, cosine, tangent, cosecant, secant, and cotangent. 3.2.2 Unit circle and angle measurement 3.2.2.1 Degrees and radians 3.2.2.2 Graphs of trigonometric functions. 3.3 Trigonometric Identities 3.3.1 Fundamental identities 3.3.1.1 Pythagorean 3.3.1.2 Reciprocal 3.3.1.3 Quotient identities 3.3.2 Co-function identities and even-odd properties. 3.3.3 Sum and difference formulas, double-angle, and half-angle formulas. 3.4 Solving Trigonometric Equations 3.4.1 Basic Trigonometric Equations 3.4.1.1 Solving equations involving basic trigonometric functions. 3.4.1.2 Using identities to simplify and solve equations. 3.4.2 Inverse Trigonometric Functions 3.4.2.1 Definition and properties. 3.4.2.2 Solving equations using inverse trigonometric functions. 3.4.3 Applications of Trigonometric Equations 3.4.3.1 Engineering problems involving periodic functions and waveforms. 3.4.3.2 Harmonic motion and oscillations	• • Written assessment • Oral assessment • Practical • Project • Third party report • Portfolio of evidence •
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	in mechanical systems. 3.5 Trigonometry in Triangles 3.5.1 Right-Angle Triangles 3.5.1.1 Solving for sides and angles using trigonometric ratios. 3.5.1.2 Applications in engineering problems such as inclined planes and forces. 3.5.2 Non-Right-Angle Triangles 3.5.2.1 Law of Sines and Law of Cosines. 3.5.2.2 Solving oblique triangles. 3.5.2.3 Applications in engineering fields of structural analysis and navigation. 3.6 Introduction to Hyperbolic Functions 3.6.1 Definitions of hyperbolic sine, cosine, tangent, and their reciprocals. 3.6.2 Graphs and properties of hyperbolic functions. 3.7 Hyperbolic Identities 3.7.1 Fundamental identities 3.7.1.1 Pythagorean-like identities 3.7.2 Sum and difference formulas, double-angle, and half-angle formulas. 3.8 Solving Hyperbolic Equations 3.8.1 Basic Hyperbolic Equations 3.8.1.1 Solving equations involving basic hyperbolic functions. 3.8.1.2 Using identities to simplify and solve equations.	
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	3.8.2 Inverse Hyperbolic Functions 3.8.2.1 Definition and properties. 3.8.2.2 Solving equations using inverse hyperbolic functions. 3.9 Applications of Hyperbolic Functions in Engineering	
4. Perform coordinates geometry	4.1 Polar equations 4.1.1 Definition of polar coordinates 4.1.2 Definition and examples of polar equations. 4.1.3 Representing curves using polar equations. 4.2 Cartesian equation 4.2.1 Definition of Cartesian coordinates. 4.2.2 Definition of a point in cartesian coordinates 4.2.3 Relationship between Cartesian and polar coordinates. 4.2.4 Definition and examples of Cartesian equations. 4.2.5 Representing lines, circles, parabolas, ellipses, and hyperbolas using Cartesian equations. 4.2.6 Conversion Between Polar and Cartesian Equations. 4.3 Graphs of polar equations 4.3.1 Plotting Polar Equations 4.3.1.1 Definition of a point in polar coordinates 4.3.1.2 Steps to graph polar equations. 4.3.1.3 Using symmetry and periodicity in polar graphs.	• Written assessment • Oral assessment • Practical • Project • Third party report • Portfolio of evidence

	4.3.2 Analyzing Polar Graphs 4.3.2.1 Identifying key features intercepts, maxima, minima 4.3.3 Applications in engineering 4.4 Normal and tangents 4.4.1 Tangents to Curves 4.4.1.1 Definition of a tangent line. 4.4.1.2 Finding the slope of a tangent to a curve at a given point. 4.4.2 Normals to Curves 4.4.2.1 Definition of a normal line. 4.4.2.2 Finding the equation of a normal to a curve at a given point. 4.4.3 Tangents and Normals in Polar Coordinates 4.4.3.1 Techniques for finding tangents and normals to curves defined by polar equations. 4.5 Loci 4.5.1 Introduction to Loci 4.5.1.1 Definition of Locus 4.5.1.2 Understanding the concept of a locus. 4.5.1.3 Importance of loci in engineering. 4.5.2 Locus in Relation to a Circle 4.5.2.1 Equations and properties of loci relative to circles. 4.5.2.2 Common loci problems involving circles 4.5.3 Applications in Engineering	
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	4.5.3.1 Using loci to solve engineering problems: robotic arm movement 4.5.3.2 Analyzing Loci of Points: tracing the path of a point on a rotating arm 4.5.3.3 Using loci to optimize mechanical systems: designing cams, robotic path planning.	
5. Carry out binomial expansion	5.1 Basic concepts of binomial theorem 5.1.1 Binomial expressions and notation. 5.1.2 Factorials and their use in binomial coefficients. 5.1.3 Binomial Coefficients 5.1.3.1 Definition and calculation using combinations (nCr). 5.1.3.2 Pascal's Triangle as a tool for finding binomial coefficients. 5.2 Binomial Expansion 5.2.1 General form of the binomial expansion expression 5.2.2 Binomial Expansion of $(a + b)^n$ where $n = 2, 3, 4, \dots$ 5.2.3 Special cases 5.2.3.1 When $(1 + x)^n$ 5.2.3.2 Negative and fractional binomial expansions using the binomial series 5.3 Applications of Binomial Expansion 5.3.1 Simplifying algebraic expressions using binomial expansion. 5.3.2 Solving polynomial equations. 5.4 Engineering Applications 5.5 Estimating values in engineering calculations.	• Written assessment • Oral assessment • Practical • Project • Third party report • Portfolio of evidence

Suggested Methods of Instruction

1. Practical
2. Demonstrations
3. Group discussion
4. Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	Engineering Mathematics by John bird 8 th edition	5 pcs	1:5
2.		Engineering Mathematics by A.K stround 8 th edition	5 pcs	1:5
3.		SMP	25	1:1
B	Learning Facilities & infrastructure			
1	Lecture/theory room	50 m ²	1	1:25
C	Consumable materials			
1.	Charts	Manila papers		
2.	marker pens	Erasable		
D	Tools and Equipment			
19.	Calculators	Scientific	25 pcs	1:1

ANALOGUE ELECTRONICS I

ISCED UNIT CODE: 0714 541 15A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/04/5/MA

UNIT DURATION: 50 HOURS

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply analogue electronics one.

UNIT DESCRIPTION

This unit describes the competencies required to apply analogue electronics. These competencies include; applying semiconductor theory, applying semiconductor diodes, applying understanding of transistors, applying special semiconductor devices and performing rectification.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply semiconductor theory	10
2.	Apply semiconductor diode	10
3.	Apply transistors	10
4.	Apply special semiconductor devices	10
5.	Perform rectification	10
TOTAL		50

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Understand semiconductor theory	1.1 Atomic structure 1.1.1. Structure of the Atom 1.1.2. Electron Configuration 1.1.3. Ions and Charge Carriers 1.2 Types of materials 1.2.1. Insulators 1.2.2. Conductors 1.2.3. Semiconductors 1.2.4. Semiconductor materials 1.3 Types of semiconductor materials 1.3.1. Intrinsic semiconductors 1.3.2. Extrinsic semiconductors	• Practical test • Project • Third Party Report • Portfolio of evidence • Written test • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	1.3.2.1. n-type extrinsic semiconductor 1.3.2.2. p-type extrinsic semiconductor 1.4 The pn junction 1.4.1. Properties of pn junction 1.4.2. Current flow in a forward biased pn junction 1.4.3. Current flow in a reverse biased pn junction 1.4.4. V-I characteristics of a pn junction	
2. Apply semiconductor diodes.	2.1 Introduction to the crystal diode 2.2 Characteristics of the crystal diode 2.2.1. Resistance of a crystal diode 2.2.2. Equivalent circuit of the crystal diode 2.3 Biasing of the crystal diode 2.3.1 Forward biasing 2.3.2 Reverse biasing 2.4 Limitations in the operating conditions of a crystal diode 2.4.1 forward current rating 2.4.2 PIV 2.4.3 power rating 2.5 Special purpose diodes 2.5.1 LED 2.5.2 Photodiode 2.5.3 Optoisolator 2.5.4 Tunnel diode 2.5.5 Varactor diode 2.5.6 Schockley diode 2.6 Application of semiconductor diodes	• Practical test • Third Party Report • Portfolio of evidence • Written test • Oral questioning
3. Apply transistors.	3.1 Bipolar junction transistors (BJTs) 3.1.1 Types and construction of BJT transistors 3.1.2 Operation of NPN and PNP transistors 3.1.3 Characteristics of BJTs, i.e., V-I, and gain 3.2 BJT configurations 3.2.1 Common emitter	• Practical test • Third Party Report • Portfolio of evidence • Written test • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	3.2.2 Common base 3.2.3 Common collector 3.3 Characteristics of BJT connections 3.4 BJT transistor load line analysis 3.4.1 DC load line 3.4.2 AC load line 3.5 BJT transistor biasing methods 3.5.1 Key terms in transistor biasing (faithful amplification, variation of transistor parameters, stabilisation) 3.5.2 Base resistor, emitter bias, collector feedback, voltage divider biasing techniques 3.6 Field Effect Transistors (FETs) – JFET & MOSFET 3.7 P and N channels of FETs 3.8 Operation of FETs 3.9 Characteristics of FETs 3.10 Biasing techniques of FETs 3.11 Application of FETs	
4. Apply special semiconductor devices.	4.1 Special semiconductor devices 4.1.1 SCR 4.1.2 LASCR 4.1.3 TRIAC 4.1.4 DIAC 4.1.5 SCS 4.1.6 UJT 4.2 Operation principle of special semiconductor devices 4.3 Schematic symbols of special semiconductor devices 4.4 Application of special semiconductor devices	• Practical test • Third Party Report • Portfolio of evidence • Written test • Oral questioning
5. Perform rectification.	5.1 Types of rectifiers 5.1.1 Half wave rectifiers 5.1.2 Full wave rectifiers (center-tap and bridge) 5.2 Classes of rectifiers 5.2.1 Uncontrolled Rectifier 5.2.2 Controlled Rectifier 5.2.3 Half-Controlled Rectifier 5.2.4 Fully-Controlled Rectifier	1 Practical test • Third Party Report • Portfolio of evidence • Written test • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	5.3 Application of rectifiers 5.4 Types of converters 5.4.1 AC to DC converter (rectifier) 5.4.2 DC to AC Converter (Inverter) 5.4.3 DC to DC Converter 5.4.4 AC to AC Converter 5.5 Application of converters	

Suggested Methods of Instruction

- Practical
- Project
- Group discussions
- Demonstration
- Visit to manufacturing and processing industries
- On-job-training
- Charts and Audio-visual presentations

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference books	Mehta, V. K., & Mehta, R. (2020). Principles of electronics (12 edition). S. Chand and Company Limited, Theraja, B. L., & Theraja, A. K. (2005). A textbook of electrical technology (1st multicolour ed., Multicolour illustrative ed., 23rd rev. multicoloured ed). S. Chand & Co. Bird, J. O. (2022). Bird's electrical and electronic principles and	10 pcs for each book	1:2.5

		technology (Seventh edition). Routledge, Taylor & Francis Group.		
2.	Software	Assorted simulation software e.g., Circuit wizard.	25	1:1
3.	Audio visual presentations	Projector	1	1:25
B	Learning Facilities & infrastructure			
4.	Lecture/theory room	60m ²	1	1:25
5.	Workshop	150m ²	1	1:25
6.	Computer laboratory	100m ²	1	1:25
C	Consumable materials			
7.	Electronic components	Breadboards, Stripboards, Jumper wires, Assorted resistors, Assorted capacitors, Assorted MOSFETs, Assorted JFETs, 555 timers, Solder wire, LEDs, Assorted BJT transistors, LDRs, OPAMPs, thermistors, 12V DC motors	25 pcs	1:1
D	Tools and Equipment			
8.	Assorted tools and equipment	Side cutters, Side cutters, Pliers, Screw driver, Crimping tools, Mult-meter, Solder guns	25 pcs	1:1
9.	Assorted electrical gadgets	Solder gun, Heat sink, Hot air guns, function generator	25 pcs	1:1
10.	Assorted measuring instruments	Digital Oscilloscope,	5 for each category	1:5
11.	Digital Multimeter,			
12.	Digital functional generator		3 pcs	1:8

13.	Laser jet printer		2 pcs	1:13
14.	Power supply	Variable power supply, 5V Power adapters, 9V Power adapters, 12V Power adapters.	10 pcs	1:3
15.	Trainers kit	Analogue training kits, PWM kit	5 pcs	1:5
16.	PCB prototyping material	Copper board, ferrite chloride solution, see-through printing paper, HASL finishing PCB	25 for each category	1:1
17.	Drilling gun		3 pcs	1:8
18.	Work stations		25	1:1

DIGITAL ELECTRONICS I

ISCED UNIT CODE: 0714 441 15A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/05/5/MA

UNIT DURATION: 50 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Digital Electronics

Unit Description

This unit describes competences required to apply digital electronics. Competences include applying knowledge of number systems, applying knowledge of binary code and applying logic gates and Boolean algebra concepts.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply knowledge of number systems	10
2.	Apply knowledge of binary codes	20
3.	Apply Logic gates and Boolean algebra concepts	20
TOTAL		50

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply knowledge of number systems	1.1 Numbers systems e.g. 1.1.1 Decimal 1.1.2 Octal 1.1.3 Hexadecimal 1.1.4 Binary 1.2 Number system representation 1.3 Conversion of number systems 1.4 Perform 1's and 2's complement	1 Observation 2 Written test 3 Practical 4 Demonstration 5 Oral questioning 6 Third party report
2. Apply knowledge of binary codes	2.1 BCD (Binary Coded Decimal) 2.2 Gray Code 2.3 Excess 3 code	1 Observation 2 Written test 3 Practical

		4 Demonstration 5 Oral questioning 6 Third party report
3. Apply Logic gates and Boolean algebra concepts	3.1 Basic logic gates 3.2 Universal logic gates operation 3.3 Special purpose gates 3.4 laws of Boolean algebra 3.5 Logic expressions simplification 3.6 K-MAPS	1 Observation 2 Written test 3 Practical 4 Demonstration 5 Oral questioning 6 Third party report

Suggested Methods of Instruction

- Role playing
- Viewing of related videos
- Discussion
- Direct Instruction

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference books	Digital Electronics: Principles, Devices and Applications By Anil K. Maini	5 pcs	1:5
2.	Installation manuals	Assorted Systems component Manufacturer's manuals and data sheets Instrumentation Handbooks	5 pcs	1:5
3.	Charts	Assorted Circuit diagrams charts	1 pcs for each	1:25
4.	Software	Assorted simulation software e.g Deeds,	25	1:1
5.	Audio visual presentations	Projector	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25

2.	Workshop	150m ²	1	1:25
3.	Computer laboratory	100m ²	1	1:25
C	Consumable materials			
1.	Assorted electronics components	ICs, resistors, capacitors	25 pcs	1:1
D	Tools and Equipment			
1.	Assorted tools and equipment	Side cutters, Side cutters, Pliers, Screw driver, Multi-meter, Oscilloscope, Solder guns, breadboards	25 pcs	1:1
2.	PPEs	Safety boots, overall	25 pcs	1:1
3.	Function generator		5 pcs	1:5
4.	Variable power supply		5 pcs	1:5
5.	Trainers kit	Assorted logic gate, combinational circuits trainer kits with component Manufacturer's manuals and data sheets	5 pcs	1:5
6.	Hot air gun		5 pcs	1:5
7.	Work stations		25	1:1

SECURITY SYSTEM INSTALLATION

ISCED UNIT CODE: 0714 451 33A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/02/5/MA

UNIT DURATION: 120 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Install security systems

UNIT DESCRIPTION

This unit covers competences required in performing security system installation.

Competences include applying health and safety measures, installing security systems, testing security system installation and maintaining security system installations.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply health and safety measures	10
2.	Install security Systems	40
3.	Test security system installation	40
4.	Maintain security system installation	30
TOTAL		120

Learning Outcomes, Content and Suggested Assessment Methods:

Learning Outcome	Content	Suggested Assessment Methods
1. Apply health and safety measures	<u>Theory Content</u> 1.1 Health and Safety Measure 1.1.1 Relevant clauses in appropriate Acts e.g. 1.1.2 Occupational safety and health act (OSHA) 1.1.3 Work injury benefits act (WIBA) 1.1.4 Safety Regulations and procedures 1.2 Work site preparation 1.3 Security systems installation tools 1.3.1 Fixing tools	1 Practical 2 Project 3 Third party report 4 Portfolio of evidence 5 Written tests Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	1.3.2 Measuring tools 1.3.3 Termination tools e.g crimping tool 1.3.4 Cutting tools 1.4 Electrical measuring instruments 1.4.1 Multimeter 1.4.2 Earth resistance meter 1.4.3 Cable testers 1.5 Marking piping and fixing tools e.g scribes, pliers 1.6 Importance of marking	
2. Install Security Systems	2.1 Types of cables in security system 2.2 Factors to consider in security system cable laying 2.3 Segregation in cable laying 2.3.1 Importance segregations 2.4 Security system installation 2.5 CCTV system 2.5.1 Identification of materials and components 2.5.1.1 Cables 2.5.1.2 Conduits, trunking etc 2.5.1.3 CCTV system components 2.5.1.4 Backup system (data and power) 2.6 Specification of tools, equipment and materials 2.6.1 Tolerance/ range 2.6.2 Make / model 2.6.3 Size 2.6.4 Class 2.7 Wiring CCTV system 2.7.1 Schematic diagram 2.7.2 Wiring diagram 2.8 Alarm systems 2.8.1 Fire alarm 2.8.2 Burglar alarm 2.9 Identification of materials and components 2.9.1 Cables 2.9.2 Conduits, trunking etc	1 Practical 2 Project 3 Third party report 4 Portfolio of evidence 5 Written tests 6 Oral questionin g

Learning Outcome	Content	Suggested Assessment Methods
	2.9.3 Alarm system components 2.9.4 Backup system(power) 2.10 Specification of tools, equipment and materials 2.10.1 Tolerance/range 2.10.2 Make/model 2.10.3 Size 2.10.4 Class 2.11 Wiring alarm system 2.11.1 Schematic diagram 2.11.2 Wiring diagram 2.12 Electric fence 2.13 identification of materials and components 2.13.1 Cables 2.13.2 Conduits, trunking etc 2.13.3 Electric fence components 2.13.4 Backup system(power) 2.14 Specification of tools, equipment and materials 2.14.1 Tolerance/ range 2.14.2 Make / model 2.14.3 Size 2.14.4 Class 2.15 Wiring electric fence system 2.15.1 Schematic diagram 2.15.2 Wiring diagram 2.16 Insulation classes of enclosures - e.g. 2.16.1 IP 44 (Ingress protection) 2.16.2 IP 55 2.16.3 IP 65 2.16.4 IP 66 2.16.5 IP 67 2.17 Cable labelling 2.18 Cable termination 2.18.1 Importance of termination 2.19 Tools used in cable termination - e.g. 2.19.1 Strip Knife 2.20 Security system integration with other components	

Learning Outcome	Content	Suggested Assessment Methods
	2.21 Housekeeping practices <u>Practical content</u> 2.22 Install security system 2.23 CCTV system 2.24 Alarm systems 2.25 Electric fence system 2.26 Coding/configuring security system 2.27 Proper disposal of waste material <u>Practical content</u> 2.28 Install security system 2.29 Coding security system 2.30 Proper disposal of waste material	
3. Test security system installation	<u>Theory content</u> 3.1 Visual inspection 3.2 Types of tests in security system e.g. 3.2.1 Insulation test 3.2.2 Short circuit test 3.2.3 Continuity test 3.2.4 Arming and disarming tests 3.2.5 Physical inspection of the system 3.3 Test results documentation 3.4 Security system commissioning <u>Practical content</u> 3.5 Test performance of system as per design specifications	1 Practical 2 Project 3 Third party report 4 Portfolio of evidence 5 Written tests 6 Oral questioning
4. Maintain security system installation	<u>Theory content</u> 4.1 Maintenance schedule preparation 4.2 System maintenance check list preparation 4.3 Maintenance tools and equipment	1 Practical 2 Project 3 Third party report

Learning Outcome	Content	Suggested Assessment Methods
	selection 4.4 Inspection and tests 4.5 Faults diagnosis 4.6 Faults rectification 4.7 Maintenance reports documentation <u>Practical Content</u> 4.8 Perform security system installation maintenance as per IEEE regulations 4.9 Inspection and tests 4.10 Faults diagnosis 4.11 Faults rectification 4.12 Maintenance reports documentation	4 Portfolio of evidence 5 Written tests 6 Oral questioning

Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	B. Scaddan Electrical installation work J. Hyde Electrical installation Principles and Practices	5 pcs	1:5
2.	Installation manuals	Equipment manuals Control panel	5 pcs	1:5

		manuals		
3.	Charts	Single line diagram Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
6.	Lecture/theory room	60m ²	1	1:25
7.	Workshop	150m ²	1	1:25
8.	Site			
C	Consumable materials			
9.	Assorted Electrical cables		5 rolls	1:5
10.	Insulation tapes		25 pcs	1:1
11.	Accessories	Switches, sockets, Junction boxes, Consumer units, Patrice boxes, Circuit breakers	25 pcs	1:1
12.	Assorted CCTV components	Cameras, Digital video recorder, power supply unit, TFT monitors,	5 pcs	1:5
13.	Security system components	Alarm panels, magnetic sensors, vibration sensors, motion detectors	5 pcs	1:5
14.	Fire Alarm system components	Addressable control panel, fire and heat detectors, call points, buzzers	5 pcs	1:5
15.	Electric fence components	Energizer, Insulators, wires, support	5 pcs	1:5
16.	Pipes and trunkings	PVC conduits, Steel conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			

17.	Hacksaws		25 pcs	1:1
18.	Striping knives		25 pcs	1:1
19.	Side cutters		25 pcs	1:1
20.	Pliers		25 pcs	1:1
21.	Tape measure		25 pcs	1:1
22.	Try Square		25 pcs	1:1
23.	Spirit level		25 pcs	1:1
24.	Assorted Screw driver		25 pcs	1:1
25.	Assorted hammers		25 pcs	1:1
26.	Crimping tools		5 pcs	1:5
27.	PPEs		25 pcs	1:1
28.	Multimeters		5 pcs	1:5
29.	Clamp meters		5 pcs	1:5
30.	Earth resistance meter		5 pcs	1:5
31.	Stocks & Dies		5 pcs	1:5
32.	Vices		5 pcs	1:5
33.	Wire fasteners		5 pcs	1:5
34.	Oscilloscope		5 pcs	1:5
35.	Pipe bending Machine		5 pcs	1:5
36.	Bending spring		5 pcs	1:5
37.	Drilling machines		5 pcs	1:5
38.	Work stations		25	1:1
39.	Installation boards		13 pcs	1:2

MODULE V

ENTREPRENEURIAL SKILLS

ISCED UNIT CODE: 0413 541 03B

TVET CDACC UNIT CODE: ENG/CU/EI/BC/04/5/MA

Relationship to occupational standards

This unit addresses the unit of competency: Apply Entrepreneurial skills.

Duration of unit: 40 hours

Unit Description:

This unit covers competences required to apply entrepreneurship. Competences include applying financial literacy skills, applying entrepreneurial concepts, identifying entrepreneurship opportunities, applying business legal aspects, developing business innovative strategies, and developing business plans.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply Financial Literacy	10
2.	Apply the Entrepreneurial Concept	5
3.	Identify Entrepreneurship Opportunities	5
4.	Apply Business Legal Aspects	5
5.	Innovate Business Strategies	5
6.	Develop A Business Plan	10
TOTAL		40

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply financial literacy	1.1 Personal finance management 1.2 Balancing between needs and wants 1.3 Budget Preparation 1.4 Saving management 1.5 Factors to consider when deciding where to save 1.6 Debt management 1.7 Factors to consider before taking a loan 1.8 Investment decisions 1.9 Types of investments 1.10 Factors to consider when investing	1 Observation 2 Project 3 Written assessment 4 Oral assessment 5 Third party report 6 Interviews

Learning Outcome	Content	Suggested Assessment Methods
	money 1.11 Insurance services 1.12 insurance products available in the market 1.13 Insurable risks	
2. Apply entrepreneurial concept	2.1 Difference between Entrepreneurs and Business persons 2.2 Types of entrepreneurs 2.3 Ways of becoming an entrepreneur Characteristics of Entrepreneurs 2.4 salaried employment and self-employment 2.5 Requirements for entry into self-employment 2.6 Roles of an Entrepreneur in an enterprise 2.7 Contributions of Entrepreneurship	1 Observation 2 Project 3 Written assessment 4 Oral assessment 5 Third party report
3. Identify entrepreneurship opportunities	3.1 Sources of business ideas 3.2 Factors to consider when evaluating business opportunity 3.3 Business life cycle	1 Observation 2 Project 3 Written assessment 4 Oral assessment 5 Third party report
4. Apply business legal aspects	4.1 Forms of business ownership 4.2 Business registration and licensing processing 4.3 Types of contracts and agreements 4.4 Employment laws 4.5 Taxation laws	1 Observation 2 Project 3 Written assessment 4 Oral assessment 5 Third party report
5. Innovate business	5.1 Creativity in business 5.2 Innovative business strategies	1 Observation 2 Project

Learning Outcome	Content	Suggested Assessment Methods
strategies	5.3 Entrepreneurial Linkages 5.4 ICT in business growth and development	3 Written assessment 4 Oral assessment 5 Third party report
6. Develop business plan	6.1 Business description 6.2 Marketing plan 6.3 Organizational/Management plan 6.5 Production/operation plan 6.6 Financial plan 6.7 Executive summary 6.8 Business plan presentation 6.9 Business idea incubation	1 Observation 2 Written assessment 3 Project 4 Oral assessment 5 Third party report

Suggested Methods of Instruction

- Direct instruction with active learning strategies
- Project (Business plan)
- Case studies
- Field trips
- Group Discussions
- Demonstration
- Question and answer
- Problem solving
- Experiential
- Team training
- Guest speakers

Recommended Resources for 25 Trainees

- 5 Case studies
- 5 Business plan templates
- 10 Computers
- 1 Overhead projectors
- Internet
- Video clips
- 5 Newspapers and Handouts
- 5 Business Journals
- 25 sets of Writing materials

ENGINEERING MATHEMATICS II

ISCED UNIT CODE: 0541 541 06A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/07/5/MA

UNIT DURATION: 80 HOURS

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply engineering mathematics

Unit Description

This unit describes the competencies required to apply a wide range of engineering mathematics. Competencies include applying calculus, applying statistics and probability, applying matrices, applying vector theory and applying complex numbers.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply Calculus	15
2.	Apply Statistics and probability	20
3.	Apply matrices	15
4.	Apply Vector Theory	15
5.	Apply Complex Numbers	15
TOTAL		80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply Calculus	1.1 Introduction to calculus 1.2 Functional notation 1.3 The gradient of a curve 1.4 Differentiation from first principles	1 Written assessment 2 Practical assessment 3 Oral Questioning

	1.5 Differentiation of $y = ax^n$ by the general rule 1.6 Differentiation of sine and cosine functions 1.7 Differentiation of e^x and $\ln x$ 1.8 Methods of differentiation 1.8.1 Differentiation of common functions 1.8.2 Differentiation of a product 1.8.3 Differentiation of a quotient 1.8.4 Function of a function 1.8.5 Successive differentiation 1.9 Some applications of differentiation 1.9.1 Rates of change 1.9.2 Velocity and acceleration 1.9.3 Turning points 1.10 Practical problems involving 1.11 Tangents and normal 1.12 Small changes 1.13 Logarithmic differentiation 1.14 Introduction to logarithmic differentiation 1.14.1 Laws of logarithms 1.14.2 Differentiation of logarithmic functions 1.14.3 Differentiation of further	
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	logarithmic functions 1.14.4 Differentiation of $[f(x)]$ 1.15 Integral Calculus 1.16 Standard integration 1.17 The process of integration 1.18 The general solution of integrals of the form ax^n 1.19 Definite integrals 1.20 Integration using algebraic substitutions 1.21 Algebraic substitutions algebraic substitutions 1.22 Further worked problems on integration using algebraic substitutions 1.23 Change of limits 1.24 Integration using trigonometric substitutions 1.25 Integration using partial fractions 1.26 The $t = \tan \theta$ substitution 1.27 Integration by parts 1.28 Numerical integration 1.29 The trapezoidal rule 1.30 The mid-ordinate rule 1.31 Simpson's rule	
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2. Apply statistics and probability	2.1 Measures of central tendency mean, mode and median 2.2 Measures of dispersion 2.3 Variance and standard deviation 2.4 Definition of probability 2.5 Laws of probability 2.6 Expectation variance and SD 2.7 Calculations involving discrete and continuous random variables. 2.8 Types of distributions 2.8.1 Binomial 2.8.2 Poisson 2.8.3 Normal 2.9 Mean, variance and SD of probability distributions 2.10 Application of probability distributions	1. Written assessment 2. Oral assessment 3. Practical 4. Project 5. Third party report 6. Portfolio of evidence •
3. Apply matrices I	3.1 Matrix 3.2 Order of a matrix 3.3 Square matrix 3.4 Compatibility in addition and 3.5 Multiplication of matrices 3.6 Multiplication of a matrix by a scalar Matrix multiplication 3.7 Identity matrix 3.8 Determinant of a 2 x 2 matrix 3.9 Inverse of a 2 x 2 matrix and 3.10 Singular matrix	1 Written assessment 2 Practical assessment 3 Oral Questioning

	3.11 Solutions of simultaneous equations in two unknowns by 3.12 Matrix method 3.13 Crammer rule	
4. Apply vector theorem	4.1 Vectors and scalar in two and three dimensions 4.1.1 Operations on vectors: Addition and subtraction 4.2 Position vectors 4.3 Resolution of vectors 4.4 Scalar and vector product 4.5 Gradient, 4.6 Curl 4.7 Divergence	1 Practical 2 Project 3 Written tests 4 Oral questioning
5. Apply complex numbers	5.1 Argand diagrams 5.2 Complex numbers operations 5.3 De Moivre's theorem	1 Practical 2 Project 3 Written tests Oral questioning

Suggested Methods of Instruction

- Practical
- Demonstrations
- Group discussion
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			

1.	Textbooks	Engineering Mathematics by John bird 8 th edition	5 pcs	1:5
2.		Engineering Mathematics by A.K stround 8 th edition	5 pcs	1:5
3.		SMP	25	1:1
B	Learning Facilities & infrastructure			
4.	Lecture/theory room	50 m ²	1	1:25
C	Consumable materials			
5.	Charts	Manila papers		
6.	marker pens	Erasable		
D	Tools and Equipment			
7.	Calculators	Scientific	25 pcs	1:1

ANALOGUE ELECTRONICS II

ISCED UNIT CODE: 0714 541 16A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/06/5/MA

DURATION: 50 hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply analogue electronics.

UNIT DESCRIPTION

This unit describes the competencies required to apply analogue electronics. These competencies include; applying amplifiers, use of oscillators and application of opto-electronics.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply amplifiers	15
2.	Use oscillators	20
3.	Apply opto-electronics	15
TOTAL		50

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply amplifiers.	1.1 Classification of amplifiers based on; 1.1.1 Stages 1.1.2 Coupling method 1.1.3 Frequency 1.2 Types of amplifiers 1.2.1 RC coupled amplifiers 1.2.2 Power amplifiers 1.2.3 Tuned amplifiers 1.3 Feedback 1.3.1 Principle of feedback 1.3.2 Positive feedback 1.3.3 Negative feedback 1.4 Operational amplifiers (OPAMPs) 1.4.1 Construction of opamps	1 Practical test 2 Third Party Report 3 Portfolio of evidence 4 Written test 5 Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	1.4.2 Characteristics of the ideal and practical opamps 1.4.3 Opamp configurations 1.4.3.1 Inverting Amplifier 1.4.3.2 Non-Inverting Amplifier 1.4.3.3 Voltage Follower (Buffer) 1.4.3.4 Summing Amplifier 1.4.3.5 Differential Amplifier 1.4.3.6 Instrumentation Amplifier 1.4.3.7 Integrator 1.4.3.8 Differentiator 1.4.3.9 Comparator 1.4.3.10 Schmitt Trigger 1.5 Application of Amplifiers	
2 Use oscillators.	2.1 Sinusoidal oscillation 2.2 Types of sinusoidal oscillations 2.2.1 Damped oscillations 2.2.2 Undamped oscillations 2.3 Components of transistor oscillators 2.4 Essential conditions for oscillations 2.5 Types of oscillators Colpitts 2.5.1 Hartley 2.5.2 Phase shift oscillator 2.5.3 Crystal oscillator 2.6 Applications of oscillators 2.7 Wave shaping and multivibrator circuits 2.8 Types of multivibrators 2.8.1 Astable 2.8.2 Monostable 2.8.3 Bistable 2.9 Passive filters 2.9.1 High pass 2.9.2 Low pass 2.9.3 Band pass 2.10 Clippers and clampers 2.11 Applications of wave shaping and multivibrator circuits	1 Practical test 2 Third Party Report 3 Portfolio of evidence 4 Written test 5 Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
3 Apply opto-electronics	3.1 Opto-electronic devices 3.1.1 LEDs 3.1.2 OLED 3.1.3 LASER diode 3.1.4 Photo transistors 3.1.5 Photo diodes 3.1.6 Optocoupler 3.1.7 LASCR 3.2 Liquid crystal displays 3.2.1 Dynamic scattering LCDs 3.2.2 Field effect scattering LCDs 3.2.3 LASERs and MASERs 3.2.4 Applications of optoelectronics	1 Practical test 2 Third Party Report 3 Portfolio of evidence 4 Written test 5 Oral questioning

Suggested Methods of Instruction

- Practical
- Project
- Group discussions
- Demonstration
- Visit to manufacturing and processing industries
- On-job-training
- Charts and Audio-visual presentations

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference books	Mehta, V. K., & Mehta, R. (2020). Principles of electronics (12 edition). S. Chand and Company Limited, Theraja, B. L., & Theraja, A. K. (2005). A textbook of electrical technology (1st multicolour ed., Multicolour illustrative ed., 23rd rev. multicoloured	10 pcs for each book	1:2.5

		ed). S. Chand & Co. Bird, J. O. (2022). Bird's electrical and electronic principles and technology (Seventh edition). Routledge, Taylor & Francis Group.		
2.	Software	Assorted simulation software e.g., Circuit wizard.	25	1:1
3.	Audio visual presentations	Projector	1	1:25
B	Learning Facilities & infrastructure			
4.	Lecture/theory room	60m ²	1	1:25
5.	Workshop	150m ²	1	1:25
6.	Computer laboratory	100m ²	1	1:25
C	Consumable materials			
7.	Electronic components	Breadboards, Stripboards, Jumper wires, Assorted resistors, Assorted capacitors, Assorted MOSFETs, Assorted JFETs, 555 timers, Solder wire, LEDs, Assorted BJT transistors, LDRs, OPAMPs, thermistors, 12V DC motors	25 pcs	1:1
D	Tools and Equipment			
8.	Assorted tools and equipment	Side cutters, Side cutters, Pliers, Screw driver, Crimping tools, Mult-meter, Solder guns	25 pcs	1:1
9.	Assorted electrical gadgets	Solder gun, Heat sink, Hot air guns, function generator	25 pcs	1:1
10.	Assorted measuring instruments	Digital Oscilloscope,	5 for each category	1:5

11.	Digital Multimeter,			
12.	Digital functional generator		3 pcs	1:8
13.	Laser jet printer		2 pcs	1:13
14.	Power supply	Variable power supply, 5V Power adapters, 9V Power adapters, 12V Power adapters.	10 pcs	1:3
15.	Trainers kit	Analogue training kits, PWM kit	5 pcs	1:5
16.	PCB prototyping material	Copper board, ferrite chloride solution, see-through printing paper, HASL finishing PCB	25 for each category	1:1
17.	Drilling gun		3 pcs	1:8
18.	Work stations		25	1:1

DIGITAL ELECTRONICS II

ISCED UNIT CODE: 0714 541 18A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/08/5/MA

UNIT DURATION: 50 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Digital Electronics

Unit Description

This unit describes competences required to apply digital electronics. Competences include applying knowledge of digital logic circuits, applying knowledge of converters (ADC and DAC) and managing computer memories.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply knowledge of digital logic circuits	15
2.	Apply knowledge of advance digital logic and converter circuits	20
3.	Manage computer memories	15
TOTAL		50

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply knowledge of digital logic circuits	1.1 Combinational logic circuits design and minimization 1.2 Logic families. 1.2.1 Bipolar Families 1.2.2 MOS Families 1.2.3 Hybrid Family 1.3 Sequential logic circuits 1.4 Flip flops	1 Observation 2 Written test 3 Practical 4 Demonstration 5 Oral questioning 6 Third party report

2. Apply knowledge of advance digital logic and converter circuits	2.1 Counters 2.2 Data handling devices 2.2.1 Decoders 2.2.2 Encoders 2.2.3 Multiplexers 2.2.4 Demultiplexers 2.2.5 Shift registers 2.3 Arithmetic circuits 2.4 Digital converters (ADC)(DAC)	1 Observation 2 Written test 3 Practical 4 Demonstration 5 Oral questioning 6 Third party report
3. Manage computer memories	3.1 Introduction to Computer Memory systems 3.2 Types of memory devices 3.2.1 RAMs 3.2.2 ROMs 3.2.3 EEPROM 3.2.4 EPROM 3.3 Memory organization. 3.4 Memory expansion	1 Observation 2 Written test 3 Practical 4 Demonstration 5 Oral questioning 6 Third party report

Suggested Methods of Instruction

- Role playing
- Viewing of related videos
- Discussion
- Direct Instruction

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference books	Digital Electronics: Principles, Devices and Applications By Anil K. Maini	5 pcs	1:5
2.	Installation manuals	Assorted Systems component Manufacturer's manuals and data sheets Instrumentation Handbooks	5 pcs	1:5
3.	Charts	Assorted Circuit diagrams	1 pcs for	1:25

		charts	each	
4.	Software	Assorted simulation software e.g Deeds,	25	1:1
5.	Audio visual presentations	Projector	1	1:25
B	Learning Facilities & infrastructure			
6.	Lecture/theory room	60m ²	1	1:25
7.	Workshop	150m ²	1	1:25
8.	Computer laboratory	100m ²	1	1:25
C	Consumable materials			
9.	Assorted electronics components	ICs, resistors, capacitors	25 pcs	1:1
D	Tools and Equipment			
10.	Assorted tools and equipment	Side cutters, Side cutters, Pliers, Screw driver, Multi-meter, Oscilloscope, Solder guns, breadboards	25 pcs	1:1
11.	PPEs	Safety boots, overall	25 pcs	1:1
12.	Function generator		5 pcs	1:5
13.	Variable power supply		5 pcs	1:5
14.	Trainers kit	Assorted logic gate, combinational circuits trainer kits with component Manufacturer's manuals and data sheets	5 pcs	1:5
15.	Hot air gun		5 pcs	1:5
16.	Work stations		25	1:1

ELECTRICAL MACHINES INSTALLATION

ISCED UNIT CODE: 0713 551 12A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/03/5/MA

UNIT DURATION: 130 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Perform Electrical Machine Installation

UNIT DESCRIPTION

This unit covers competences required in performing electrical machine installation. Competences include conducting site survey, installing electrical machine, testing electrical machine installation, maintaining electrical machine installation, and commissioning electrical machine installations.

Summary of Learning Outcomes

By the end of this unit of learning the trainee will be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Conduct site survey	10
2.	Install electrical machines	50
3.	Test electrical machine installation	30
4.	Maintain electrical machine installation	30
5.	Commission electrical machine installation	20
TOTAL		130

Learning Outcomes, Content and Suggested Assessment Methods:

Learning Outcome	Content	Suggested Assessment Methods
1. Conduct site survey	1.1 Relevant clauses in appropriate Acts e.g. 1.1.1 Occupational Safety and Health Act (OSHA) 1.1.2 Work Injury Benefits Act (WIBA) 1.1.3 Environment Management and Coordination Act (EMCA) 1.1.4 Relevant regulations and standards: 1.1.5 IEEE regulations	1 Observation 2 Portfolio of Evidence 3 Project 4 Written assessment 5 Practical assessment 1 Oral assessment

	1.1.6 BS 7671 regulations 1.1.7 KEBS standards 1.1.8 KPLC/EPRA by-laws 1.1.9 County by-laws 1.2 Common hazards and sources of danger and their preventive measures e.g. burns, cuts, electric shock, electric fire, falling from heights, falling objects, noise, dust, chemicals etc 1.3 Safe and correct handling, use, maintenance and storage of different types of PPE 1.4 Security issues e.g. theft, vandalism 1.5 Utilities available 1.5.1 Water 1.5.2 Electricity 1.5.3 Communication e.g. Phones 1.6 Preparation of work plan 1.6.1 Preparation of machine layout drawing based on manufactures manual 1.6.2 Identification of scope of installation work 1.6.3 Identify installation team 1.7 Cable management 1.7.1 Cable trays 1.7.2 Cable ducts 1.7.3 Bus bars 1.7.4 Trunking BOQ preparation 1.8 BOQ 1.9 Survey report May include; client/details, technical lead details, sketches, collected data, available utilities, site photos,	
2. Install electrical machine	<u>Theory content</u> 2.1 Safety procedures 2.2 Preparation of wiring diagrams 2.3 Materials, tools and equipment's 2.4 Machine support 2.5 Types of machine layout e.g. 2.5.1 Fixed position layout 2.5.2 Product layout 2.5.3 Process layout 2.5.4 Combinational 2.5.5 Machine mounting	1 Observation 2 Portfolio of Evidence 3 Project 4 Written assessment 5 Practical assessment 6 Oral assessment

	2.6 Types of machines 2.6.1 DC machine 2.6.2 AC machine 2.7 Cables and conductors 2.7.1 Cabling in machine 2.7.2 Current capacity 2.7.3 Type of the cable 2.7.4 Cable termination 2.7.5 Cables labelling 2.8 Installation route 2.8.1 Surface 2.8.2 Underground 2.8.3 Concealed 2.9 Mounting structures/housing 2.9.1 Structural strength 2.9.2 Size 2.10 Control gear 2.10.1 DOL 2.10.2 Star delta 2.10.3 Forward reverse 2.10.4 Disconnect switches 2.10.5 Circuit breaker 2.10.6 Isolator 2.11 Installation conditions e.g. temperature, humidity, moisture, dust 2.12 Conduits, trunks, and support systems 2.13 Machine enclosures 2.14 Housekeeping practices. 2.14.1 Disposal of waste 2.14.2 Cleaning 2.14.3 Tools storage 2.14.4 recycling and reusing 2.15 IEEE regulations <u>Practical content</u> 2.16 Construct and mount electrical machine support 2.17 Install electrical machine control gear 2.18 Install electrical machine 2.19 Terminations 2.20 Housekeeping	
3. Test electrical machine	<u>Theory content</u> 3.1 Visual inspection	1 Practical 2 Project

installation	3.2 Machine installation firmness 3.3 Type of tests 3.3.1 Earth continuity tests 3.3.2 Insulation resistance tests 3.3.3 Earth electrode resistance test 3.3.4 Short circuit 3.3.5 Open circuit test 3.3.6 Continuity test 3.3.7 On load and offload tests 3.4 Testing instruments e.g. 3.4.1 Multimeter 3.4.2 Insulation resistance tester 3.4.3 Earth electrode resistance tester 3.4.4 Ohmmeter 3.5 Importance of installation testing 3.6 Test results documentation <u>Practical content</u> 3.7 Perform tests 3.7.1 Earth continuity tests 3.7.2 Insulation resistance tests 3.7.3 Earth electrode resistance test 3.7.4 Short circuit 3.7.5 Open circuit test 3.7.6 Continuity test 3.7.7 On load and offload tests	3 Third party report 4 Portfolio of evidence 5 Written tests 6 Oral questioning
4. Maintain Electrical Machine Installation	4.1 Safety procedures 4.2 Meaning of terms 4.3 Maintenance 4.3.1 Types and procedures 4.3.2 Periodic service 4.3.3 Preventive 4.3.4 Breakdown 4.3.5 Corrective 4.3.6 Scheduling maintenance based on service manuals 4.3.7 Identification and documentation of maintenance tools, materials and equipment 4.3.8 Specifications of identified	1 Practical 2 Projects 3 Third party report 4 Portfolio of evidence 5 Written tests 6 Oral questioning

	tools, materials and equipment against safety standards 4.4 Fault diagnosis/troubleshooting 4.5 System isolation points 4.6 System tests e.g. continuity test, functionality test, polarity test etc. 4.7 Maintenance report and records 4.7.1 Fault diagnosis/troubleshooting 4.7.2 System Tests e.g. continuity test, functionality test, polarity test etc. 4.8 Maintenance report and records	
5. Commission electrical machine installation	5.1 Commissioning 5.1.1 End user training 5.1.2 Handing over operating manuals 5.1.3 Report preparation Issuance of certificates	1 Practical 2 Projects 3 Third party report 4 Portfolio of evidence 5 Written tests 6 Oral questioning

Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	V.K Mehta Principles of Electrical Machines	5 pcs	1:5

		B. Scaddan Electrical installation work J. Hyde Electrical installation Principles and Practices		
2.	Installation manuals	Electrical machine manuals	5 pcs	1:5
3.	Charts	Single line diagram Motor starting circuits Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
6.	Lecture/theory room	60m ²	1	1:25
7.	Workshop	150m ²	1	1:25
8.	Site			
C	Consumable materials			
9.	Electrical wires	1.5mm ² (red, black green)	5 rolls	1:5
		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
10.	Insulation tapes		25 pcs	1:1
11.	Accessories	Push buttons, relays, Timers, contactors, Thermo overloads, DIN	25 pcs	1:1

		rail, Circuit breakers, TPN		
12.	Pipes and trunkings	PVC conduits, Steel conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			
13.	3-Phase motors		5 pcs	1:5
14.	Synchronous Motors		5 pcs	1:5
15.	Single Phase motors		5 pcs	1:5
16.	Electric Generator		5 pcs	1:5
17.	DC motors		5 pcs	1:5
18.	Hacksaws		25 pcs	1:1
19.	Striping knives		25 pcs	1:1
20.	Side cutters		25 pcs	1:1
21.	Pliers		25 pcs	1:1
22.	Tape measure		25 pcs	1:1
23.	Try Square		25 pcs	1:1
24.	Spirit level		25 pcs	1:1
25.	Assorted Screw driver		25 pcs	1:1
26.	Assorted hammers		25 pcs	1:1
27.	Crimping tools		5 pcs	1:5
28.	PPEs		25 pcs	1:1
29.	Multimeters		5 pcs	1:5
30.	Clamp meters		5 pcs	1:5
31.	Earth resistance meter		5 pcs	1:5
32.	Stocks & Dies		5 pcs	1:5
33.	Vices		5 pcs	1:5
34.	Oscilloscope		5 pcs	1:5
35.	Pipe bending Machine		5 pcs	1:5
36.	Bending spring		5 pcs	1:5
37.	Drilling machines		5 pcs	1:5
38.	Work stations		25	1:1
39.	Installation boards		13 pcs	1:2

SOLAR PV SYSTEMS INSTALLATION

ISCED UNIT CODE: 0713 451 32A

TVET CDACC CODE: ENG/CU/EI/CR/04/5/MA

UNIT DURATION: 80 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Install PV Solar System

Unit Description

This unit covers competences required to install solar PV system. Competences include conducting solar PV site survey, sizing solar PV system installation, installing solar PV components, testing solar PV system installation, commissioning solar PV system and maintaining solar PV system

Summary of Learning Outcomes

S/NO	Learning Outcome	Duration (Hours)
1.	Conduct Solar PV site survey	5
2.	Size solar PV system installation	20
3.	Install Solar PV Components	40
4.	Test solar PV system installation	5
5.	Commissioning solar PV system	5
6.	Maintain Solar PV system	5
TOTAL		80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Conduct Solar PV site survey	<u>Theory Content</u> 1.1 Relevant clauses in appropriate Acts e.g. 1.1.1 Occupational safety and health act (OSHA) 1.1.2 Work injury benefits act (WIBA) 1.1.3 Environment management and coordination Act (EMCA) 1.2 Relevant regulations:	• Practical • Project • Third party report • Portfolio of evidence • Written tests

	1.2.1 IEEE regulations 1.2.2 KPLC by-laws 1.2.3 County by-laws 1.3 End user needs 1.4 Site conditions assessment 1.4.1 Shading analysis 1.4.2 Irradiance 1.4.3 Orientation 1.4.4 Sun hours / insolation 1.5 Type of installation 1.5.1 Solar Stand-alone systems 1.5.2 On-grid 1.5.3 Hybrid systems 1.6 Installation mapping 1.7 Site survey report <u>Practical Content</u> 1.8 Safety procedures 1.9 Conduct site survey 1.10 Site survey report preparation	• Oral questioning
2. Size solar PV system	<u>Theory Content</u> 2.1 Factors to consider in solar PV system sizing 2.2 Load requirements 2.3 Size and rating of solar PV system 2.2.1 Panel Ratings 2.2.2 Charge controller 2.2.3 Batteries 2.2.4 Inverter 2.2.5 Cables (Voltage drop) 2.2.6 Fuses and circuit breakers 2.3 Mechanical structures: 2.3.1 Types of mounting structures e.g. metallic, wooden, plastic, etc. 2.4 BOQ preparation 2.5 Verification of Solar PV materials <u>Practical content (Case Study)</u> 2.6 Sizing of the Solar PV system 2.7 BOQ preparation	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
3. Install Solar PV Components	<u>Theory Content</u> 3.1 Components of solar system 3.1.1 Solar Panels: 3.1.1.1 Types of solar panel i.e. Amorphous/thin film, Monocrystalline, Polycrystalline	• Practical • Project • Third party report • Portfolio of evidence • Written tests

	3.1.1.2 Methods of solar panel connection i.e. parallel, series, etc. 3.1.1.3 Solar Panel Mounting structures and positioning; - Slanting angle 3.1.2 Types and working principle of; - 3.1.2.1 Charger controller 3.1.2.2 Inverters 3.1.2.3 Solar batteries 3.1.3 Cables 3.2 Integration of solar PV system with other sources of power (Change over switch) 3.3 Care and Maintenance of solar PV System components 3.4 As-built drawings preparation <u>Practical content</u> 3.5 As-built drawings preparation 3.6 Mechanical structures mounting 3.7 Install solar PV components 3.8 Wiring system installation 3.9 Accessories installation 3.10 Housekeeping practices	• Oral questioning
4. Test solar PV system installation	<u>Theory content</u> .1 Visual inspection .2 Types of tests in solar PV system .3 Continuity test .4 Insulation resistance test .5 Polarity test .6 Earth resistance tests .7 Earth loop impedance .8 Solar PV system commissioning <u>Practical Content</u> .9 Perform solar installation tests as per IEEE regulations .10 Solar PV system commissioning	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
5. Commissioning solar PV system	6.1 Importance of solar system commissioning 6.2 Commissioning 6.2.1 Commissioning team/panel 6.2.2 Commissioning procedure 6.2.3 Safety procedures during commissioning 6.2.4 Functionality test e.g. Short circuit test, Open circuit voltage, current at maximum power points, output power, etc. 6.2.5 Performance test	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

	6.2.6 Preparation documents 6.2.7 Commissioning reports 6.2.8 End-user training 6.2.9 Project Handover	
7. Maintain Solar PV system	<u>Theory Content</u> 7.1 Definition of maintenance terms 7.2 Maintenance schedule preparation 7.3 Preparation of maintenance checklist 7.4 Solar PV system testing 7.5 Maintenance tools, equipment and materials preparation 7.6 Maintenance procedures/activities 7.7 Maintenance records documentation <u>Practical Content</u> 7.8 Perform solar installation maintenance as per IEEE regulations 7.9 Maintenance records documentation	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning

Suggested Methods of Delivery

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	Solar Electric Handbook by Solar energy International B. Scaddan Electrical installation work J. Hyde Electrical installation Principles and Practices	5 pcs	1:5

2.	Installation manuals	Solar PV manuals	5 pcs	1:5
3.	Charts	Single line diagram Solar PV layout Circuit diagrams Colour codes	1 pc for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
5.	Lecture/theory room	60m ²	1	1:25
6.	Workshop	160m ²	1	1:25
7.	Site			
C	Consumable materials			
8.	Electrical wires	1.5mm ² (red, black green)	5 rolls	1:5
		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
9.	Insulation tapes		25 pcs	1:1
10.	Accessories	Switches, sockets, Junction boxes, Consumer units, Lamp holders, Patrice boxes, Circuit breakers	25 pcs	1:1
11.	Solar Panels		10	1:3
12.	Charge controller		10	1:3
13.	Batteries		10	1:3
14.	Inverters		10	1:3
15.	Solar connectors	Male and female	20	1:2
16.	Crodile clips		20	1:2

17.	Mounting racks		10	1:3
18.	Pipes and trunkings	PVC conduits, Steel conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			
19.	Hacksaws		25 pcs	1:1
20.	Striping knives		25 pcs	1:1
21.	Side cutters		25 pcs	1:1
22.	Pliers		25 pcs	1:1
23.	Tape measure		25 pcs	1:1
24.	Try Square		25 pcs	1:1
25.	Spirit level		25 pcs	1:1
26.	Assorted Screw driver		25 pcs	1:1
27.	Assorted hammers		25 pcs	1:1
28.	Crimping tools		5 pcs	1:5
29.	PPEs		25 pcs	1:1
30.	Multimeters		5 pcs	1:5
31.	Clamp meters		5 pcs	1:5
32.	Earth resistance meter		5 pcs	1:5
33.	Stocks & Dies		5 pcs	1:5
34.	Vices		5 pcs	1:5
35.	Oscilloscope		5 pcs	1:5
36.	Pipe bending Machine		5 pcs	1:5
37.	Bending spring		5 pcs	1:5
38.	Drilling machines		5 pcs	1:5
39.	Work stations		25	1:1
40.	Installation boards		13 pcs	1:2
41.	Hydrometer		5 pcs	1:5
42.	Solar irradiance meter		1 pc	1:25
43.	Grid tie Synchroniser		1	1:25
44.	Battery capacity tester		1	1:25

MODULE VI

ENGINEERING MATHEMATICS III

ISCED UNIT CODE: 0541 541 07A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/01/6/MA

Unit duration: 100 hours.

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Engineering mathematics

Unit Description: This unit describes the competencies required by a technician in order to apply technician mathematics. It enables the learner to; apply Power Series, solve Partial differential equations, ordinary differential Equations and apply Laplace transforms.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hour)
1.	Apply Power Series	20
2.	Solve Partial differential equations	30
3.	Ordinary differential Equations	30
4.	Apply Laplace transforms	20
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Solve Partial differential equations	1.1 Definition of partial derivatives 1.1.1 Derivation of a function of two or more variables 1.1.2 Application of partial derivatives 1.1.3 Stationary points of functions of two variables.	• Written assessment • Oral assessment • Practical • Project • Third party

		report Portfolio of evidence
2. Ordinary differential Equations	2.1 Formation and solution of 1st and 2 nd order differential equations 2.1.1 1 st order variable separable 2.1.2 1 st order homogeneous 2.1.3 1 st order linear 2.2 2 nd order by the determination of coefficients. 2.3 D-operator method 2.3.1 Simultaneous differential Equations 2.3.2 Series Method 2.4 Applications of differential equations	- Written assessment - Oral assessment - Practical - Project - Third party report - Portfolio of evidence
3. Apply Laplace transforms	3.1 Definition of Laplace transform of a function 3.1.1 Derivation of Laplace transforms of simple functions 3.1.2 Laplace transforms of elementary functions 3.1.3 Inference of linearity properties 3.1.4 Use of list of standard transforms 3.2 Properties of Laplace transforms 3.2.1 Derivation of Laplace transforms of $F(t) = e^{at}$ 3.2.2 Statement of initial and final value theorems 3.2.3 Derivation of the Laplace transforms of the first and second derivatives from definition 3.3 Determination of the inverse Laplace transform of simple transforms 3.3.1 Definition of inverse Laplace transform 3.3.2 Determination of inverse	- Written assessment - Oral assessment - Practical - Project - Third party report - Portfolio of evidence

	Laplace transforms of simple forms 3.3.3 Determination of partial fractions for expressions up to degree three in the denominator 3.4 Determination of inverse Laplace transforms of partial fractions and a list of standards 3.5 Differential equations and simultaneous equations with constant coefficients and given initial conditions 3.6 Application of theorem for the Laplace transforms 3.7 Evaluations of Laplace transform using a small stock of transform key pair 3.8 Application of Laplace transforms to impulse response problems	
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Suggested Delivery Methods

- Demonstration
- Discussions
- Practical
- Project work
- Direct instruction

List of Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			

1.	Textbooks	Engineering Mathematics by K.A. Stroud	5 pcs	1:5
		Advanced Engineering Mathematics by Erwin Kreyszig	5 pcs	1:5
B	Learning Facilities & infrastructure			
2.	Lecture/theory room	60m ²	1	1:25
3.	Computer	Operating System: 64-bit Windows 11 or 10 version 1809 or above Processor: 2.5 GHz (3+ GHz recommended), Memory: 8 GB (32GB recommended) Disk space: 10 GB Display: 1920 x 1080 resolution Display Card: 2 GB GPU (8 GB recommended) and DirectX 11 compliant (DirectX 12 recommended)	25 pcs	1:1
4.	Projector		1	1:25
5.	Interactive screen	Specifications: 77-inch interactive whiteboard with touch and pen functionality.	1	1:25
C	Software			
6.	MATLAB	License: Educational	Installed	1:1

		licenses available. Features: Matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, interfacing with programs in other languages.	in 25 computers	
7.	GeoGebra	License: Free educational software. Interactive geometry, algebra, statistics, and calculus applications	Installed in 25 computers	1:1
D	Consumables			
8.	Pens, pencils, rulers and paper	Whiteboard markers, 2H pencils, plastic rulers, A2 white papers	Enough	

ELECTRICAL PRINCIPLES I

ISCED UNIT CODE: 0713 541 10A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/02/6/MA

UNIT DURATION: 80 HOURS

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply electrical principles

UNIT DESCRIPTION

This unit describes competences required to apply electrical principles in their work. It involves apply basic electrical machines, apply three phase power supply, apply transients in dc circuits.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hour)
1.	Apply electrical machines	20
2.	Apply three phase power supply	30
3.	Apply transients in DC Circuits	30
TOTAL		80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply electrical machines	1.1 Electrical machines identification 1.2 electrical machines operations 1.2.1 DC motors 1.2.2 DC generators 1.2.3 AC single phase motors	• Practical • Project • Third party report • Portfolio of evidence

	1.2.4 AC three phase machines: Induction, Synchronous 1.2.5 Transformer 1.3 Derivation of generator e.m.f equation 1.4 Electrical machine control 1.5 Calculations involving electrical machines 1.6 Applications of electrical machines	• Written tests • Oral questioning
2. Apply three phase power supply	2.1 Principles of three phase power generation 2.2 Connections of three phase power supply 2.2.1 Star and delta connection 2.3 Calculations involving three phase power supply connections 2.4 Measurements of three phase power supply 2.4.1 One wattmeter method 2.4.2 Two wattmeter method 2.4.3 Three wattmeter method	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
3. Apply transients in Electrical DC Circuits	3.1 Derivation of growth and decay equations in R-L and R-C circuits. 3.1.1 Sketching of Growth and decay curves in R-L and R-C circuits 3.1.2 Calculations involving Growth and decay in R-L and R-C circuits based on the time constants. 3.2 Application of the effect of	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning

	time constant in switching inductive and capacitive loads 3.3 Analysis of Passive and active filters	
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Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
6.	Textbooks	J. Bird Electrical and Electronic Principles V.K. Mehta & R. Mehta Basic Electrical Engineering	5 pcs	1:5
7.	Installation manuals	Electronic components datasheets	5 pcs	1:5

8.	Charts	Circuit diagrams Colour codes	1 pcs for each	1:25
9.	Scientific Calculators		25	1:1
10.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
1.	Workshop	150m ²	1	1:25
C	Consumable materials			
1.	Connector wires	Jumper wires,	5 pkts	1:5
2.	Insulation tapes		25 pcs	1:1
2.	Circuit boards	Bread board, copper strip boards	25 pcs	1:1
3.	Assorted electronic components	Resistors, diodes, capacitors, transistors, ICs, Transformers, Inductors, Batteries	25 pcs	1:1
4.	Soldering wires		5 rolls	1:5
D	Tools and Equipment			
1.	Striping knives		25 pcs	1:1
2.	Side cutters		25 pcs	1:1

3.	Pliers		25 pcs	1:1
4.	Assorted Screw driver		25 pcs	1:1
5.	Crimping tools		5 pcs	1:5
6.	PPEs		25 pcs	1:1
7.	Multimeters		5 pcs	1:5
8.	Oscilloscope		5 pcs	1:5
9.	Function generator		5 pcs	1:5
10.	Spectrum analyser		5 pcs	1:5
11.	Variable power supply		5 pcs	1:5
12.	Solder guns		25 pcs	1:1
13.	Hot air gun		5 pcs	1:5
14.	Work stations		25	1:1

MICRO-CONTROL SYSTEMS

ISCED UNIT CODE: 0714 541 19A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/03/6/MA

Duration: 80 Hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply micro-control systems.

UNIT DESCRIPTION

This unit covers competences required to apply micro-control systems. Competences include establishing complexity of the task, selecting micro control device, selecting micro control programming software, performing controller programming, interfacing micro control system, performing controller system test and inspection

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hour)
1.	Establish complexity of the task	10
2.	Select micro control device	10
3.	Select micro control programming software	20
4.	Perform controller programming	20
5.	Interface micro control system	10
6.	Perform controller system test and inspection	10
TOTAL		80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Establish complexity of	1.1 Introduction to process control systems	Practical Test

the task	1.2 Term and definitions 1.3 Types of process control systems 1.4 Process Control Components 1.5 Sensors and transducers 1.5.1 Inductive sensors 1.5.2 Capacitive sensors 1.5.3 Resistive sensors 1.6 Actuators 1.7 Electric motors 1.8 Hydraulic and Pneumatic actuators 1.9 Electromechanical actuators 1.10 Indicators 1.11 Signal processing 1.12 Signal conditioning Display 1.13 Need for process control system 1.14 Introduction to microcontroller technology 1.14.1 Term and definitions 1.14.2 Block diagram of microcontroller system 1.14.3 Microcontroller Architecture 1.14.4 Memory Organization 1.14.5 Peripherals 1.14.6 System clock 1.14.7 Communication channels 1.14.7.1 Serial communication 1.14.7.2 UART	Project Third party report Portfolio of evidence Written tests Oral Questioning
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	1.14.7.3 SPI	
	1.14.7.4 I2C	
	1.14.7.5 Ethernet	
	1.14.7.6 CAN	
	1.14.7.7 USB	
	1.14.7.8 Parallel Communication	
	1.14.7.9 Parallel Data Bus	
	1.14.7.10 PIA	
	1.14.7.11 PPI	
	1.14.7.12 PCI	
	1.14.7.13 Wireless communication	
	1.14.7.14 Bluetooth/BLE	
	1.14.7.15 GPRS	
	1.14.7.16 WiFi	
	1.14.7.17 PIO	
	1.14.7.18 Interrupt Controller	
	1.14.7.19 Timers and Counters	
	1.14.7.20 ADC/DAC	
	1.14.7.21 PWM	
	1.14.7.22 Watch Dog Time	
	1.14.7.23 Real-time clock	
	1.14.7.24 Capture/Compare/PWM (CCP) Module	
	1.14.7.25 Touch Sensing	
	1.14.7.26 LCD controller	

2. Select micro control device	2.1 Factor to consider when selecting a microcontroller 2.2 Power rating 2.3 Communication Protocols. 2.4 Processing Speed 2.5 Interfacing 2.6 Scalability 2.7 Portability 2.8 Time to market	Practical Test Project Third party report Portfolio of evidence Written tests Oral Questioning
3. Select micro control programming software	3.1 Operating System Requirements: 3.2 Install microcontroller-related software (e.g., Arduino IDE, VSCode) 3.3 User Interface and Communication Systems 3.4 Select communication modules (e.g., Ethernet for industrial automation, Bluetooth for home systems). 3.5 Licensing and support of the software. 3.6 Software Configuration	Practical Test Project Third party report Portfolio of evidence Written tests Oral Questioning
4. Perform controller programming	4.1 Introduction to hardware programming 4.2 Terms and definitions 4.3 Programming concepts 4.4 Program design methods 4.5 Top-Down Design 4.6 Bottom-Up Design 4.7 Modular Design 4.8 Structured Programming 4.9 Object-Oriented Design (OOD)	Practical Test Project Third party report Portfolio of evidence Written tests Oral Questioning

	4.10 Event-Driven Design 4.11 Data-Driven Design 4.12 Functional Design 4.13 Component-Based Design 4.14 Types of hardware programming languages 4.14.1 Object-Oriented Languages (OOP) 4.14.2 Functional Programming Languages 4.14.3 Scripting Languages 4.14.4 Hardware programming methods 4.14.5 Bare-Metal Programming 4.14.6 Register-Level Programming 4.14.7 Interrupt-Driven Programming 4.14.8 Real-Time Programming 4.14.9 In-Circuit Programming (ICP) 4.14.10 In-System Programming (ISP) 4.15 Event-Driven Programming 4.16 Hardware Abstraction Layer (HAL) Programming 4.16.1 Low-Power and Sleep Mode Programming 4.16.2 Program Development: 4.16.3 Memory addressing (Variables, Data pointers and registers) 4.16.4 Functions and Methods 4.16.5 Classes and structures 4.16.6 Loop structures, conditionals, and branching 4.16.7 Compilation and Debugging	
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	4.16.8 Pre-processing 4.16.9 Compilation 4.16.10 Assembly 4.16.11 Linking 4.16.12 Binary file 4.16.13 Hex file 4.16.14 Program Documentation 4.16.15 Document the program thoroughly with comments explaining the functionality of each section.	
5. Interface micro control system	5.1 Types of I/O 5.2 Analogue vs digital 5.3 Active vs passive 5.4 Sourcing vs sinking 5.5 Types of interfacing ports 5.6 Pin Headers (Male or Female) 5.7 Jumper Wires 5.8 IDC Connectors 5.9 DB9 Connector 5.10 FTDI USB to Serial Cable 5.11 FPC/FFC Connectors 5.12 JST Connectors 5.13 Screw Terminal Connectors 5.14 USB Type A/B/C, Micro-USB, Mini-USB 5.15 JTAG Port 5.16 10-Pin or 20-Pin IDC Connector 5.17 RJ45 Connector	Practical Test Project Third party report Portfolio of evidence Written tests Oral Questioning

	5.18 RS-232 Port	
	5.19 Wi-Fi/Bluetooth Module Ports	
6. Perform controller system test and inspection	6.1 Importance of testing 6.2 Meaning of terms 6.2.1 Inspection 6.2.2 Testing 6.3 Types of tests e.g. 6.3.1 Electrical tests 6.3.2 Functionality tests 6.4 Test Signal Conditioning Elements 6.5 Test Signal Processing elements 6.6 Test Data presentation elements 6.7 Testing tools and equipment 6.7.1 Multimeter 6.7.2 Oscilloscope 6.7.3 Signal generators 6.7.4 Troubleshooting techniques 6.7.5 Performance characteristics 6.7.6 Statics and dynamics 6.8 Error and loading effects 6.8.1 Accuracy 6.8.2 Precision 6.8.3 Linearity 6.8.4 Sensitivity 6.8.5 Resolution 6.8.6 Response time	Practical Test Project Third party report Portfolio of evidence Written tests Oral Questioning

	6.8.7 Drift 6.8.8 Calibration 6.8.9 Functional integration 6.8.10 Need for end user training 6.8.11 Commissioning documents 6.8.12 Safety documentation 6.8.13 Handover report	
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Suggested Methods of Instruction

Practical

Project

Group discussions

Demonstration

Visit to manufacturing and processing industries

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	Susnea, I., & Mitescu, M. (2005). Microcontrollers in Practice. Springer. https://buku.app/book/144497	Online	1:1
2.	Installation manuals and datasheets	ATMEL, Cortex-M3, Intel 8051, Arduino Nano, Arduino Mega, ESP32, Raspberry pi A4	5 pcs for each	1:5
B	Learning Facilities & Infrastructure			

3.	Lecture/Theory room	60m2	1	1:25
4.	Workshop	150m2	1	1:25
5.	Projector		1 pc	1:25
6.	Interactive display screen	75 inches	1 pc	1:25
7.	Computers with Internet access	Processor: Intel Core i5/i7 or equivalent RAM: 8GB or higher Storage: 256GB SSD or higher	25 pcs	1:25
9.	Microcontroller	ATMEL328P/CORTEX-M3/INTEL-8051	25 pcs for each category	1:1 for each category
9.	LCD panels	1.3" Inch I2C IIC OLED LCD Module 4pin- Blue, 16x2 LCD Display Module(Blue) 5V	5 pcs/5 pcs/5 pcs	1:5 for each category
10.	Limit switches	Mechanical, optical, inductive, capacitive	25 pcs each category	1:1 for each category
11.	Industrial pressure sensors	0-10 bar, 4-20mA output	25 pcs	1:1
12.	Proximity sensors	NPN/PNP, 10-30V DC	25 pcs	1:1
13.	Industrial temperature sensors	RTD/PT100, -50 to 200°C	25 pcs	1:1
14.	Flow sensors	4-20mA, 0-10V, RS485	25 pcs	1:1
15.	Level sensor	Ultrasonic 4-20mA, 0-10V	25 pcs	1:1

16.	Power supply units	Output Voltage: 24V DC Output Current: 5A or higher Input Voltage: 220V AC	10 pcs	1:2.5
17.	Adjustable wrenches	Metric and standard sizes	10 pcs	1:2.5
18.	Sets of combinational wrenches	Metric and standard sizes	5 sets	1:5
19.	Assorted size socket wrenches	Metric and standard sizes	5 sets	1:5
20.	Assorted sizes of screw drivers	Insulated, flathead	5 sets	1:5
21.	Assorted sizes of Allen keys	Metric and standard sizes	5 sets	1:5
22.	Wire Strippers	Suitable for 0.5 mm ² to 4 mm ² wires	10 pcs	1:2
26.	Cordless drills	16 V	5 pcs	1:5
27.	Relays	24V DC coil, 10A contacts	25 pcs	1:1
28.	Solenoid valves	24V DC, 1/4" or 1/2" connections	25 pcs	1:1
29.	DC motors	24V DC	25 pcs	1:1
30.	DC motor drives	12V/24V	25 pcs	1:1
31.	Push buttons	Voltage: 24V DC coil/230V AC coil Contact Configuration: 1 NO or 1 NC Illumination: Non-illuminated or LED illuminated Color Options: Red, Green, Yellow, Blue, Black	100 pcs	1:4

32.	Select switches	Voltage:24V DC coil/230V AC coil Contact Configuration: 2 NO, 2 NC, or 1 NO + 1 NC Positions: 2-position or 3-position Illumination: Non-illuminated or LED illuminated Color Options: Black, Red, Green	100 pcs	1:4
33.	Digital multimeters	With voltage, current, and resistance measurement	25 pcs	1:1
37.	Network switches	Cisco 24 ports	2	1:12.5
38.	Routers	TP link	10	1:2.5
C	Consumable materials			
	Writing materials	Pens, pencils, paper	enough	
	Signal cables	0.5 mm ² to 1.5 mm ² , shielded	200 metres for each category	50m:1 for each category
	Jumper cables		200 metres for each category	50m:1 for each category
	Communication cables	Cat6 Ethernet cables	200 m	50m:1
	First aid kit	Full kits	5 pcs	1:5
D	Software			
	Programming and simulation software	Arduino IDE, Mbed	Installed in 25	1:1

			Computers	
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ELECTRICAL POWERLINES INSTALLATION

ISCED UNIT CODE: 0713 451 31A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/01/6/MA

UNIT DURATION: 140 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Install power systems

Unit description

This unit covers competences required in installing power system. Competences include conducting site survey, constructing electrical power lines, testing electrical power line and maintaining electrical power lines.

Summary of Learning Outcomes

By the end of this unit, the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Conduct site survey	10
2.	Design electrical powerlines	10
3.	Construct power lines	40
4.	Test electrical power lines	40
5.	Maintain electrical power lines	40
TOTAL		140

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Conduct site survey.	1.1 Definition of terms; 1.1.1 Site survey 1.1.2 Transmission line. 1.2 Relevant clauses in appropriate Acts e.g. 1.2.1 Occupational safety and health act (OSHA) 1.2.2 Work injury benefits act (WIBA) 1.2.3 Environment management and coordination Act (EMCA) 1.3 Relevant regulations: 1.3.1 IEEE regulations 1.3.2 KPLC by-laws 1.3.3 County by-laws	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	1.4 Types of surveys e.g. 1.4.1 Overhead line survey 1.4.2 Way leave survey 1.4.3 Cadastral survey 1.4.4 Topographical survey 1.5 Tools and equipment 1.6 Powerline elements 1.6.1 Conductors 1.6.2 Insulators 1.6.3 Line supports 1.6.4 Cross arms 1.6.5 D-iron 1.6.6 Damper 1.6.7 Bolt and nuts 1.7 Pole points marking 1.8 Conduct survey 1.9 Prepare survey report	
2. Design electrical power lines	2.1 Power lines design 2.1.1 Determination of requirements e.g voltage levels, power outputs, transmission distance, and current type. 2.1.2 Selection of transmission lines e.g overhead, underground and hybrid. 2.1.3 Design considerations e.g conductor selection and spacing, tower design and support structures, efficiency and power losses 2.2 Types of electrical transmission line designs 2.2.1 DC or AC 2.2.2 Short, medium or long 2.2.3 Single phase or multiphase 2.2.4 Single circuit or multi-circuit 2.2.5 Underground or overhead 2.3 Transmission line components 2.3.1 Conductors	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	2.3.2 Insulators 2.3.3 Cross arms 2.3.4 Line supports 2.3.5 Transformers 2.3.6 Lightning arrestors 2.3.7 Danger tags/plates 2.3.8 Connectors (line taps) 2.3.9 Earth conductors 2.3.10 Earthing wires 2.3.11 Fuses 2.3.12 Circuit breakers 2.3.13 Surge protectors 2.3.14 Voltage regulators 2.3.15 Isolators 2.4 Line parameters 2.4.1 Resistance 2.4.2 Inductance 2.4.3 Capacitance 2.5 Corona 2.5.1 Theory of corona formation 2.5.2 Factors affecting corona 2.5.3 Effects of corona 2.5.4 Methods or reducing corona effects 2.5.5 Critical disruptive voltage 2.5.6 Visual critical voltage 2.5.7 Power loss due to corona 2.6 Electrical transmission line drawings 2.7 BOQ Preparation. 2.8 Contract documents.	
3. Construct electrical power lines	3.1 Health and Safety Procedures 3.2 List and assemble tools, equipment and materials 3.3 Meaning of power line support 3.4 Types of powerlines supports 3.4.1 Wooden poles 3.4.2 Concrete (RRC)	• Practical • Project • Third party report • Portfolio of evidence • Written tests

Learning Outcome	Content	Suggested Assessment Methods
	3.4.3 Steel 3.5 Line fittings mounting 3.6 Line protection devices 3.7 Types of transmission line conductors 3.8 Line termination 3.8.1 Lugs 3.8.2 Droppers 3.9 Carry out Earthing 3.10 Housekeeping 3.11 Perform erection of power line support 3.12 Mount power line components 3.13 Terminate power line 3.14 Earthing 3.15 Housekeeping	• Oral questioning
4. Test electrical power lines	4.1 Visual inspection 4.2 Line integrity 4.2.1 Tension 4.2.2 Sag 4.2.3 Span size 4.2.4 Clearance/wayleave requirements 4.3 Earth resistance test 4.4 Perform power line tests 4.5 Inspection	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
5. Maintain electrical power lines	5.1 Maintenance schedules 5.2 Electrical equipment and system inspection 5.3 Maintenance materials and tools preparation. 5.4 Maintenance activities 5.5 Electrical power line tests 5.6 Maintenance records 5.7 Perform maintenance activities 5.8 Carry out system tests	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning

Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	V.K. Mehta Principles of Power system J.B. Gupta A course in Electrical power	5 pcs	1:5
2.	Installation manuals	Kenya power Electrical Safety Rules	5 pcs	1:5
3.	Charts	Single line diagram Circuit diagrams Colour codes Maps Substation layout	1 pcs for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
3	Lecture/theory room	60m ²	1	1:25
4	Workshop	150m ²	1	1:25
5	Site			
C	Consumable materials			
6	Electrical conductors		1 drums	1:25
7	Insulation tapes		25 pcs	1:1
8	Accessories	Insulators, surge diverters, switch gear,	25 pcs	1:1

		Stay, Earth rods		
9	Power line support		5 pcs	1:5
10	Transformer		1 pc	1:25
D	Tools and Equipment			
11	Hacksaws		25 pcs	1:1
12	Striping knives		25 pcs	1:1
13	Side cutters		25 pcs	1:1
14	Pliers		25 pcs	1:1
15	Tape measure		25 pcs	1:1
16	Try Square		25 pcs	1:1
17	Assorted hammers		25 pcs	1:1
18	Crimping tools		5 pcs	1:5
19	PPEs		25 pcs	1:1
20	Clamp meters		5 pcs	1:5
21	Earth resistance meter		5 pcs	1:5
22	Drilling machines		5 pcs	1:5
23	Fastening tools	Assorted spanners and wrenches, Lifting and tensioning tools	5 pcs	1:5
24	Climbers		5 pcs	1:5

MODULE VII

ENGINEERING MATHEMATICS IV

ISCED UNIT CODE: 0541 541 08A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/04/6/MA

Unit duration: 100 hours.

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Engineering mathematics

Unit Description: This unit describes the competencies required by a technician in order to apply technician mathematics. It enables the learner to; apply Fourier Series, apply multiple Integrals, apply complex variables, apply numerical methods and apply matrices II.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply Fourier Series	20
2.	Apply multiple Integrals	20
3.	Apply Complex Variables	20
4.	Apply numerical methods	20
5.	Apply matrices II	20
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods

1. Apply Fourier series	1.1 Definition of Fourier series for a function 1.1.1 Determination of Fourier series for period 2π to T 1.2 Determination of Fourier series of non-periodic functions 1.3 Determination of the Fourier series for non-periodic functions over a given range 1.4 Determination of the Fourier series for odd and even functions and half-range series	• Written assessment • Oral assessment • Practical • Project • Third party report • Portfolio of evidence
2. Apply multiple integrals	2.1 Definition of double and triple integrals 2.2 Using double integrals to find areas and volumes. 2.3 Apply double and triple integrals in polar, cylindrical and spherical coordinates. 2.4 Use of triple integrals in solving problems.	• Written assessment • Oral assessment • Practical • Project • Third party report • Portfolio of evidence
3. Apply matrices II	3.1 Definition of Eigen values and Eigen Vectors (Eigen values, Eigenvectors, characteristics polynomials, distinct Eigen values, Normalized Eigen Vectors) 3.2 Calculation of Eigen values and Eigen Vectors 3.3 Explanation of function of matrices (similarity transformation and properties, exponential and meaning, logarithms of matrices) 3.4 Definition of Jordan form of a matrix 3.5 State transition matrix (properties of continuous time transition matrix for a linear time varying system)	7. Written assessment 8. Oral assessment 9. Practical 10. Project 11. Third party report 12. Portfolio of evidence •

4. Apply complex variables	4.1 Functions of complex variable 4.1.1 Derivatives of analytic functions 4.2 Define analytic (regular) functions 4.2.1 Cauchy Riemann equation	• Written assessment • Oral assessment • Practical • Project • Third party report • Portfolio of evidence
5. Apply numerical methods	5.1 Define numerical methods 5.1.1 Application of iterative methods to solve equations 5.2 Definition of interpolation and extrapolation 5.3 Application of interpolation and extrapolation	• Written assessment • Oral assessment • Practical • Project • Third party report • Portfolio of evidence •

Suggested Delivery Methods

- Demonstration
- Discussions
- Practical
- Project work
- Direct instruction

List of Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			

9.	Textbooks	Engineering Mathematics by K.A. Stroud	5 pcs	1:5
		Advanced Engineering Mathematics by Erwin Kreyszig	5 pcs	1:5
B	Learning Facilities & infrastructure			
10.	Lecture/theory room	60m ²	1	1:25
11.	Computer	Operating System: 64-bit Windows 11 or 10 version 1809 or above Processor: 2.5 GHz (3+ GHz recommended), Memory: 8 GB (32GB recommended) Disk space: 10 GB Display: 1920 x 1080 resolution Display Card: 2 GB GPU (8 GB recommended) and DirectX 11 compliant (DirectX 12 recommended)	25 pcs	1:1
12.	Projector		1	1:25
13.	Interactive screen	Specifications: 77-inch interactive whiteboard with touch and pen functionality.	1	1:25
C	Software			
14.	MATLAB	License: Educational	Installed	1:1

		licenses available. Features: Matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, interfacing with programs in other languages.	in 25 computers	
15.	GeoGebra	License: Free educational software. Interactive geometry, algebra, statistics, and calculus applications	Installed in 25 computers	1:1
D	Consumables			
16.	Pens, pencils, rulers and paper	Whiteboard markers, 2H pencils, plastic rulers, A2 white papers	Enough	

ELECTRICAL PRINCIPLES II

ISCED UNIT CODE: 0713 541 11A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/05/6/MA

UNIT DURATION: 100 HOURS

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply electrical principles

UNIT DESCRIPTION

This unit describes competences required to apply electrical principles in their work. It involves conduct system earthing and protection, apply illumination principles, apply two port networks and apply electromagnetic field.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Conduct System Earthing and protection	20
2.	Apply illumination principles	30
3.	Apply Two Port networks	20
4.	Apply Electromagnetic field	30
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Conduct system earthing and protection	1.1 Application of System and equipment protection principles	• Practical • Project

	1.1.1 Protection zones 1.1.2 Protection systems 1.2 Perform Protection system design 1.2.1 Protection system Drawings 1.2.2 Protection system Device sizing 1.2.3 Protection system Location 1.3 Design Earthing systems 1.3.1 TT 1.3.2 TNC 1.3.3 TNCS 1.3.4 IT 1.3.5 TNS 1.4 Perform Test on an earthing system 1.4.1 Earth resistance test 1.4.2 Earth loop impedance test 1.5 Identification of various types of lightning strikes based on Benjamin Franklin 1.6 Perform Lightning system design 1.6.1 Lightning arrestors 1.6.2 Lightning design drawing 1.6.3 Size of lightning system	• Third party report • Portfolio of evidence • Written tests • Oral questioning
2. Apply illumination principles	2.1 Application of Laws of lighting 2.2 Calculations on Light requirements as laws of lighting 2.3 Selection of Electric luminaires 2.4 Design of lighting schemes	• Practical • Project • Third party report • Portfolio of evidence

		• Written tests • Oral questioning
3. Use two port networks	3.1 Analysis of basic passive networks based on black box technique 3.2 Determination of characteristic impedance 3.3 Derivation of ABCD constants 3.4 Application of cascaded networks as per network parameters. 3.5 Types of transmission lines and their applications	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning
4. Apply electromagnetic field theory	4.1 Identification of Electromagnetic radiation sources as per EN 300386 v1.6.1 4.2 Identification of detectors of Electromagnetic radiations as per EN 300386 v1.6 4.3 Application of electromagnetic waves as per EN 300386 v1.6.1 4.4 Identification of electromagnetics Laws based on Maxwell's equation. 4.5 Calculations of 4.6 Electromagnetic wave parameters based on Maxwell's equation. 4.6.1 Wavelength 4.6.2 Velocity 4.6.3 Frequency 4.7 Behaviours and effects of Electromagnetic waves 4.8 Identification of electrostatics terms.	• Practical • Project • Third party report • Portfolio of evidence • Written tests • Oral questioning

	4.9 Identification of Magneto statics terms 4.10 Identification of Electrodynamics laws 4.11 Identification of Energy conservation theorem as per the Internal energy 4.12 Calculation of Electromagnetic Energy flow as per the Maxwell's equations. 4.13 Calculation of Energy flow in an antenna as per the E-H propagation.	
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Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussions
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
11.	Textbooks	J. Bird Electrical and Electronic Principles	5 pcs	1:5

		V.K. Mehta & R. Mehta Basic Electrical Engineering		
12.	Installation manuals	Electronic components datasheets	5 pcs	1:5
13.	Charts	Circuit diagrams Colour codes	1 pcs for each	1:25
14.	Scientific Calculators		25	1:1
15.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
5.	Workshop	150m ²	1	1:25
C	Consumable materials			
1.	Connector wires	Jumper wires,	5 pkts	1:5
2.	Insulation tapes		25 pcs	1:1
6.	Circuit boards	Bread board, copper strip boards	25 pcs	1:1
7.	Assorted electronic components	Resistors, diodes, capacitors, transistors, ICs, Transformers, Inductors,	25 pcs	1:1

		Batteries		
8.	Soldering wires		5 rolls	1:5
D	Tools and Equipment			
15.	Striping knives		25 pcs	1:1
16.	Side cutters		25 pcs	1:1
17.	Pliers		25 pcs	1:1
18.	Assorted Screw driver		25 pcs	1:1
19.	Crimping tools		5 pcs	1:5
20.	PPEs		25 pcs	1:1
21.	Multimeters		5 pcs	1:5
22.	Oscilloscope		5 pcs	1:5
23.	Function generator		5 pcs	1:5
24.	Spectrum analyser		5 pcs	1:5
25.	Variable power supply		5 pcs	1:5
26.	Solder guns		25 pcs	1:1
27.	Hot air gun		5 pcs	1:5
28.	Work stations		25	1:1

CONTROL SYSTEMS

ISCED UNIT CODE: 0714 541 12A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/06/6/MA

DURATION: 100 Hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply control systems.

UNIT DESCRIPTION

This unit covers the competencies required to apply understanding of control systems. Competencies include applying basic concepts of control systems, system modelling, apply system performance Apply system compensation and apply servo systems.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply basic concepts of control systems	20
2.	Apply illumination principles	20
3.	Apply system performance	20
4.	Apply system compensation	20
5.	Apply servo systems	20
TOTAL		100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply basic concepts of control systems	1.1 Types of control systems 1.1.1 Man-made system 1.1.2 Natural system 1.1.3 Hybrid system 1.1.4 Controlled variable 1.2 Open and closed loop systems 1.3 Feedback 1.3.1 Positive feedback	• Practical • Portfolio of evidence • Third party report • Oral questioning • Written tests

	1.3.2 Negative feedback 1.3.3 Effects of feedback on overall gain, sensitivity, stability and noise. 1.4 Block diagrams 1.4.1 Basic elements of block diagram 1.4.2 Block diagram representation of electrical systems 1.4.3 Block diagram algebra 1.4.3.1 Series connection 1.4.3.2 Parallel connection 1.4.3.3 Feedback connection 1.5 Simplification of block diagrams 1.5.1 Canonical forms 1.5.2 Transfer functions 1.5.3 Superposition 1.6 Signal flow graphs 1.6.1 Basic elements of signal flow graphs 1.6.2 Construction of signal flow graphs 1.7 Simplification of system loops 1.7.1 Masons rule 1.7.2 Complex loop 1.7.3 Loop reduction 1.8 Conversion of block diagrams to signal flow graphs 1.8.1 Nodes 1.8.2 Sinks	
2. Apply system modelling	2.1 Need for modelling 2.2 Mathematical models 2.2.1 Differential equation model 2.2.2 Transfer function model 2.2.3 State space model 2.3 Derivation of transfer functions for simple networks 2.3.1 Electrical 2.3.2 Mechanical 2.3.3 Laplace transforms, $j\omega$, D-operations 2.4 Electrical-mechanical analogy 2.4.1 Force- Voltage 2.4.2 Force – current 2.4.3 Translational vs Rotational 2.5 Modelling of practical systems 2.5.1 Generators 2.5.2 Temperature control systems 2.5.3 Calculations	• Practical • Portfolio of evidence • Third party report • Oral questioning • Written tests

3. Apply system performance	3.1 Time response analysis 3.1.1 Transient response 3.1.2 Steady state response 3.2 Test signals 3.2.1 Unit step 3.2.2 Unit ramp 3.2.3 Unit parabolic 3.2.4 Unit impulse 3.3 Dynamic response for 1st and 2nd order systems 3.3.1 Response terms 3.3.2 Standard 2nd order equation 3.3.3 Response graphs 3.3.4 Derivation of dimensionless 2nd order equation 3.4 Damping methods 3.4.1 Velocity feedback 3.4.2 Error rate 3.4.3 Viscous friction damping 3.4.4 Effects of damping ratio 3.4.5 Calculation of limiting values 3.5 Time domain specifications 3.5.1 Delay time 3.5.2 Rise time 3.5.3 Peak time 3.5.4 Peak overshoot 3.5.5 Settling time 3.6 Types of system stability 3.6.1 Relative stability 3.6.2 Absolute stability 3.6.3 Asymptotic Stability 3.6.4 Marginal Stability 3.6.5 Bounded input-bounded output 3.7 System stability analysis methods 3.7.1 Routh's stability criterion 3.7.2 Nyquist diagrams 3.7.3 Bode plots 3.7.4 Nichol's chart 3.7.5 Root locus 3.8 Process control strategies 3.8.1 Proportional (P), 3.8.2 Integral (I), 3.8.3 Derivative (D), 3.8.4 Proportional plus Integral (PI),	• Practical • Portfolio of evidence • Third party report • Oral questioning • Written tests •
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	3.8.5 Proportional plus Derivative (PD), 3.9 Proportional plus Integral plus Derivative (PID)	
4. Apply system compensation	4.1 Need for compensation 4.2 Compensating networks transfer functions 4.3 Lead compensator 4.4 Lag compensator 4.5 Lead-lag compensator 4.6 Designing a Bode compensator 4.7 Compensation using 3-term controller	• Practical • Portfolio of evidence • Third party report • Oral questioning • Written tests
5. Apply servo systems	5.1 Servo mechanisms 5.1.1 Position 5.1.2 Speed 5.1.3 Acceleration 5.2 AC and DC Servo amplifiers 5.2.1 Linear 5.2.2 Pulse-width modulation 5.2.3 Digital 5.3 Operation of stepper motors 5.3.1 Construction 5.3.2 Control circuits 5.3.3 Calculations 5.3.4 Interfacing 5.3.5 Applications 5.4 Characteristics curves of ac and dc servo motors 5.4.1 Torque Vs Speed curve 5.4.2 Torque Vs Current curve 5.4.3 Efficiency Vs Load curve 5.5 Calculations	• Practical • Portfolio of evidence • Third party report • Oral questioning • Written tests •

Suggested Methods of Instruction

- Projects
- Practical
- Demonstration
- Group Discussions
- Direct Instruction
- Illustration

Recommended Resources for 25 trainees

S/No .	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	S.K.B Pearson Control Systems Theory and Application Control systems by Yousef Zadek	5 pcs of each	1:5
2.	Installation manuals	Electrical machine manuals	5 pcs	1:5
3.	Charts	Single line diagram Motor starting circuits Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Software	PLC software Arduino IDE Matlab		
5.	Projector and screen	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
2.	Workshop	150m ²	1	1:25

3.	Computer Laboratory	100m ²	1	1:25
C	Consumable materials			
1.	Electrical wires	1.5mm ² (red, black green)	1 rolls	1:5
		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
2.	Cables	USB, Ethernet, UART, 4-20mA current loop cable	5 pcs	1:5
3.	Insulation tapes		25 pcs	1:1
4.	Accessories	Sensors, Actuators, Limit switches, Push buttons, Timers, Relays, input/output modules, keyboard	25 pcs	1:1
5.	Pipes and trunkings	PVC conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			
1.	PLC module		5	1:5
2.	Microcontroller tool kit		5	1:5
3.	PPEs		25 pcs	1:1
4.	Multimeter		5 pcs	1:5

5.	Oscilloscope		5 pcs	1:5
6.	Servomotors		5 pcs	1:5

ELECTRICAL SYSTEMS AUTOMATION

ISCED UNIT CODE: 0714 551 36A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/02/6/MA

UNIT DURATION: 160 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Automate Electrical systems

Unit Description

This unit covers competences required in automating electrical systems. Competences include; designing automation system, installing automation components, programming automation system, commissioning automation system, monitoring automation system and maintaining automation system.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Design automation system	20
2.	Install automation components	40
3.	Program automation system	50
4.	Commission Automation System	20
5.	Monitor automation system	10
6.	Maintain automation system	10
TOTAL		160

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Design Automation system	1.1 Identifying the need for automation 1.1.1 Meaning of Automation system 1.1.2 Factors considered in choosing automation 1.1.3 Factors to consider in designing automation system 1.2 System analysis conducted 1.2.1 Functionality assessment 1.2.2 Methodology of system analysis 1.2.3 Determining set parameters e.g. performance criteria, constrains and expected outcomes 1.2.4 Recording of findings 1.3 Determining automation technology. 1.3.1 Types of automation technologies 1.3.1.1 PLC 1.3.1.2 Microcontroller 1.3.1.3 SCADA systems 1.3.1.4 Human Machine Interface 1.4 Developing system design plan 1.4.1 Types of system designs 1.4.2 Procedures of developing system plans 1.4.3 Tools 1.4.4 Drawings 1.5 Carry out Modelling, simulation and prototyping 1.5.1 Developing automation models and prototypes 1.5.2 Prototyping techniques 1.5.3 Conduct simulations 1.6 Preparation of BOQ 1.6.1 Understanding BOQ 1.7 Quantities and cost of materials	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
2. Install automation components	2.1 Safety procedures for installing automation components and systems 2.1.1 Relevant clauses in appropriate Acts e.g. WIBA 2.1.2 Environment management and coordination Act (EMCA) 2.1.3 Relevant regulations: 2.1.4 IET regulations 2.1.5 Causes of accidents 2.1.6 Types of PPE handling, use, maintenance and storage of different types of PPE 2.1.7 Classes of fires and firefighting equipment 2.1.8 First aid procedures 2.2 Assembling tools, materials and equipment 2.2.1 Tools e.g. 2.2.1.1 Cable strippers 2.2.1.2 Pliers 2.2.1.3 Screw drivers 2.2.1.4 Hammers 2.2.1.5 Chisels 2.2.1.6 Electrician knives 2.2.1.7 Crimping tools 2.2.1.8 Bending springs 2.2.1.9 Steel tapes 2.2.1.10 Hack saws 2.2.1.11 Drills 2.2.2 Equipment e.g. 2.2.2.1 Stock and die 2.2.2.2 Vice 2.3 Materials e.g. 2.3.1 PCB 2.3.2 Breadboards 2.3.3 Networking cables 2.4 Installation of conduits, trunks, enclosures and support systems 2.4.1 IET regulations 2.4.2 Automation layout 2.4.3 Laying of cable routes	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	2.5 Installation of automation components 2.5.1 IET regulations on automation 2.5.2 Industrial sensors 2.5.3 Types of actuators 2.5.4 Machines starters 2.5.5 PLC 2.5.6 Microcontroller 2.5.7 Scada systems 2.5.8 Human Machine Interface 2.6 Installing cables and conductors 2.6.1 Types of cables 2.6.2 Cable sizes 2.6.3 Right choice of cables 2.7 Cable lugging, glanding and termination 2.7.1 Meaning of terminologies 2.7.2 Types of cable joints, cable lugs 2.8 Cable labelling 2.8.1 Cables are labelled as per IET standards. 2.9 Performing Electrical Tests 2.9.1 IET regulations 2.9.2 Continuity tests 2.9.3 Insulation resistance test 2.10 Polarity test	
3. Program automation system	3.1 Meaning of programming in automation 3.2 Developing PLC programs 3.2.1 Types of PLC programming languages 3.2.1.1 Ladder diagram 3.2.1.2 Instruction List 3.2.1.3 Functional Block diagram 3.3 Developing Microcontroller programs 3.3.1 Types of programming languages 3.3.1.1 C/C++	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	3.3.1.2 Python 3.4 Configuration of components and their parameters 3.5 Testing and debugging of the program 3.6 Uploading program to automation systems. 3.6.1 Procedure of uploading and downloading 3.7 Configuring control parameters of automation system	
4. Commission automation system	4.1 Conducting unit test 4.1.1 Meaning of unit test 4.1.2 Regulations/standards - OEMS, IEE and KEBS standards. 4.1.3 Testing parameters 4.1.4 Procedure of unit testing 4.2 Conducting functional tests 4.2.1 Meaning of functional test 4.2.2 Testing parameters 4.2.3 Procedure of functional testing 4.3 Conducting system integration tests 4.3.1 Meaning of system integration tests 4.3.2 Regulations/standards - OEMS. 4.3.3 Testing parameters 4.3.4 Procedure of testing 4.4 Maintaining documentation 4.4.1 Purpose of documentation 4.4.2 Types of documentation 4.5 Conducting end user training 4.6 Conducting final inspection and validation 4.6.1 Safety standards relating to final inspection and validation. 4.6.2 Regulation requirements 4.6.3 Checklist relating to final inspection	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	4.7 Handing over automation system to the end user 4.7.1 Handover documentation 4.7.2 Final acceptance	
5. Monitor automation system	5.1 Monitoring and diagnostic tools 5.1.1 Meaning 5.1.2 Purpose 5.1.3 Developing monitoring and diagnostic tools 5.1.4 Types of monitoring and diagnostic tools eg. 5.1.4.1 HMI 5.1.4.2 Data logger 5.1.4.3 SCADA 5.2 Integrating monitoring and diagnostic tools 5.2.1 Methods of integration 5.2.2 Procedure of integrating 5.3 Extracting data from monitoring tools 5.3.1 Data types 5.3.2 Data extraction Techniques 5.3.3 Procedure of data extraction 5.4 Storing data 5.4.1 Data loggers 5.4.2 Storage systems 5.4.3 Data security 5.5 Analysing data 5.5.1 Analysis techniques 5.5.2 Key Performance Indicators 5.5.3 Data analysis tools 5.6 Carrying out System optimization 5.6.1 Identify optimization areas 5.6.2 Optimization strategies 5.6.3 Application of optimization strategies	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
6. Maintain automation system	6.1 Preparing maintenance schedule 6.1.1 Types of maintenance 6.1.2 Maintenance schedule 6.2 Preparing system maintenance check list 6.3 Selection of maintenance tools and equipment 6.4 Carrying out inspection and tests 6.5 Diagnosing faults 6.5.1 Types of faults in automation system 6.5.2 Procedure of fault diagnosis 6.6 Rectifying faults 6.6.1 Repair activities 6.6.2 Replacement 6.7 Preparing maintenance report	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Suggested Methods of Instruction

- Practical
- Project
- Demonstration
- Direct instruction
- Group Discussion

Recommended Resources for 25 trainees

S/No.	Item	Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	S.K.B Pearson Control Systems Theory and Application. Fundamentals of automation and industrial control systems by Ayman Aly El Naggar. Fundamentals of Industrial Automation by Dobrivojic Popovic and Vijay P.	5 pcs	1:5
2.	Installation manuals	Electrical machine manuals Siemens PLC manual	5 pcs	1:5
3.	Charts	Single line diagram Motor starting circuits Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Software	PLC SCADA Arduino IDE language		
5.	Power point presentations	For trainer's use	1	1:25

B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
2.	Workshop	150m ²	1	1:25
C	Consumable materials			
1.	Electrical wires	1.5mm ² (red, black green)	1 rolls	1:5
		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
2.	Cables	USB, Ethernet, UART, 4-20mA current loop cable	5 pcs	1:5
3.	Insulation tapes		25 pcs	1:1
4.	Accessories	Sensors, Actuators, Limit switches, Push buttons, Timers, Relays, input/output modules, keyboard	25 pcs	1:1
5.	Pipes and trunkings	PVC conduits, Mini trunking	25 pcs	1:1
D	Tools and Equipment			
1.	PLC module		5	1:5
2.	Microcontroller tool kit		5	1:5
3.	Hacksaws		25 pcs	1:1

4.	Striping knives		25 pcs	1:1
5.	Side cutters		25 pcs	1:1
6.	Pliers		25 pcs	1:1
7.	Tape measure		25 pcs	1:1
8.	Try Square		25 pcs	1:1
9.	Spirit level		25 pcs	1:1
10.	Assorted Screw driver		25 pcs	1:1
11.	Assorted hammers		25 pcs	1:1
12.	Crimping tools		5 pcs	1:5
13.	PPEs		25 pcs	1:1
14.	Multimeter		5 pcs	1:5
15.	Clamp meters		5 pcs	1:5
16.	Earth resistance meter		5 pcs	1:5
17.	Bending spring		5 pcs	1:5
18.	Drilling machines		5 pcs	1:5
19.	Work stations		25	1:1
20.	Installation boards		13 pcs	1:2

MODULE VIII

RESEARCH METHODS

ISCED UNIT CODE: 0111 541 21A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/07/6/MA

UNIT DURATION: 80 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Research Methods

Unit Description

This unit covers the competencies required to apply research methods. Competencies include; identifying research problem, conducting literature review, developing research methodology, analyzing collected data and preparing research report.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Identify research problem	10
2.	Conduct literature review	10
3.	Develop Research Methodology	20
4.	Analyse collected data	20
5.	Prepare research report	20
TOTAL		80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Identify research problem	1.1 Methods of identifying research problem 1.2 Gathering background information 1.3 Research variables 1.4 Research objectives	• Practical • Demonstration • Projects • Written tests • Oral test

	1.5 Formulating research questions 1.6 Significance of research 1.7 Establishment of scope of study	
2. Conduct literature review	2.1 Sources of literature review 2.1.1 Primary 2.1.2 Secondary 2.2 Keywords and phrases 2.3 Ethical research guidelines on referencing and citation 2.4 Organization and reporting of collected literature	• Practical • Demonstration • Projects • Written tests • Oral test
3. Develop Research Methodology	3.1 Types of research designs 3.1.1 Descriptive 3.1.2 Correlational 3.1.3 Experimental 3.1.4 Longitudinal 3.1.5 Cross-sectional 3.2 Identification of study population 3.3 Sampling techniques 3.3.1 Simple random 3.3.2 Systematic 3.3.3 Stratified 3.3.4 Clustered 3.4 Research proposal and budget preparation	• Practical • Demonstration • Projects • Written tests • Oral test
4. Analyze collected data	4.1 Methods of data collection 4.1.1 Surveys and questionnaires 4.1.2 Interviews 4.1.3 Focus groups 4.1.4 Observations 4.1.5 Case studies 4.2 Research guidelines on data collection 4.3 Data cleaning 4.4 Data analysis tools 4.4.1 Statistical software 4.4.2 Qualitative analysis software 4.4.3 Spreadsheet software	• Practical • Demonstration • Projects • Written tests • Oral test

	4.5 Data presentation 4.5.1 Data visualization tools 4.5.2 Presentation software	
5. Prepare research report	5.1 Discussion of research findings 5.2 Drawing of conclusions based on findings 5.3 Recommendations 5.4 Referencing systems 5.4.1 APA 5.4.2 MLA 5.4.3 Harvard 5.4.4 IEEE 5.5 Appendices 5.6 Research report presentation	• Practical • Demonstration • Projects • Written tests • Oral test

Suggested Methods of Instruction

- Practical
- Lab and field projects
- Demonstrations
- Group discussions
- Direct instructions
- Role playing
- Interactive lectures
- Individual assignments
- Viewing of related videos

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference books	• Research Methodology: A Step-by-Step Guide for Beginners by Ranjit Kumar	5 pcs	1:5

		Research Methodology: Tools And Techniques By Dr. Prabhat Pandey and Dr. Meenu Mishra Pandey		
2.	Charts	Research design flow chart	1 pc for each	1:25
3.	Software	Data analysis software i.e SPSS, Microsoft Excel Data presentation software i.e. Microsoft powerpoint, Tableau, Canva, Microsoft Excel. Planning and management software. i.e. Google Calendar	25	1:1
4.	Audio visual presentations	Projector	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
2.	Computer laboratory	100m ²	1	1:25
3.	Research Centre	100m ²	1	1:25
C	Consumable materials			
1.	Field research materials	Sampling supplies i.e. soil, bags, notebook, battery and fuel	25 pcs	1:1
2.	Social research materials	Surveys and questionnaires	25 pcs	1:1
3.	Laboratory	Chemical reagents, glassware	25	1:5

	research materials	and plastic ware		
D	Tools and Equipment			
1.	Assorted research and data collection tools and equipment	Camera, Digital storage device, computer,	25 pcs	1:1

ELECTRICAL PROJECT SUPERVISION

ISCED UNIT CODE: 0713 551 22A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/08/6/MA

UNIT DURATION: 60 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: Supervise Electrical Project

UNIT DESCRIPTION

This unit covers competences required to manage an electrical project. Competences include preparing work schedule, preparing safety work plan, leading project teams, supervising materials tools and equipment, monitoring project implementation and commissioning electrical project.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Prepare work schedule	10
2.	Prepare safety work plan	10
3.	Lead project teams	10
4.	Supervise materials tools and equipment	10
5.	Monitor project implementation	10
6.	Commission electrical project	10
TOTAL		60

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Prepare work schedule	1.1 Determination of scope of work 1.1.1 Meaning of terms	• Practical demonstration • Projects • Written tests • Oral test

Learning Outcome	Content	Suggested Assessment Methods
	1.1.2 Project work planning 1.1.3 Elements of project work planning 1.1.4 Factors to consider in project planning 1.1.5 Project objectives 1.1.6 Project lifecycle 1.2 Identification of project work resources 1.2.1 Finance 1.2.2 Personnel 1.2.3 Consultancy 1.2.4 Materials 1.2.5 Tools 1.2.6 Storage facilities 1.2.7 Buildings 1.3 Identification of project work tasks 1.3.1 Objectives 1.3.2 Activities 1.3.3 Timeframe 1.3.4 Performance indicators 1.3.5 Resource required 1.3.6 Outputs 1.3.7 Resource allocation 1.4 Identification of project logistics 1.4.1 Budget 1.4.2 Finance	

Learning Outcome	Content	Suggested Assessment Methods
	1.4.3 Resource loading and scheduling 1.4.4 Resource constrains and tradeoff 1.4.5 Resource monitoring and control 1.4.6 Progress reports 1.4.7 Transportation and delivery 1.5 Develop a project organization structure 1.5.1 Top management 1.5.2 Supervisors 1.5.3 Workers 1.5.4 Suppliers 1.5.5 Financiers 1.6 Develop project policies 1.6.1 Cost control policy 1.6.2 Communication policy 1.6.3 Compliance and legal policy 1.6.4 Ethics and integrity policy 1.6.5 Risk management policy 1.6.6 Safety policy 1.6.7 Quality assurances policy 1.6.8 Environmental policy 1.6.9 Procurement and vender management	

Learning Outcome	Content	Suggested Assessment Methods
	policy 1.7 Determine project time span 1.7.1 Gantt charts 1.7.2 Work breakdown structure (WBS) 1.7.3 Network diagrams: critical path analysis, program evaluation and review technique (PERT) 1.7.4 Task sequencing 1.8 Project crashing	
2. Prepare safety work plan	2.1 Meaning of terms 2.2 Identification of risks are 2.2.1 Understanding risks in electrical projects 2.2.2 Operational risks 2.2.3 Short term strategic risks 2.2.4 Long term strategic risks 2.3 Conduct risk analysis 2.3.1 EHS Standards 2.3.2 SWOT analysis 2.4 Determine risk mitigation plans 2.4.1 Design and technical risks mitigation 2.4.2 Procurement and supply chain mitigation 2.4.3 Operational risks mitigation 2.4.4 Safety risk mitigation 2.4.5 Regulatory and risk mitigation 2.4.6 Financial risk mitigation 2.4.7 Scheduling and time mitigation 2.4.8 Environmental risk mitigation 2.4.9 Communication risk mitigation 2.5 Continuous monitoring and	• Practical demonstration • Projects • Written tests • Oral test

Learning Outcome	Content	Suggested Assessment Methods
	review	
3. Lead project team	3.1 Identification of project team 3.1.1 Project manager 3.1.2 Lead engineer 3.1.3 Construction manager 3.1.4 Procurement manager 3.1.5 Quality control manager 3.2 Develop Job descriptions of the team 3.2.1 Recruitment of staff 3.2.2 Qualifications 3.2.3 Competencies 3.2.4 Experience 3.3 Communicate Objectives of the project 3.3.1 Project policy 3.3.2 Communication channels 3.3.3 Importance of objectives 3.4 Delegate the project activities 3.4.1 Performance contracts 3.4.2 Effective delegation 3.4.3 Concept of authority, responsibility and accountability 3.4.4 Importance of delegations 3.4.5 Challenges 3.5 Adherence to OSHA and EHS 3.5.1 Monitoring and evaluation 3.5.2 Importance of safety and healthy practices 3.6 Training of project team 3.6.1 Types of trainings 3.6.2 Administrative trainings 3.6.3 Financial management 3.6.4 Communication 3.6.5 Safety 3.7 Direct teams on the expected output 3.7.1 Results oriented management (ROM) 3.8 Results agreement management (RAM)	• Practical demonstration • Projects • Written tests • Oral test

Learning Outcome	Content	Suggested Assessment Methods
4. Supervise Materials, tools, and equipment.	4.1 Identification of tools, materials and equipment 4.1.1 Types of electrical tools, materials and equipment 4.1.2 Quality aspect 4.1.3 Condition of tools 4.2 Perform auditing of tools, materials and equipment 4.2.1 Audit criteria 4.2.2 Importance of auditing 4.2.3 Types of audits 4.2.4 Disposal Act 4.3 Develop tools, material and equipment inventory system 4.3.1 Resources 4.3.2 Types of inventory systems 4.3.3 Responsibility 4.4 Classification of tools, materials and equipment 4.4.1 Types of classifications 4.4.2 Importance of tool classification 4.4.3 Responsibility 4.4.4 Functionality and operation 4.5 Maintenance of tools, materials and equipment 4.5.1 EHS standards 4.5.2 Maintenance schedules 4.5.3 Maintenance trainings 4.5.4 Importance of maintenance 4.5.5 Types of tools, equipment and material maintenance	• Practical demonstration • Projects • Written tests • Oral test
5. Monitor project implementation	5.1 Meaning of terms 5.2 Preparation of monitoring tools 5.2.1 Importance of monitoring 5.2.2 Types of monitoring 5.2.3 Schedules 5.2.4 Resources 5.2.5 Gantt charts 5.3 Monitoring project activities	• Practical demonstration • Projects • Written tests • Oral test

Learning Outcome	Content	Suggested Assessment Methods
	5.3.1 Responsibilities 5.3.2 Schedules of monitoring 5.3.3 Reallocation of duties and resources 5.3.4 Report writing 5.4 Assessment of quality of work 5.4.1 Assessment methods 5.4.2 Lead person 5.4.3 Importance of quality assessment 5.4.4 Quality policy 5.4.5 Consultancy services 5.5 Planning for short range action 5.6 Evaluation of project progress 5.7 Preparing monitoring reports	
6. Commission electrical project	6.1 Constitute a commissioning panel 6.1.1 Leaders and guests 6.2 Develop a program with requirements 6.2.1 Commissioning procedure 6.2.2 End-user training 6.2.3 Completion certificate 6.2.4 Handover documents 6.2.5 User manuals 6.2.6 Post commissioning support 6.3 Adherence to safety procedures 6.3.1 OSHA standards. 6.3.2 Conduct functionality tests 6.4 Preparation of documents 6.4.1 Issuance of certificates 6.4.2 Work commencement certificate 6.4.3 Test certificates 6.4.4 Inspection certificate 6.4.5 Completion certificate 6.5 Prepare report and share with relevant parties 6.5.1 Report writing 6.5.2 Organization procedures 6.6 Conduct end-user training	• Practical demonstration • Projects • Written tests • Oral test

Learning Outcome	Content	Suggested Assessment Methods
	6.7 Project handover	

Suggested Methods of Instruction

1. Practical
2. Project
3. Demonstration
4. Direct instruction

List of Recommended Resources for 25 trainees

S/No.	Item	Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	R. Tricker Electrical project management B. Scaddan Electrical installation work J. Hyde Electrical Installation Principles and Practices	5 pcs	1:5
2.	Charts	Critical path Chart	1 pcs each	1:25
3.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			

1.	Lecture/theory room	60m ²	1	1:25
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ELECTRICAL MEASUREMENT AND FAULT DIAGNOSIS

ISCED UNIT CODE: 0713 541 20A

TVET CDACC UNIT CODE: ENG/CU/EI/CC/09/6/MA

UNIT DURATION: 80 HOURS

Relationship to Occupational Standards

This unit addresses the unit of competency: **Perform Electrical Measurements and Fault Diagnosis**

UNIT DESCRIPTION

This unit covers competences required to perform electrical measurement and fault diagnosis. Competences include applying electrical measurement instruments, applying waveform analyzing instruments, applying sensors and transducers and calibrating measurement instruments.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Apply electrical measurement instruments	10
2.	Apply waveform analysing instruments	30
3.	Apply sensors and transducers	30
4.	Calibrate measurement instruments.	10
TOTAL		80

Learning Outcome	Content	Suggested Assessment Methods
1. Apply electrical measurement instruments	1.1 Health and safety procedures e.g. PPEs, Hazards, Accidents etc. 1.2 Electrical Quantities e.g. 1.2.1 AC/DC Current 1.2.2 AC/DC Voltage 1.2.3 Power 1.2.4 Resistance	• Practical demonstration • Written assessment • Report writing • Project • Portfolio of evidence

Learning Outcome	Content	Suggested Assessment Methods
	1.2.5 Capacitance 1.2.6 Inductance 1.3 Electrical Units e.g. 1.3.1 Amperes 1.3.2 Volts 1.3.3 Watts 1.3.4 Ohms 1.3.5 Farads 1.3.6 Henrys 1.4 Measurement Standards 1.4.1 International 1.4.2 Primary 1.4.3 Secondary 1.4.4 Working 1.5 Electrical Measuring Instruments e.g. 1.5.1 Voltmeter 1.5.2 Ammeters 1.5.3 Ohmmeter 1.5.4 Multimeter etc 1.6 Converters e.g. Analogue to Digital converters, Digital to Analogue converters etc. 1.7 Measurement presentation mechanisms e.g. Display, Recording 1.8 Performance characteristics 1.8.1 Static characteristics e.g. precision, resolution repeatability, reproducibility, drift etc 1.8.2 Dynamic characteristics e.g. fidelity, measuring lag, measuring error etc	• Third-party reports

Learning Outcome	Content	Suggested Assessment Methods
2 Apply waveform analyzing instruments	2.1 Health and safety procedures e.g. PPEs, Hazards, Accidents etc. 2.2 Types of Waveforms e.g. Sinusoidal wave, step, ramp, impulse etc. 2.3 Waveform Analysis Instruments 2.3.1 Type's e.g. Oscilloscope Spectrum analysis, wave analysers etc 2.3.2 Construction of waveform analysis equipment	• Practical demonstration • Written assessment • Report writing • Project • Portfolio of evidence • Third-party reports
3 Apply sensors and transducers	3.1 Definition of terms 3.2 Transducers/Sensors 3.2.1 Types/principles of operation transducers/sensors, e.g. restive, capacitive e.t.c 3.2.2 Sensors/transducer e.g. displacement, force, flow pressure etc. 3.3 Signal processors 3.3.1 Definition of terms 3.3.2 Importance 3.3.3 Methods used in signal processing e.g. amplification, Attenuation Filtration 3.4 Data analysis and documentation 3.5 Signal presentation	• Practical demonstration • Written assessment • Report writing • Project • Portfolio of evidence • Third-party reports
4 Calibrate measurement instruments	4.1 Health and safety procedures e.g. PPEs, Incident reporting, emergency procedures training and awareness etc. 4.2 Calibrators 4.5.1 Meaning 1.2.1 Types e.g. signal generator, oscilloscope, Data logger etc 4.3 Steps in calibration 4.4 Measurement of initial values 4.4.1 Integration of measuring	• Practical demonstration • Written assessment • Report writing • Project • Portfolio of evidence • Third-party reports

Learning Outcome	Content	Suggested Assessment Methods
	instrument into calibrating instrument e.g. compatibility, settings configuration etc 4.4.2 Source the physical variables from calibrator 4.4.3 Measuring current physical variable 4.4.4 Measurement error calculation e.g. parallax, relative, environmental, instrumental, random etc. 4.4.5 Resetting the readings of calibrated instrument e.g. to zero error, span error 4.5 Calibration documentation 4.5.1 Traceability table 4.5.2 Repeatability table 4.5.3 Calibration Certificate 4.5.4 Equipment tag	

Suggested Methods of Instruction

- Discussions
- Demonstration by trainer
- Practice by the trainee
- Field trips
- On-job-training

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			

5.	Textbooks	A Course in Electrical and Electronic Measurements and Instrumentation by A.K. Sawhney, Puneet Sawhney,	5 pcs	1:5
6.	User manuals		5 pcs	1:5
7.	Measuring instruments manuals	Voltmeter Multimeters Ammeters Megaohmmeter	5pcs	1:5
8.	Charts	Single line diagram Circuit diagrams Colour codes	1 pc for each	1:25
9.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
2.	Workshop	160m ²	1	1:25
3.	Site			
C	Consumable materials			
1.	Electrical wires	1.5mm ² (red, black	5 rolls	1:5

		green)		
		2.5mm ² (red, black green)	5 rolls	1:5
		4.0 mm ² (red, black green)	3 rolls	1:10
		6.0 mm ² (red, black green)	2 rolls	1:12
		10 mm ² (red, black green)	2 rolls	1:12
2.	Insulation tapes		25 pcs	1:1
3.	Accessories	Breadboard, strip board, resistors, capacitors, inductors, transformers, power supply units, function generator, signal tracers, diagnostic software, sensors etc	25 pcs each	1:1

POWER ELECTRONIC CIRCUITS FABRICATION

ISCED UNIT CODE: 0714 551 35A

TVET CDACC UNIT CODE: ENG/CU/EI/CR/03/6/MA

UNIT DURATION: 150 hours

Relationship to Occupational Standards

This unit addresses the unit of competency: Fabricate power electronic circuits

Unit description

This unit covers competences required in fabricating power electronic circuits. Competences include designing power converter circuit, fabricating power converter circuit, testing power converter circuit and maintaining power converter circuit.

Summary of Learning Outcomes

By the end of the unit the trainee should be able to:

S/NO	Learning Outcome	Duration (Hours)
1.	Design power converter circuits	20
2.	Fabricate power converter circuit	80
3.	Test power converter circuit	30
4.	Maintain power converters	20
TOTAL		150

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Design power converter circuits	1.1 Identifying the need for power conversion 1.2 Carrying out system analysis 1.3 Determining power conversion technology 1.3.1 Power converter devices 1.3.1.1 Power diode 1.3.1.2 SCR 1.3.1.3 Power BJT 1.3.1.4 Power MOSFET 1.3.1.5 IGBT 1.4 Types of AC to DC Conversion 1.4.1 Uncontrolled Rectification	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	1.4.1.1 Single phase 1.4.1.2 Three phase 1.4.2 Controlled rectification 1.4.2.1 Single phase 1.4.2.2 Three phase 1.4.3 Voltage multipliers 1.4.4 DC to AC Conversion 1.4.4.1 Voltage source inverters 1.4.4.2 Current source inverters 1.4.4.3 Application of inverters 1.4.5 AC to AC Conversion 1.4.5.1 Cyclo-converters 1.4.5.2 AC voltage controllers 1.4.6 DC to DC Conversion 1.4.6.1 DC choppers types 1.4.6.2 Application of dc choppers 1.5 Conducting modelling, prototyping and simulation 1.5.1 Modelling techniques 1.5.2 Circuit designs 1.5.3 Electronic CAD 1.6 Preparing Bill of Quantities	
2. Fabricate power converter circuit	2.1 Adhering to health and safety procedures 2.1.1 OSHA regulations 2.1.2 PPEs 2.1.3 Types of hazards 2.1.4 Safety procedures 2.1.5 warnings sign 2.2 Assembling Tools, materials and equipment 2.2.1 Tools e.g. 2.2.1.1 Cable strippers 2.2.1.2 Pliers 2.2.1.3 Screw drivers 2.2.1.4 Electrician knives 2.2.1.5 Crimping tools 2.2.1.6 Measuring tape 2.2.1.7 Soldering Gun 2.2.1.8 Solder sucker 2.2.1.9 Side cutter 2.2.1.10 Multimeter 2.2.1.11 Hack saws 2.2.1.12 Precision kit	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	2.2.2 Equipment e.g. 2.2.2.1 Oscilloscope 2.2.2.2 Vice 2.2.2.3 Workstation bench 2.2.2.4 Etching equipment 2.2.3 Materials e.g. 2.2.3.1 Relays 2.2.3.2 Diodes 2.2.3.3 Power BJT 2.2.3.4 Power MOSFET 2.2.3.5 SCR 2.2.3.6 IGBT 2.2.3.7 Transformers 2.2.3.8 PCB 2.2.3.9 Strip board 2.2.3.10 Power supply 2.2.3.11 Switches 2.2.3.12 Breadboards 2.2.3.13 Etching chemicals 2.3 Identifying Printing technique 2.3.1 Types of printing techniques 2.4 Printing of circuit board layout 2.4.1 Printing materials 2.4.2 Printing techniques 2.4.2.1 Inkjet 2.4.2.2 Laser 2.4.2.3 3D printing 2.5 Etching of PCB 2.5.1 Chemicals used 2.5.2 Procedure of etching 2.6 Mounting of components on PCB. 2.7 Soldering components 2.8 Powering power converter circuit	
3. Test power converter circuit	3.1 Conducting unit test 3.1.1 Meaning of unit test 3.1.2 Testing parameters 3.1.3 Procedure of unit testing 3.2 Conducting output tests 3.2.1 Testing output parameters- voltage, current, frequency 3.2.2 Functional test 3.2.3 Procedure of functional testing 3.3 Conducting fault diagnosis.	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Learning Outcome	Content	Suggested Assessment Methods
	3.3.1 Types of faults in power conversion circuit 3.3.2 Procedure of fault diagnosis 3.4 Rectifying faults 3.4.1 Repair activities 3.5 Replacement	
4. Maintain power converters	4.1 Preparing maintenance schedule 4.1.1 Types of maintenance 4.1.2 Maintenance schedule 4.2 Preparing system maintenance check list 4.3 Selection of maintenance tools and equipment 4.4 Carrying out inspection and tests 4.5 Diagnosing faults 4.5.1 Types of faults in power electronic circuits 4.5.2 Procedure of fault diagnosis 4.6 Rectifying faults 4.6.1 Repair activities 4.6.2 Replacement 4.7 Preparing maintenance report	• Practical • Projects • Third party report • Portfolio of evidence • Written tests • Oral questioning

Suggested Methods of Instruction

- Practical
- Project
- Demonstration
- Direct instruction

List of Recommended Resources for 25 trainees

S/No.	Item	Specifications	Quantity	Recommended
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				Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	R L. Boylestad Electronic devices and circuit theory V.K Mehta Principles of Electronics	5 pcs	1:5
2.	Installation manuals	Electronic components datasheets	5 pcs	1:5
3.	Charts	Circuit diagrams Colour codes	1 pcs for each	1:25
4.	Power point presentations	For trainer's use	1	1:25
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25
2.	Workshop	150m ²	1	1:25
3.	Computer laboratory	100m ²	1	1:25
C	Consumable materials			
1.	Connector wires	Jumper wires,	5 pkts	1:5
2.	Insulation tapes		25 pcs	1:1
3.	Circuit boards	Bread board, copper strip boards	25 pcs	1:1
4.	Assorted electronic components	Resistors, diodes, capacitors, transistors,	25 pcs	1:1

		ICs, Transformers, Inductors,		
5.	Soldering wires		5 rolls	1:5
D	Tools and Equipment			
1.	Striping knives		25 pcs	1:1
2.	Pliers		25 pcs	1:1
3.	Assorted Screw driver		25 pcs	1:1
4.	Crimping tools		5 pcs	1:5
5.	PPEs		25 pcs	1:1
6.	Multimeters		5 pcs	1:5
7.	Oscilloscope		5 pcs	1:5
8.	Function generator		5 pcs	1:5
9.	Spectrum analyser		5 pcs	1:5
10.	Variable power supply		5 pcs	1:5
11.	Trainers kit	Digital electronics, PWM kit	5 pcs	1:5
12.	PCB prototyping machine		2	1:13
13.	Solder guns		25 pcs	1:1
14.	Hot air gun		5 pcs	1:5
15.	Drilling machines		5 pcs	1:5
16.	Work stations		25	1:1